

MGEOSE_31

M.Sc., Geography

Semester - III



WATERSHED MANAGEMENT

Department of Geography
School of Sciences
Tamil Nadu Open University
Chennai – 600 015.

MASTER OF SCIENCE IN GEOGRAPHY

WATERSHED MANAGEMENT

MGEOSE-31

SEMESTER-III

Department of Geography,
School of Sciences
Tamil Nadu Open University
577, Anna Salai, Saidapet, Chennai - 600 015
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May 2023

Name of Programme:	M.Sc Geography
Course Code:	MGEOSE-31
Course Title:	Watershed Management
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May 2023 (First Edition)

ISBN No: 978-93-5753-015-6

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Vice Chancellor

Date:09.03.2023

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MGEOSE 31: Watershed Management
Syllabus Details

Block 1: Introduction to Watershed management

1. Philosophy and Concept of Watershed management
2. Components of watershed: land, water and vegetation
3. Physical and Socio-Economic characteristics of watershed
4. Watershed Delineation and codification

Block 2: Function and Structure of Watershed

5. Morphological Characteristic of Watershed
6. Climatic characteristics of Watershed

Block 3: Watershed Analysis:

7. Land capability classification of watershed
8. Soil resource evaluation
9. Slope analysis and vegetation cover

Block 4: Integrated Watershed Management

10. Objectives of IWM
11. Land and Water conservation practices
12. Importance of land use planning in watershed development
13. Water harvesting and afforestation, population, and livestock development.

Block 5: Development and Management of Watershed

14. Participatory Rural Appraisal in Watershed Programme: basic principle, fundamentals, and Different Tools Employed in PRA
15. Empowerment of Women and other gender issues
16. Problems and propriety of Watershed Management.

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Block 1

Introduction to Watershed management

Unit 1: Philosophy and Concept of Watershed management

Unit 2: Components of watershed: land, water, and vegetation

Unit 3: Physical and Socio-Economic characteristics of watershed

Unit 4: Watershed Delineation and codification

Unit 1

Philosophy and Concept of Watershed management

Structure

Learning Objectives

Overview

1.1 Concept and Definition of Watershed

1.2 Scope of Watershed Management

1.3 Watershed Management: Indian and Global Perspective

1.4 Timeline of Watershed Management Programmes in India

1.5 Problems and Prospects in Watershed Management

1.5.1 Problems and Constraints in Watershed Management

1.5.2 New Prospects and Opportunities Associated with Watershed

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to understand the following,

- Concept and Definition and Scope of Watershed
- Watershed Management: Indian and Global Perspective
- Timeline of Watershed Management Programmes in India

Overview

A watershed is a natural hydrologic entity that encompasses a 'specific area' stretch of land surface where from rainfall or run off flows to a 'specific defined drain' be it a channel/nullah, small stream or river. Watersheds are considered appropriate units for both surveys and assessment of soil and land resources as well as for planning and implementation of various developmental programmes. Owing to optimum interaction and synergetic effect of land and water resources within a watershed, watershed approach,

over the period, has become popular in various developmental programmes in the country.

The process of delineation of watersheds has a scientific and rational approach. However, for presenting them as units for planning and development, formulation of an appropriate framework for watersheds is a pre-requisite. Such framework shall not only follow a hierarchical system of delineating bigger hydrologic units into watersheds but also include a codification system so that each watershed or units of interventions can be identified as an 'unique' individual entity with clear identifiable linkage with higher hierarchical units, that is, basin, catchments, subcatchment.

The concept of stream order is often followed in geomorphic analysis of natural drainage systems. However, framework of watersheds requires a different approach also indicating macro and micro level of delineation so that programmatic interventions units are of manageable sizes. Each of the big drainage systems needs to be divided and subdivided through stages.

The framework applied in this document employs popular terms like basin, catchment, subcatchment, watershed for denoting various stages of delineation from macro to micro levels.

The size of sub-watersheds is, however, delineated keeping in view of its relative importance in land and water resource development programmes. On the other hand, the size of smallest hydrologic units which are below sub-watersheds are generally governed by viability and working feasibility of implementation of various developmental programmes.

1.1 Concept and Definition of Watershed

The word 'watershed' has different meanings. In British English it means a ridge line or a line which shows slopes in two different directions on its either side. A ridge line is also a line connecting the points of highest elevation in a terrain. Therefore, ridge line is also known as 'watershed line' or a 'surface water divide'. In colloquial language the word 'watershed' is used to describe a path breaking event.

The word 'watershed' is used as a synonym for 'catchment' or 'basin' wherein rainwater or storm water gets collected from an area enclosed by a ridge line. This water eventually flows through the various drainage channels which merge with one another to form one or rarely more than one outfall(s) of a stream. Thus, 'watershed' is defined as an area enclosed within a watershed line. In this course, the word 'watershed' is used for a small basin or a small catchment representing a hydrological unit which

drains all its rainwater into a stream. Therefore, it is independent in terms of its water in general and surface water in particular.

To distinguish a watershed -which generally implies a small catchment or a basin, Bali (1980) suggested an upper area limit of 2,000 km^2 for a watershed. This classification is an extension of the classification suggested by Rao (1975) for large river basins -with an area greater than 20,000 km^2, medium river basins –with an area between 2,000 and 20,000 km^2 and small river basins commonly referred to as watersheds.

Bali's classification of watersheds was probably reflected in the watershed classification by the All-India Soil and Land Use Survey (AISLUS) in 1990. According to this classification, watersheds are further classified into 5 categories based on their areas as 'macro-watersheds - with area between 500 and 2,000 km^2, 'sub-watersheds' –with area between 100 and 500 km^2, 'milli-watersheds' –with area between 10 and 100 km^2, 'mini-watersheds' –with area between 1 and 10 km^2 as well as 'micro-watersheds' –with area less than 1 km2.

A watershed is a physical entity consisting of the natural elements in it such as plants of various sizes and types which grow over various types of soil or rock layers. Additionally, watershed also comprises of all the artificial elements such as roads, bridges, tunnels, buildings, and burrow holes etc. which are mostly introduced in it by human beings and sometimes by other animals. In the next section, we shall discuss about the scope of watershed management.

1.2 Scope of Watershed Management

As we have already seen in the previous section, watersheds represent small basins. By delineating the ridgelines in a medium or a large river basin, the entire basin can be subdivided into a number of watersheds, each with an area within 2,000 km^2. Because of their compact size, it is always easier to manage watersheds rather than a river basin. In a well-managed watershed, all the natural resources such as soil, water, vegetation, etc. are conserved.

Vegetation or plants play a vital role in conserving the natural resources of a watershed such as soil and water. The underground components of the plants such as roots spread within the soil and thereby stabilize and reinforce the soil. This generally leads to soil conservation. The water infiltrates below the ground through the voids in the soil as well as through the interface between the root surface and the soil. The terrestrial components of plants such as stems, branches and leaves prevent the soil

below it from getting directly exposed to sunlight as well as to the impact of raindrops. Thus, a significant part of the momentum and energy in rainwater is absorbed and thereby inducing/ accelerating the downward movement of rainwater through stem flow and infiltration. On one hand this process creates water bodies like the groundwater reservoirs and rivers, which are good sources of water and nutrients required for plant growth. On the other hand, this process also substantially reduces the soil erosion and the surface flow velocity of storm water.

Additionally, there will be release of ample amount of oxygen, generation of colorful and fragrant flowers, fresh leaves as well as fruits through the process of photosynthesis. This makes the entire watershed very pleasant for human beings, migratory birds, flying insects as well as all other animals. The fruits and leaves also serve as food for human beings and animals.

A watershed containing large amounts of vegetation is considered as a healthy watershed. It is also called a well-managed or a 'green watershed'. It has no or very limited soil erosion and also it has large reserves of groundwater as well as surface water. In general, it has most of its natural resources conserved.

Thus, the scope of watershed management involves all the actions and programs aimed at achieving an overall balance between utilization and conservation of natural resources in a watershed. It represents a sustainable approach for resource conservation through watershed management. In the next section, the Indian and global perspective to watershed management is discussed.

1.3 Watershed Management: Indian and Global Perspective

India has the second highest population of over 1.2 billion among all the nations [i.e., 17.1% of the world population], a seventh highest land area of 3.29 million km^2 among all the nations [i.e., 2.4% of the world area] and an annual river flow of 1869 km^3 out of an annual rainfall of about 4000 km^3 [i.e., 4% of the world water]. The rainfall distribution is highly uneven spatially with the highest annual rainfall of 11,690 mm in the north-eastern state of Meghalaya and the least annual rainfall of 150 mm in the western part of the north-western state of Rajasthan. The number of rainy days [i.e., number of days with a minimum recorded daily rainfall of 2.5 mm] varies from 5 to 150. The rainfall distribution is also very uneven temporally with about 75% of the annual rainfall occurring only in the four monsoon months of June to September. The average annual rainfall is 1160 mm which is

slightly higher than the global average of 1110 mm. In the year 2010, the annual per capita water availability was estimated at 1588 m^3, which is considered as water stressed [i.e., between 1,000 and 1,700 m^3] as per the international norms. The per capita water availability was 5200 m^3 during the year 1951. The annual potential evapo-transpiration (PET) varies from 1,500 to 3,500 mm.

Although India has a well-developed precipitation pattern in the form of monsoons and an equally well-developed drainage network consisting of 14 large river basins, 44 medium river basins and hundreds of small river basins, there is a huge stress on water and land resources due to continuous overexploitation. This has led to many adverse hydro-meteorological impacts like large scale soil erosion, excessive lowering of water table, extensive river/ ground water pollution due to municipal/industrial wastewaters, widespread loss of forests/ grass lands/ crop lands/ wetlands/ water bodies, silting of existing water bodies, frequent occurrence of floods/ droughts, alarming reduction in Himalayan glaciers etc. All these phenomena have generally made the Indian perspective in watershed management very vulnerable to climatic and anthropogenic factors. Thus, achieving sustainable water resources development and integrated watershed management are two major challenges in the Indian context.

In spite of this alarming scenario, there are hundreds of Best Management Practices (BMPs) is adopted both in the government sector and the non-government sector over the entire length and breadth of India, which have been the bright spots in water and land resources management. These BMPs employ technologies which are either traditional or modern or a combination of both. Some of these BMPs which were effectively implemented in different parts of India, are as follows:

1) An effective implementation of the ban on tree cutting policy by the local government authorities in the north-eastern state of Sikkim resulted in an increase in the forest cover from 44% in 1995-'96 to 47.59% in 2009.

2) During 2000 to 2006, voluntary work by hundreds of people led by a spiritual saint near Jalandhar in the north Indian state of Punjab, resulted in the near total cleaning and rejuvenation of 35 km of Kali Bein River, which was heavily polluted by industrial effluents and garbage.

3) Over a 20-year period starting from 1974, a severely drought prone village of Ralegan Siddhi in the western Indian state of Maharashtra –even with an annual rainfall of about 200 mm, had transformed into a village with ample drinking water, food and fodder. This was possible due to the

adoption of ridge to valley approach in watershed management through social forestry, grassland development, continuous contour trenching, loose boulder structures, brushwood dams, nulla bunds, percolation tanks, underground dams, gabion bunds, check dams, farm ponds, staggered trenches for arresting soil erosion and ban on free grazing.

Global perspective on watershed management is having many similarities and some differences with the Indian perspective. Moreover, there are even bigger spatial and temporal variations in water/ pollutant distribution. It is also very much affected by soil erosion, excessive lowering of water table, extensive river/ ground water pollution due to municipal/industrial wastewaters, widespread loss of forests/ grass lands/ crop lands/ wetlands/ water bodies, silting of existing water bodies, frequent occurrence of floods/ droughts, alarming reduction in glaciers etc. These phenomena have resulted in major constraints due to water scarcity and land scarcity. However, in majority of the developed world and in many parts of the developing world, sufficient watershed management activities have been initiated in the government and non-governmental sectors.

The impact of these watershed management programs is varied ranging from failures with undesirable environmental and socio-economic consequences to significant benefits. To make the watershed management programs sustainable, land and water resources need to be managed together with an interdisciplinary approach. There is also a strong need to develop regional training and networking programs at all levels, especially when government agencies are not fulfilling their role in watershed management. The emergence of citizen-based watershed organizations in the United States and many other countries is a very positive development.

1.4 Timeline of Watershed Management Programmes in India

The actual timeline of watershed management programmes in India starts from the 1950s during the First Five Year Plan, with the establishment of a number of Soil Conservation Research Demonstration and Training Centres (SCRDTCs) by the Ministry of Agriculture (MoA) of the Government of India (GoI). In 1956, 42 small [i.e., less than 1 km^2] experimental watersheds were established for monitoring the impact of land use changes and conservation measures on surface hydrology, soil loss reduction and biomass productivity improvement. In 1961-'62, the MoA, GoI sponsored a scheme for soil conservation in the catchments of River Valley Projects (RVPs) for preventing siltation in major reservoirs.

In 1974, all the SCRDTCs were reorganized under the Central Soil and Water Conservation Research and Training Institute (CSWCRTI), Dehradun. A real breakthrough was achieved by CSWCRTI when watershed technologies were demonstrated under natural field settings using community driven approaches through four model Operational Research Projects (ORPs) in different regions of the country. The world famous Sukhomajri model in Haryana was also one of them. The Ministry of Rural Development (MoRD), GoI launched major nationwide watershed development programs like the Drought Prone Area Programme (DPAP) in 1973-'74 and Desert Development Programme (DDP) in 1977-'78. The MoA, GoI launched watershed programs in 10 catchments under the Flood Prone Rivers (FRP) Project.

During 1983, encouraged by the success in the earlier four model ORPs, CSWCRTI, Dehradun developed 47 model watersheds in the country in association with the Central Research Institute for Dryland Agriculture (CRIDA), Hyderabad. MoRD, GoI also started adopting watershed approach in 1987. The Planning Commission, GoI also started adopting integrated watershed approach in 1987-'88 for its Western Ghats Development Programme (WGDP) and Hill Area Development Programme (HADP) covering 16,000 km2 area in Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu. In 1989-'90, the Ministry of environment and Forests (MoEF) initiated National Afforestation and Eco-development Projects (NAEP) scheme following integrated watershed approach.

In the 1990s, many watershed development programs externally funded by the World Bank, European Economic council (EEC), Danish International Development agency (DANIDA), some Indo-German, Indo-Swiss and Japanese organizations were undertaken in various parts of India. Around the same time, MoA initiated a massive project on National Watershed Development Programme for Rainfed areas (NWDPRA) in 1991. In 1995, MoRD launched another big project called Integrated Wastelands Development Project (IWDP) with well formulated guidelines.

In 2001, the Planning Commission, GoI drew up an ambitious plan of treating 88.5 Mha of degraded/ rainfed lands in India, by the end of the 13th Five Year Plan in 2022 involving a huge financial investment of Rs. 72,750 crores. To strengthen the participating institutes, MoRD revised the watershed development guidelines as 'Haryali' [i.e., greenery] guidelines in 2003. The GoI established the National Rainfed Area Authority (NRAA) under the Planning Commission in 2006. MoA also started the projects on Reclamation of Alkali soils (RAS), Watershed Development Project for

Shifting Cultivation Areas (WDPSCA), Indo-German Bilateral Project (IGBP) and World Bank assisted Sodic Land Reclamation Project (SLRP). MoRD has initiated watershed projects under Mahatma Gandhi National Rural employment Guarantee Act (MGNREGA), Investment Promotional Scheme (IPS) and Technology Development Extension and Training (TDET), Wastelands Development Task Force (WDTF). Till March 2005, an area of 28.53 Mha was treated at an investment of Rs. 1,457 crores by MoA, MoRD, MoEF, out of a total degraded land of 146.82 Mha – as per the estimates of the National Bureau of Soil Survey & Land Use Planning (NBSS&LUP), Nagpur. From 2008, the new watershed projects are being implemented as per the latest common guidelines for watershed development projects, developed by the NRAA.

1.5 Problems and Prospects in Watershed Management

Watershed management encounters many problems and constraints. Some of the major problems and constraints in watershed management are listed below.

1.5.1 Problems and Constraints in Watershed Management

a. Land degradation in rain fed areas due to soil erosion from runoff is one of the major problems. In India it was estimated that the soil erosion in the 1990s was almost double that of soil erosion in the 1980s. Rainfall uncertainty and poor economic conditions act as a major constraint and thus prevents the farmers in rainfed areas from making investments. This leads to improper watershed management.

b. Equitable benefit sharing of watershed management within the farming communities as well as within the different locations of watershed is a huge problem. Generally, women, marginal farmers and landless laborers gain very little or nothing at all from the watershed management activities. Several case studies in water scarce states of Gujarat and Madhya Pradesh in India have showed that overdevelopment of water harvesting structures in the upstream portion of watersheds had significantly reduced the inflows into the downstream reservoirs. On the other hand, it is also noticed that building of large reservoirs resulted in the submergence and hardship in the upstream parts and benefits for people in the downstream parts of the same watershed or a neighboring watershed generally having an urban or an industrial area.

c. Acute shortage of water in general and drinking water especially in summer has been observed in many watersheds with inadequate

watershed management which may result in severe/ recurrent droughts. It may often result in limited and temporary food productivity gains. (d) Many a times, common lands do not get treated adequately and re-vegetation does not take place as expected in spite of the watershed management programs. As a result of this, domestic/ ecosystem water needs and livestock water/ fodder needs are either inadequately addressed or are made to suffer due to increased water withdrawals by other uses or due to overgrazing.

d. Problems exist or new problems crop up due to improper understanding of the interaction between biophysical and socio-economic processes in watershed management.

e. Conflict among various government ministries such as those related to agriculture [with emphasis on food production], rural development [with emphasis on employment generation & poverty alleviation], forests [with emphasis on maintaining biodiversity & wildlife], as well as conflict between government bureaucracy and elected representatives in their zeal to control funds, is a major problem in watershed management programs -which requires to be resolved on a priority basis.

f. It is hard to conduct meaningful impact assessment studies on watershed management programs for lack of baseline data for monitoring and comparison of the current conditions. The whole exercise of watershed management is undertaken without properly estimating the water supply scenarios under drought/ normal/ surplus years as well as without proper demand management especially during drought years.

g. Large areas inhabited with tribal population lack facilities to harvest water and to stabilize their food/ crop/ fodder production due to reduced forest yields, deterioration in land quality, lack of tribal agriculture policy and population pressure. This leads to a sustained misery, socio-political unrest, and insurgency among the tribal population.

1.5.2 New Prospects and Opportunities Associated with Watershed

Management In spite of the above-mentioned problems and constraints as well as some other problems and constraints, watershed management is associated with new prospects and opportunities. Some of them are listed below:

a. There is a need to produce more and better food without further undermining the environment/ ecology, especially the land, water,

forests, wildlife, and atmosphere. This may include adoption of best management practices (BMPs) such as organic farming, de-silting for reservoir capacity restoration as well as for crop productivity increase, sprinkler and/ or drip irrigation to avoid excess use of water, no tree felling policy, afforestation and arboriculture through high oxygen yielding & other medicinal plants etc.

b. There is a need to ensure that gains due to groundwater recharge are not dissipated by excess groundwater extraction. To achieve this, groundwater over-extraction should be avoided through public awareness and through regulation.

c. There is a need to consider the downstream impacts of intensive upstream water conservation. For this, watershed associations with representations from all the stakeholders in the watershed should be made operational. These associations can take decisions in the best interest of all the people concerned.

d. Decreasing the costs at which the gains are achieved and thereby increasing the modest benefit-cost ratio should offer new prospect and opportunity in watershed management. To realize this, low-cost technologies which may involve local materials, labour at practically no cost, technologies which are traditional and time tested should be employed to generate more benefits spread over the entire watershed among all the stakeholders.

e. Increasing all sections of people's participation beyond the project implementation stage to ensure sustainable watershed management should be a top priority. Only this can ensure progress on a sustained basis overcoming the hydro-geological, socio-political and other uncertainties.

f. Many successful watershed management programs -especially in India, have been implemented on a small scale in a few villages by collaborated efforts among the government departments, non-governmental organizations (NGOs) and research organizations. They represent sporadic BMPs. Hence there is a need to scale up the watershed management activities over large areas which could include remote and/or difficult terrains, so that many problems affecting our agricultural, rural and forest sectors can be effectively addressed.

g. Since there have been no or very few institutions built for research & development on collective management of watersheds, there is a need to build centers of advanced learning employing the modern tools of

remote sensing, geographic information systems, decision support systems, computer-based planning tools, poverty & socio-economic analysis etc.

h. There is a need to preserve and improve common pool resources (CPRs) of land, water, fodder, forest, fisheries, wildlife and agriculture which significantly contribute towards people's livelihood especially in the rural areas.

i. There is a need to minimize migration to urban areas by creating opportunities in agriculture, natural disasters like floods/ droughts, forest/ mountain economies and by arresting fall in agricultural prices, gap in urban/ rural wages, gaps in urban/ rural employment opportunities.

Let us sum up

The word 'watershed' is used as a synonym for 'catchment' or 'basin' wherein rainwater or storm water gets collected from an area enclosed by a ridge line. This water eventually flows through the various drainage channels which merge with one another to form one or rarely more than one outfall(s) of a stream. Thus, 'watershed' is defined as an area enclosed within a watershed line. In this course, the word 'watershed' is used for a small basin or a small catchment representing a hydrological unit which drains all its rainwater into a stream. Therefore, it is independent in terms of its water in general and surface water in particular.

The process of delineation of watersheds has a scientific and rational approach. However, for presenting them as units for planning and development, formulation of an appropriate framework for watersheds is a pre-requisite. Such framework shall not only follow a hierarchical system of delineating bigger hydrologic units into watersheds but also include a codification system so that each watershed or units of interventions can be identified as an 'unique' individual entity with clear identifiable linkage with higher hierarchical units, that is, basin, catchments, subcatchment.

Glossary

Wwatershed: It's a land area that channels rainfall and snowmelt to creeks, streams, and rivers, and eventually to outflow points such as reservoirs, bays, and the ocean.

Catchment area: A river drains the water collected from a specific area, which is called its catchment area.

River Basin: An area drained by a river and its tributaries is called a drainage basin.

Check your progress

1. What is a Watershed?

A watershed – the land area that drains to one stream, lake or river – affects the water quality in the water body that it surrounds. Like water bodies (e.g., lakes, rivers, and streams), individual watersheds share similarities but also differ in many ways. Every inch of the World is part of a watershed – in other words, all land drains into a lake, river, stream or other water body and directly affects its quality. Because we all live on the land, we all live in a watershed — thus watershed conditions are important to everyone.

2. What is a Healthy Watershed?

A healthy watershed is one in which natural land cover supports:

- dynamic hydrologic and geomorphologic processes within their natural range of variation,
- habitat of sufficient size and connectivity to support native aquatic and riparian species, and
- physical and chemical water quality conditions able to support healthy biological communities.

3. What is Watershed Management?

Watershed management is a term used to describe the process of implementing land use practices and water management practices to protect and improve the quality of the water and other natural resources within a watershed by managing the use of those land and water resources in a comprehensive manner.

4. What is watershed management and its importance?

Watershed Management is an adaptive, comprehensive, integrated multi-resource management planning process that seeks to balance healthy, ecological, economic and cultural/social conditions, within a watershed. It serves to integrate the planning of land and water.

Suggested Readings

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Unit 2

Components of watershed: land, water, and vegetation

Structure

Learning Objectives

Overview

2.1 Land resources Management

2.1.1 Need For Land Management

2.1.2 Objectives of Land Management

2.1.3 Sustainable Land Management

2.1.4 Principles of Sustainable Land Management

2.1.5 Land Management Examples

2.2 Water Resources management

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2.2.2 Integrated water resources management

2.2.3 Managing water in urban settings

2.3 Forest Resources management

2.3.1 Forest Management Planning

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2.3.3 Forest Management with Satellite Technologies

2.3.4 Sustainable Forest Management

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to learn about the following,

- Need, objectives, principles and sustainable Land resources Management.
- Sustainable, integrated and Urban Water Resources management
- Planning, practice technology and sustainable Forest Resources management

Overview

Major components of Integrated Watershed Management are

- Management of Land resources (soil conservation);
- Management of water resources (e.g., rainwater harvesting);
- Management of Vegetation/biomass (reforestation, agroforestry, etc.);
- Management of human resources (qualification of resource managers and users, etc.)

Management of these resources proceeds on the level of watersheds, Additionally, Watershed Management comprises the following components:

- soil management (land use planning, for example);
- Pasture management (area closure, for example);
- Crop yield management (e.g., storage management, marketing);
- Livestock management (e.g., marketing, dairy production);
- Rural energy management (e.g., decentralised use of renewable energy resources);
- Farm and off-farm activities add value (e.g., creation of infrastructure).

In this unit you can learn about the land, water and vegetation. Let us see one by one.

2.1 Land resources Management

Land management refers to land's correct usage and maintenance as per its features and requirements. Land management necessitates decision-making and is influenced by the function it fulfils, such as agricultural

production, habitation, recreational, mineral extraction, and the character and features of the land. Land management has several facets. Although it is applied primarily to farming and agricultural cultivation, it encompasses other activities such as habitation, urban and industrial expansion, transportation and roads, recreation activities, mines, and environmental conservation.

2.1.1 Need For Land Management

Modern farming necessitates several fertilisers, herbicides, and chemicals to fulfil high output targets. It is reliant on highly mechanised field activities. This causes many environmental issues, the most evident are soil and water contamination and rising soil erosion.

Non-agricultural uses, on the other hand, are becoming increasingly significant in industrialised nations, occupying a growing amount of land. These applications are concentrated in urban and industrial growth, highways and transportation, mines, forestry, and conservation areas. Care and land restoration have become critical issues with increasing environmental preservation concerns.

These prioritise erosion and pollution prevention, the implementation of environmentally friendly farming methods and manufacturing activities, and, eventually, the prohibition of the most harmful items and repair and rehabilitation of contaminated places. The ultimate objective should be to have sustainable land use with healthy soil.

2.1.2 Objectives of Land Management

Ideas of wisdom are inextricably tied to good land management. This necessitates the application of ancient hereditary knowledge and the assimilation of new technical expertise. Land management, and particularly agricultural land management, has three primary goals:

- A financial or production goal that focuses on optimisation in long-term output maximisation.
- A conservation goal of preserving the accessible characteristics, capabilities, and results.
- A reclamation component to remove limitations, avert deterioration, and rehabilitate or recover the land’s qualities and use potential.
- Excellent management necessitates the continuity of good practices and the ability to adapt to changing situations.

2.1.3 Sustainable Land Management

Land offers an environment for farming systems and is also necessary for enhanced environmental management such as supplier processes for greenhouse gases, nutrient cycles, contaminant alteration and filtration, and water transmission and cleansing as components of the hydrological processes. Sustainable land management aims to balance the complementary objective of providing ecological, financial, and social possibilities for current and subsequent populations while conserving and improving the quality of the land resource (soil, water, and air).

Sustainable land management involves technology, strategies, and actions that strive to integrate socioeconomic considerations with environmental impacts to:

- Maintain and improve output
- Minimise production risk and increase soil capacity to protect from environmental pollutants
- Safeguard natural resource potential and prevent soil and water quality deterioration
- Be financially feasible
- Be socially acceptable while ensuring access to advantages of better land management

2.1.4 Principles of Sustainable Land Management

We can only accomplish sustainability via the combined efforts of those directly responsible for resource management.

This necessitates a policy framework that allows farmers and other local decision-makers to profit from excellent land-use decisions while holding them accountable for improper land uses.

It is vital to integrate economic and environmental concerns holistically to accomplish the goals of sustainable land management. This necessitates those environmental issues be prioritised alongside economic success when assessing the effects of development initiatives and that credible metrics of environmental performance be created.

There is now an immediate requirement to address the global problem of producing more food to support the world's fast-growing population while protecting the land's biological production capacity, strength, and ecological management systems.

If correctly developed and performed, sustainable land management will guarantee that agriculture becomes a component of the environmental solution instead of an issue.

2.1.5 Land Management Examples

Land degradation is a broad and expanding issue that can just be addressed by joint and integrated methods and focused initiatives and actions.

As a consequence of land degradation, agricultural productivity in Sub-Saharan Africa is declining each year. Various efforts include a collective mechanism for tackling bottlenecks, enhancing the efficiency of economic and non-economic resources, and consequently, mainstreaming and funding successful nationally-led Service Level Management policies.

The usage of barren land to create parks for children is an intelligent alternative to reduce land wastage.

Creating agricultural sites near water resources increases crop production and reduces water transportation. This would be an effective land management method.

The conservation of forests and forest land protection is another example of land management. It preserves nature and ensures an ecological balance of the earth.

2.2 Water Resources management

'Water is an important and critical resource in India.' Water is a very important and critical resource in India because water is a precondition of life. All living things need water to survive. The erratic rainfall has added to its critical nature. Water is needed not only for domestic purposes but also for agriculture, industries, power generation, navigation and disposal of municipal sewage. The growing urbanization has resulted in increased consumption of water.

Freshwater is the water that can be used by living beings such as water for drinking, irrigation etc. Three-fourths of the world is covered with water.2.5 % of the total water available on the earth's surface is fresh water which can be used by living beings.

Water Resources Management (WRM) is **the process of planning, developing, and managing water resources, in terms of both water quantity and quality, across all water uses**. It includes the institutions, infrastructure, incentives, and information systems that support and guide water management.

Water scarcity affects more than 40% of the global population. Water-related disasters account for 70% of all deaths related to natural disasters. The World Bank helps countries ensure sustainability of water use, build climate resilience, and strengthen integrated management.

Ideally, water resource management planning has regard to all the competing demands for water and seeks to allocate water on an equitable basis to satisfy all uses and demands. As with other resource management, this is rarely possible in practice so decision-makers must prioritise issues of sustainability, equity, and factor optimisation (in that order!) to achieve acceptable outcomes. One of the biggest concerns for water-based resources in the future is the sustainability of the current and future water resource allocation.

2.2.1 Sustainable water management

At present, only about 0.08 percent of all the world's fresh water is accessible. And there is ever-increasing demand for drinking, manufacturing, leisure and agriculture. Due to the small percentage of water available, optimizing the fresh water we have left from natural resources has been a growing challenge around the world.

Much effort in water resource management is directed at optimizing the use of water and in minimizing the environmental impact of water use on the natural environment. The observation of water as an integral part of the ecosystem is based on integrated water resources management, based on the 1992 Dublin Principles.

Sustainable water management requires a holistic approach based on the principles of Integrated Water Resource Management, originally articulated in 1992 at the Dublin (January) and Rio (July) conferences. The four Dublin Principles, promulgated in the Dublin Statement are:

- Fresh water is a finite and vulnerable resource, essential to sustain life, development and the environment;
- Water development and management should be based on a participatory approach, involving users, planners and policy-makers at all levels;
- Women play a central part in the provision, management and safeguarding of water;
- Water has an economic value in all its competing uses and should be recognized as an economic good.

Implementation of these principles has guided reform of national water management law around the world since 1992.

Further challenges to sustainable and equitable water resources management include the fact that many water bodies are shared across boundaries which may be international (water conflict) or intra-national (Murray-Darling basin).

2.2.2 Integrated water resources management

Integrated water resources management (IWRM) has been defined by the Global Water Partnership (GWP) as "a process which promotes the coordinated development and management of water, land and related resources, in order to maximize the resultant economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems".

Some scholars say that IWRM is complementary to water security because water security is a goal or destination, whilst IWRM is the process necessary to achieve that goal.

IWRM is a paradigm that emerged at international conferences in the late 1900s and early 2000s, although participatory water management institutions have existed for centuries.[60] Discussions on a holistic way of managing water resources began already in the 1950s leading up to the 1977 United Nations Water Conference.] The development of IWRM was particularly recommended in the final statement of the ministers at the International Conference on Water and the Environment in 1992, known as the Dublin Statement. This concept aims to promote changes in practices which are considered fundamental to improved water resource management. IWRM was a topic of the second World Water Forum, which was attended by a more varied group of stakeholders than the preceding conferences and contributed to the creation of the GWP.

In the International Water Association definition, IWRM rests upon three principles that together act as the overall framework:

- Social equity: ensuring equal access for all users (particularly marginalized and poorer user groups) to an adequate quantity and quality of water necessary to sustain human well-being.
- Economic efficiency: bringing the greatest benefit to the greatest number of users possible with the available financial and water resources.
- Ecological sustainability: requiring that aquatic ecosystems are

acknowledged as users and that adequate allocation is made to sustain their natural functioning.

In 2002, the development of IWRM was discussed at the World Summit on Sustainable Development held in Johannesburg, which aimed to encourage the implementation of IWRM at a global level. The third World Water Forum recommended IWRM and discussed information sharing, stakeholder participation, and gender and class dynamics.

IWRM practices depend on context; at the operational level, the challenge is to translate the agreed principles into concrete action.

Operationally, IWRM approaches involve applying knowledge from various disciplines as well as the insights from diverse stakeholders to devise and implement efficient, equitable and sustainable solutions to water and development problems. As such, IWRM is a comprehensive, participatory planning and implementation tool for managing and developing water resources in a way that balances social and economic needs, and that ensures the protection of ecosystems for future generations. In addition, in light of contributing the achievement of Sustainable Development goals (SDGs). IWRM has been evolving into more sustainable approach as it considers the Nexus approach, which is a cross-sectoral water resource management. The Nexus approach is based on the recognition that "water, energy and food are closely linked through global and local water, carbon and energy cycles or chains."

Water's many different uses — for agriculture, for healthy ecosystems, for people and livelihoods — demands coordinated action. An IWRM approach is consequently cross-sectoral, aiming to be an open, flexible process, and bringing all stakeholders to the table to set policy and make sound, balanced decisions in response to specific water challenges faced.

An IWRM approach focuses on three basics and aims at avoiding a fragmented approach of water resources management by considering the following aspects:

- Enabling Environment: A proper enabling environment is essential to both ensure the rights and assets of all stakeholders (individuals as well as public and private sector organizations and companies), and also to protect public assets such as intrinsic environmental values.
- Roles of Institutions: Institutional development is critical to the formulation and implementation of IWRM policies and programmes. Failure to match responsibilities, authority and capacities for action are all major sources of difficulty with implementing IWRM.

- Management Instruments: The management instruments for IWRM are the tools and methods that enable and help decision-makers to make rational and informed choices between alternative actions.

Some of the cross-cutting conditions that are also important to consider when implementing IWRM are:

- Political will and commitment
- Capacity development
- Adequate investment, financial stability and sustainable cost recovery
- Monitoring and evaluation

IWRM should be viewed as a process rather than a one-shot approach - one that is long-term and iterative rather than linear in nature. As a process which seeks to shift water development and management systems from their currently unsustainable forms, IWRM has no fixed beginnings or endings.

Furthermore, there is not one correct administrative model. The art of IWRM lies in selecting, adjusting and applying the right mix of these tools for a given situation.

2.2.3 Managing water in urban settings

Typical urban water cycle depicting drinking water purification and municipal sewage treatment systems.

Integrated urban water management (IUWM) is the practice of managing freshwater, wastewater, and storm water as components of a basin-wide management plan. It builds on existing water supply and sanitation considerations within an urban settlement by incorporating urban water management within the scope of the entire river basin.[65] IUWM is commonly seen as a strategy for achieving the goals of Water Sensitive Urban Design. IUWM seeks to change the impact of urban development on the natural water cycle, based on the premise that by managing the urban water cycle as a whole; a more efficient use of resources can be achieved providing not only economic benefits but also improved social and environmental outcomes. One approach is to establish an inner, urban, water cycle loop through the implementation of reuse strategies. Developing this urban water cycle loop requires an understanding both natural, pre-development, water balance and the post-development water balance. Accounting for flows in the pre- and post-development systems is an important step toward limiting urban impacts on the natural water cycle.[66]

IUWM within an urban water system can also be conducted by performance assessment of any new intervention strategies by developing a holistic approach which encompasses various system elements and criteria including sustainability type ones in which integration of water system components including water supply, waste water and storm water subsystems would be advantageous.[67] Simulation of metabolism type flows in urban water system can also be useful for analysing processes in urban water cycle of IUWM.

2.3 Forest Resources management

Forest management is vital for commercial forestry. Woodland is an essential ecosystem's part: it provides clean air and water, is home to wild animals, and contains resources that we consume every day. Many people get the impression that this natural environment does not require human intervention. So, here is a question: why is forest management important for commercial activity? Without it, woodlands deteriorate and lose the qualities necessary for production purposes. In neglected form, such an environment is favorable for diseases and insects. There is also the danger of overpopulation, resulting in an increased rival for nutrients, water, and light. Woodland fires pose a severe threat to the environment, animals, and people. Landowners can maintain the optimal state of the ecosystem with the help of forest management. Modern technologies make this process even more efficient.

Forest Management

The purpose of forest management is to achieve environmental, social, and economic goals effectively. It can be

- the security of the forest and its inhabitants,
- the preservation of the ecosystem's health,
- the improvement of wood goods and services production.

Traditionally, national authorities control this process because woodlands usually cover a large extent, and their control requires severe administrative, financial, legal, and scientific decisions.

The importance of forest management is also about maintaining the health of our planet. Uncontrolled deforestation and pollution damage the environment. However, a unified system at the global level is not advisable because different regions have specific socio-economic conditions. It imposes additional responsibility on local governments and private enterprises to meet public needs, while maintaining a healthy environment.

2.3.1 Forest Management Planning

Due to planning, landowners determine the strategy for many years to come. In general, the following benefits of a forest management plan can be distinguished:

- making decisions on forest management and transferring information about its current state to other participants in the process;
- collecting an evidence base on compliance with legal obligations established for this territory control.
- obtaining licensing and grant payments.

Forest management planning can be detailed or general. In the first case, landowners draw up short-term steps to solve current problems. In the second, they plan long-term decisions that determine the development of woodland over years.

The forest management plan is in the form of a working document. First, it should define the goals of a landowner. They can be long-term or short-term. Maps are an integral plan's part. They display the boundaries of the woodland, the location of groups of trees, and landscape features. Although documented, the project should be periodically reviewed and adjusted as the situation changes. To fully reveal the forest management plan benefits, you should use the specialists services: foresters, soil scientists, biologists, etc.

2.3.2 Forest Management Practices

As noted above, each plan contains specific activities determined by the landowner's goals. Let's consider the main types of forest management practices.

Timber Harvesting

This practice enhances the health of the woodland and wildlife and for industrial and economic purposes, such as paper production or the organization of tourist routes. There are the following timber harvesting methods:

- clear-cutting.
- selective logging: group selection and single tree selection;
- shelterwood.
- seed tree cutting.

Each method is suitable for specific forest management situations and has advantages and disadvantages. With the help of specialists, landowners should independently determine the best option for their goals.

Timber Thinning

Under natural conditions, trees compete for vital resources. As a result, fast-growing trees survive, and slow-growing ones usually die. Moreover, the productivity of such a woodland can be low, as all trees grow more slowly due to competition. Proper forest management solves this problem with timber thinning. Specialists thin out the forest by cutting down a certain percentage of the trees. Thus, the remaining plants grow more active and become stronger.

This forest management practice has two types. Commercial timber thinning implies partial or complete coverage of tree felling costs. It requires carefully calculating the production process so that the wood retains its marketable qualities. Pre-commercial thinning is much easier to implement, but it does not cover the landowner's cost and is suitable for felling small trees not required for lumber production.

Prescribed Burning

This forest management practice is essential as it prevents fire hazards. Moreover, the prescribed burning enables the growth of plants that animals feed on and decreases the disease and insect infestation danger. Also, this method encourages the growth of some trees, such as longleaf pine. Still, prescribed burnings should be performed following strict rules to prevent massive wildfires.

Reforestation

Woodland is a renewable resource, so the process of growing, cutting down, and reforesting can be an endless cycle. There are two ways of this practice.

Natural regeneration involves the germination of plants from seeds that fell into the ground naturally and from the fallen trees' stumps. Artificial reforestation is a more costly forest management practice but has several benefits. Thus, landowners can determine the types of trees grown, introduce better seeds, and control the distance between seedlings.

Forest Succession

Woodland succession is a successive change of plant and animal communities naturally or with the help of forest management. Natural succession occurs in woodlands affected by natural disasters or human

activity. This practice can meet the needs of many animals. For example, rabbits use such areas for food and shelter, and birds prey on rodents that live in early succession.

Forest Monitoring

Satellite-driven software for forest stand monitoring and sustainable forest management. Forest Management to Prevent Fires

Woodland fires are one of the most common natural disasters (discover the largest wildfires in 2022). Every year they destroy over a million acres of land and thousands of homes and properties worldwide.

Woodland fires are of three main types:

- Crown fires are the riskiest and most challenging to control. They spread over the tops of the trees. Therefore, they can cover large areas with a strong wind in the shortest possible time.
- Ground fires flare up under fallen leaves and peat moss. The main danger is that they are challenging to detect early.
- Surface fires only destroy the terrain cover. This disaster is the most manageable to put out, so they are considered the least dangerous.

A combination of woodland management practices can significantly reduce the likelihood of a fire or contain it, minimizing damage. Forest fire management techniques have the following activities:

- Maintaining the tree’s health to increase their resistance to fire through the application of fertilizers and planting of various tree species;
- controlled combustion to lower the flammable materials;
- creation of firebreaks: open land areas that prevent the fire spread.

Also, extremely useful is woodland thinning. This forest management practice creates more space for healthy plants and reduces the danger by removing diseased and vulnerable trees. Moreover, a thinned woodland is not only less likely to be exposed to large-scale fire, but it recovers much faster after a disaster.

Proper forest management to prevent fires depends on the characteristics of the woodland and the financial means that the landowner has to carry out preventive measures. However, adaptive forest management and the integrated use of relevant practices is the best solution.

2.3.3 Forest Management with Satellite Technologies

Monitoring diseases, pests, and trees' health is essential for woodland management. Modern technologies optimize this process by providing landowners with accurate data and allowing them to control the process in real-time, including remotely.

Forest management software Forest Monitoring provides all the necessary tools to analyze woodlands and implement sustainable production practices. Thus, satellite data allow remote monitoring of deforestation and changes in woodlands over a certain period (monthly, quarterly, and annually). Analyzing weather trends based on historical data is equally crucial to woodland management. It allows you to plan long-term agricultural work. With EOSDA Forest Monitoring, you can also track tree crop moisture levels and receive water stress reports to predict the risk of fires.

2.3.4 Sustainable Forest Management

Even though the woodland is a renewable resource, it requires proper management methods. Otherwise, landowners can suffer significant economic losses and cause severe damage to the environment. At the same time, the development of sustainable forest management is critical for the entire planet. Climate change threatens both the environment and all of humanity. Mismanagement of natural resources reduces our ability to produce food in the future, leading to global famine. Under these conditions, landowners, woodland managers, logging companies, and the food industry switch to sustainable production methods, including sustainable forest management. Satellite technologies facilitate this transition and provide practical tools to achieve this goal.

Social Forestry:

Social forestry offers an alternative form of forest management. It is an approach that balances the needs of local people with multiple external interests. Promotes sustainable forest use and management among communities who live near and in close proximity to forests

Empowers communities by increasing awareness, building capacity, working with locals to establish policies, and respecting their rights and knowledge systems

Gives communities access to forest resources and benefits in exchange for their participation in sustainable forest management

Classification of Social Forestry

The objectives of social forestry adopted by the NCA were to fulfill the basic and economic needs of the community.

The National Commission on Agriculture (1976) has classified social forestry into three categories. These are:

- Urban Forestry
- Rural Forestry, and
- Farm Forestry

Urban Forestry

Urban forestry pertains to the raising and management of trees on public and privately owned lands in and around urban centers such as green belts, parks, roadside avenues, industrial and commercial green belts, etc.

The Ministry is implementing the Nagar Van Yojana the scheme initiated for the creation of Nagar Van (city forest) in urban areas under the National Fund of the Compensatory Afforestation Fund Management and Planning Authority (CAMPA).

Development of Urban Forests will contribute to India's decision to sequester 2.5 to 3 billion tonnes of carbon dioxide equivalent in the country's forests. They will also function as urban lungs.

Rural Forestry

Rural forestry lays emphasis on the promotion of agro-forestry and community forestry.

Agro-forestry

Agro-forestry is the raising of trees and agricultural crops on the same land inclusive of waste patches. It combines forestry with agriculture, thus, altering the simultaneous production of food, fodder, fuel, timber, and fruit.

Sub-Mission on Agroforestry (Har Medh Par Ped) Scheme was launched in 2016-17 to encourage tree plantation on farmland along with crops/ cropping systems to help the farmers get additional income and make their farming systems more climate resilient and adaptive.

The Agro-forestry scheme envisages a lot of quantifiable benefits like additional income, protection from crop damage, and higher carbon sequestration along with climate change adaptation and mitigation.

Community Forestry

Planting trees on public or community land, such as schoolyards, roadsides, canal banks, village pastures, and temple grounds, is known as community forestry.

The goal of community forestry programs is to benefit the entire community. Community forestry provides a means under which the people of landless classes can associate themselves with tree raising and thus, get those benefits that otherwise are restricted for landowners.

According to a study that was published in the journal Nature Sustainability, community-forest management led to a 37 percent relative decrease in deforestation and a 4.3 percent relative decrease in poverty.

Farm Forestry

Farm forestry is the term used to describe the practice of farmers growing trees on their fields for both commercial and non-commercial purposes. Non-commercial farm forestry refers to the practice of growing trees in a variety of places, including the margins of agricultural fields, grasslands and pastures, the land surrounding dwellings, and cow sheds.

Farm Forestry can take many different forms, such as timber belts, plantations, woodlots, widely spaced tree plantings, and the sustainable management of existing native plant stands.

Let us sum up

Land degradation is a broad and expanding issue that can just be addressed by joint and integrated methods and focused initiatives and actions.

Integrated water resources management (IWRM) has been defined by the Global Water Partnership (GWP) as "a process which promotes the coordinated development and management of water, land and related resources, in order to maximize the resultant economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems.

The importance of forest management is also about maintaining the health of our planet. Uncontrolled deforestation and pollution damage the environment. However, a unified system at the global level is not advisable because different regions have specific socio-economic conditions. It imposes additional responsibility on local governments and private enterprises to meet public needs, while maintaining a healthy environment.

Glossary

Agro-forestry is the raising of trees and agricultural crops on the same land inclusive of waste patches.

Community Forestry: Planting trees on public or community land, such as schoolyards, roadsides, canal banks, village pastures, and temple grounds, is known as community forestry.

Farm forestry is the term used to describe the practice of farmers growing trees on their fields for both commercial and non-commercial purposes.

Check your progress

1. What are the elements of land management?

The process of land use planning and its implementation, which is land use management as defined above, hinges on three elements: the stakeholders, the quality or limitations of each component of the land unit, and the viable land use options in the area.

2. Why is water resource management important?

Water management helps in developing efficient irrigation practices for the betterment of agriculture in the country. Proper utilization of water in our homes too can save this precious resource. Water management teaches us to use a limited amount of water whenever required.

3. What is the importance and management of forest resources?

Over 1.6 billion people depend on forests for food or fuel, and some 70 million people worldwide - including many Indigenous communities - call forests home. Forests provide us with oxygen, shelter, jobs, water, nourishment and fuel.

Suggested Readings

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Unit 3

Physical and Socio-Economic characteristics of watershed

Structure

Learning Objectives

Overview

3.1 Physical Characteristics of watershed

3.1.1 Geology:

3.1.2 Geomorphology characteristics

3.1.3 Hydrology

3.1.4 The Hydrologic Cycle

3.5 Soil

3.1.6 Water Shapes the Watershed

3.1.7 Vegetation and Land Use

3.2 Socio-Economic characteristics of watershed

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to understand the

- Physical Characteristics of watershed
- Socio-Economic characteristics of watershed

Overview

Watershed is a land area that channels rainfall and snowmelt to creeks, streams, and rivers, and eventually to outflow points such as reservoirs, bays, and the ocean. Basin area, basin length, basin slope, and basin shape are the physical characteristics of watersheds, significantly affecting the characteristics of runoff and other hydrologic processes. Major socio-

cultural & economic factors include Land tenure; Capital; Labour; Perception & beliefs; Gender. Land tenure & watershed management: Land tenure is the terms and conditions on which land & other natural resources (e.g., trees and water) are held & used. In this unit you will learn about the different types of physical and economic characteristics of watershed.

3.1 Physical Characteristics of watershed

3.1.1 Geology:

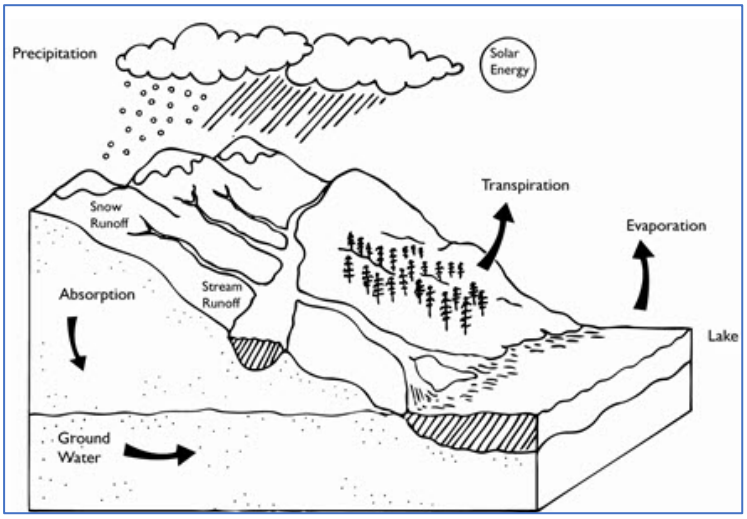
The geology of a watershed is important because it influences topography, direction of water flow, shape of the drainage basin, stream bed materials, water quality and biological productivity. Geologic forces cause the earth's surface to rise or fall and, along with weathering, determine the topography of the watershed. Geology affects where and how water moves in a watershed. For example, karst areas like the Ozark region have geology of limestone or dolomite that have openings where water has dissolved the rock along cracks and between layers. These openings act as direct routes to funnel water very quickly to the water table. These routes not only affect the quantity of water in streams and groundwater but can allow pollutants from the land surface to quickly contaminate the groundwater system.

Geology determines the types of bedrock formations, or parent materials, and the associated soils of the watershed. The parent materials and the soils produced from them help determine the water quality, biological productivity, and aquatic life of a stream. By supplying the materials that make up the bottoms and banks of channels, parent material and soils also affect how vulnerable a stream is to erosion. Parent materials also provide controls on how stream channels can develop. Bedrock limits how far down into the earth a stream can easily dig its channel. Bedrock can also control the lateral, or sideways, movement of the stream.

3.1.2 Geomorphology characteristics

Geomorphology deals with landforms in a watershed. A survey of landforms will result in a better understanding of the erosion process, hazards. For instance, a valley at youth stage will have more active erosion than one at old stage. High stream density usually means quick surface runoff and flash floods, etc. This kind of information, together with rock types and structures, permits proper selection of sites for dams and roads as well as estimation of peak flows and timing, etc. In addition to collection of descriptive landform information, there are some quantitative analysis methods which can be used for comparison or interpretation.

A watershed is a basic unit of hydrological behavior. On the land surface, it is a geographical unit in which the hydrological cycle and its components can be analyzed. Usually, a watershed is defined as the area that appears, based on topography, to contribute all the water that passes through a given point of a stream. A watershed embraces all its natural and artificial (man-made) features, including its surface and subsurface features, climate and weather patterns, geologic and topographic settings, soils and vegetation characteristics, and land use (shown in figure 3.1). A watershed carries



water “shed” from the land after rain falls and snow melts. Drop by drop, water is channeled into soils, groundwater, creeks, and streams, making its way to larger rivers and eventually the sea.

Fig. 3.1. A Watershed Illustration. (Source: Rees, 1986)

Watershed geomorphology refers to the study of the characteristics, configuration and evolution of landforms and properties; developing physical characteristics of the watershed. It comprises of the characteristics of land surface as well as the characteristics of the channels within the watershed/basin boundary. These properties of watersheds significantly affect the characteristics of runoff and other hydrological processes. The principal watershed characteristics are:

1. Basin Area
2. Basin Slope
3. Basin Shape
4. Basin Length

Basin shape is reflected by several watershed parameters as are given below.

- Form Factor
- Shape Factor
- Circularity Ratio
- Elongation Ratio
- Compactness Coefficient

Along with the surface characteristics of a watershed, the channel characteristics are important in transiting the runoff water from the overland region to channels (streams) and from the channel of one order (primary) to the other higher order (e.g., river stream). The most common and important channel characteristics of the watersheds are:

- Channel Order
- Channel Length
- Channel Slope
- Channel Profile
- Drainage Density

The quantification of these physical and geomorphologic properties of watershed/basin are important for estimating the watershed hydrologic processes.

Watershed geomorphology refers to the physical characteristics of the watershed. Basin area, basin length, basin slope, and basin shape are the physical characteristics of watersheds, significantly affecting the characteristics of runoff and other hydrologic processes.

3.1.3 Hydrology

Hydrology is the science that deals with the properties, distribution, and circulation of water in the atmosphere, on the land and underground. Short-term and longterm climatic conditions affect how much precipitation is available to shape and develop the features of a watershed. The amount, type and timing of precipitation directly affect erosion and deposition in a watershed. Hydrology of a watershed is greatly affected by how much precipitation and temperature vary over time.

3.1.4 The Hydrologic Cycle

The circulation of water from the oceans to atmosphere, to land, to lakes and streams and back to the ocean and atmosphere is known as the hydrologic cycle, or the water cycle (Figure 4). Water movement through and between these various places is irregular and cyclic. The quantities of

water falling as precipitation, evaporating from exposed surfaces, transpired by plants, stored as groundwater, or contained in stream channels, ponds and lakes varies from day to day, season to season, and year to year. This variation helps produce dynamic aquatic and terrestrial ecosystems.

The four basic parts of the hydrologic cycle are precipitation, evapotranspiration, storage, and runoff. Precipitation includes rain, hail, snow, and other forms of moisture falling from the atmosphere to the earth. Evapotranspiration returns water vapor to the atmosphere through evaporation from surface water and through transpiration by plants. Most of the precipitation falling on Missouri is intercepted by depressions or vegetation and is then evaporated or transpired back to the atmosphere. Water that remains in wetlands and lakes or soaks through the ground and infiltrates into groundwater is called storage. Runoff, or overland flow, is water that hits the earth and flows over land and into channels. Base flow or groundwater flow is the precipitation that percolates deep into the soil and is slowly released over weeks or months to a stream channel or spring. Base flows sustain many streams and springs during long, dry periods.

3.5 Soil

The debris deposited by the glaciers forms the parent material for the soil throughout the Watershed. The almost infinite variety of combinations of mineral materials located in many conditions of topography and climate have resulted in a great number of soil types of varying fertility. Sandy and loamy soils are common throughout the basin.

The rate storm water infiltrates through soil have important implications with regard to storm water management. When infiltration occurs slowly, precipitation tends to flow over the ground surface during intense rain events and quickly enter storm sewers, ditches, creeks, and other water bodies. Water flows with higher energy, resulting in erosion, flooding, and impaired water quality. There are three types of waterborne erosion: sheet, rill, and gully. Sheet erosion occurs when rainfall hits the ground and runs across its surface in a large sheet, picking up loose soil particles. Little to none of the water infiltrates. Rill erosion occurs when precipitation cuts small drainage pathways into the surface of the land, giving the precipitation little time to infiltrate. Gully erosion occurs when rills become much larger and deeper. Rills can be easily obliterated by normal tillage practices, whereas gullies cannot. Soil erosion susceptibility is greatest for loose soils on steep slopes. This Watershed has many soils that are susceptible to all three types of erosion.

3.1.6 Water Shapes the Watershed

Precipitation and runoff sculpt the land surface. Each watershed has its own runoff patterns and erosion rates that depend on geology, vegetation, land use and topography. Changes in runoff can increase erosion and sedimentation of the land surface. Excessive runoff can cause downstream flooding and accelerated erosion of the land. Too much runoff can also cause destructive amounts of streambed and streambank erosion as channels adjust over time to carry more water. High runoff can also increase the amounts of soil, sand, gravel, nutrients, and pollutants flushed into streams.

3.1.7 Vegetation

The more vegetation there is in a watershed to intercept and transpire water, the less runoff there will be. The amount and type of vegetation in a watershed influences the rates of runoff and erosion in that watershed. Plants reduce the impact of raindrops on soil and slow the speed of water flowing across the surface of the land, allowing more water to infiltrate into the soil. A forested or native grass watershed typically delivers its runoff slowly, so soil and channel erosion are usually not severe.

Runoff rates and amounts are usually excessive in watersheds with too much timber clearing, row cropping, grazing, paving, urbanization, and other types of development. Runoff increases in these situations because less water is allowed to infiltrate to groundwater or return to the atmosphere through transpiration. Unnaturally high runoff rates produce more frequent, larger, and faster-rising floods. Likewise, more runoffs can increase erosion of the land surface. More flooding can increase streambed and streambank erosion as channels adjust to the new channel forming flows.

Many conditions and activities in a watershed can influence a stream. Uncontrolled erosion and runoff from construction sites; row cropping of highly erodible land; construction of houses, buildings and parking lots that prevent infiltration and speed runoff of precipitation; removal of riparian vegetation; improper disposal or excessive application of industrial, agricultural, or household chemicals; point and nonpoint sources of nutrients and runoff from strip mines are some ways receiving streams can be degraded. These and other factors contribute to poor water quality, unstable stream channels and unhealthy biological communities. Setting up erosion control practices on construction areas and farmland, limiting the amount of impervious area associated with urban developments and maintaining or establishing areas of timber and grasses are important efforts

toward reducing the negative effects on streams caused by our uses of the land.

Land Use

Land use defines the exploitation (natural and human interactions) characteristics of watersheds which affect the various hydrological processes within the watershed. The watershed classification based on the land use can be given as below.

- Agricultural
- Urban
- Mountainous
- Forest
- Desert
- Coastal or marsh, or
- Mixed - a combination of two or more of the previous classifications

Agricultural Watershed: Agricultural watershed is the watershed in which agricultural activities (crop cultivation) is dominant. It experiences perhaps the most dynamically significant land-use change. This usually leads to increased infiltration, increased erosion, and/or decreased runoff. Depression storage is also increased by agricultural operations. When the fields are barren, falling raindrops tend to compact the soil and infiltration is reduced. There is lesser development of streams in agricultural watersheds. The small channels formed by erosion and runoff in the area are obliterated by tillage operations. The soil structure is altered by regular application of organic and/or inorganic manure. This, in turn, leads to changed infiltration characteristics.

Urban Watershed: These are the watershed areas having maximum manipulation for the convenience of human being. These are dominated by buildings, roads, streets, pavements, and parking lots. These features reduce the infiltrating land area and increase imperviousness. As drainage systems are artificially built, the natural pattern of water flow is substantially altered. For a given rainfall event, interception and depression storage can be significant but infiltration is considerably reduced. As a result, there is pronounced increase in runoff and pronounced decrease in soil erosion. Thus, an urban watershed is more vulnerable to flooding if the drainage system is inadequate. Once a watershed is urbanized, its land use is almost fixed and its hydrologic behavior changes due to changes in precipitation.

Mountainous Watershed: Because of higher altitudes, such watersheds receive considerable snowfall. Due to steep gradient and relatively less porous soil, infiltration is less and surface runoff is dominantly high for a given rainfall event. The areas downstream of the mountains are vulnerable to flooding. Due to snow melt, water yield is significant even during spring and summer.

Forest Watershed: These are the watersheds where natural forest cover dominates other land uses. In these watersheds, interception is significant, and evapotranspiration is a dominant component of the hydrologic cycle. The ground is usually littered with leaves, stems, branches, wood, etc. Consequently, when it rains, the water is held by the trees and the ground cover provided greater opportunity to infiltrate. The subsurface flow becomes dominant and there are times when there is little to no surface runoff. Because forests resist flow of overland water, the peak discharge is reduced. Complete deforestation could increase annual water yield by 20 to 40 %.

Desert Watershed: There is little to virtually no vegetation in desert watersheds. The soil is mostly sandy and little annual rainfall occurs. Stream development is minimal. Whenever there is rainfall, most of it is absorbed by the porous soil, some of it evaporates, and the remaining runs off only to be soaked in during its journey. There is limited groundwater recharge due to occurrence of less rainfall in these watersheds.

Coastal Watershed: The watersheds in coastal areas may partly be urban and are in dynamic contact with the sea. Their hydrology is considerably influenced by backwater from wave and tidal action of the sea. Usually, these watersheds receive high rainfall, mostly of cyclonic type, do not have channel control in flow, and are vulnerable to severe local flooding. In these watersheds, the water table is high, and saltwater intrusion threatens the health of coastal aquifers, which usually are a source of the fresh water supply.

Marsh or Wetland Watershed: Such lands are almost flat and are comprised of swamps, marshes, water courses, etc. They have rich wildlife and plenty of vegetation. As water is no limiting factor to satisfy evaporative demand, evaporation is dominant. Rainfall is normally high and infiltration is minimal. Most of the rainfall becomes runoff. The flood hydrograph peaks gradually and lasts for a long time.

Mixed Watershed: These are the watersheds, where multiple land use/land cover exists either because of natural settings or due to a combination of natural and human interaction activities. In these watersheds, a combination

of two or more of the previous classifications occurs and none of the single characteristics dominate the area. In India, most of the watersheds are of mixed nature of characteristics, where agriculture, forest, settlements (urban and rural) etc. land use occurs.

3.2 Socio-Economic characteristics of watershed

Land use of watersheds is hugely controlled by the socioeconomic characteristics of the area. Socioeconomic characteristics comprise:

- Demography: Size, structure, and composition of the community
- Literacy: Awareness status
- Socio-cultural aspects: Social, cultural, religious belief
- Land ownership: Landholding size
- Economic aspect: Income generating source, dependency.
- Peoples need: Willingness, demand, and supply.
- Institutional aspect: Legislations, incentives, and local institutions

Being the Interdisciplinary sector, Watershed management should incorporate the integrated approach. Integrated watershed management (IWM) is the process of developing and implementing a plan of action involving the manipulation of natural, agricultural, and human resources in a watershed to achieve resource objectives while considering the social, economic, and institutional factors operating within a river basin or other relevant region.

Let us sum up

A watershed is an area of land that drains or “sheds” water into a specific waterbody. Everybody in water has a watershed. Watersheds drain rainfall and snowmelt into streams and rivers. These smaller bodies of water flow into larger ones, including lakes, bays, and oceans. Gravity helps to guide the path that water takes across the landscape.

Glossary

Isochrones: The isochrone is a contour joining points of equal concentration time of the water in a watershed. The farthest isochrones from the outlet section represent the time passed for which the whole watershed surface contributes to the flow towards the outlet section after a uniform rainfall.

Channel Slope: The channel slope is determined as the elevation difference between the endpoints of the main channel divided by the channel length.

Channel Profile: It includes the point of origin of the stream called the head, the point of termination called the mouth, and a decreasing gradient of the stream channel towards the mouth.

Drainage Density: Drainage density (Dd) is the measure of closeness of drainage spacing. It is the indication of drainage efficiency of overland flow and the length of overland flow as well as the index of relative proportions. It is defined as the length of drainage per unit area.

Check your progress

1. What are the characteristics of watershed management?

Size, shape; physiography, slope, climate, drainage, land use, vegetation, geology and soils, hydrology and hydrogeology, socio-economic characteristics, basic data on watersheds.

2. What are the principal watershed characteristics?

The principal watershed characteristics are:

- Basin Area
- Basin Slope
- Basin Shape
- Basin Length

3. What are the channel characteristics of watershed?

The most common and important channel characteristics of the watersheds are:

- Channel Order
- Channel Length
- Channel Slope
- Channel Profile
- Drainage Density

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.

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Unit 4

Watershed Delineation and Codification

Structure

Learning Objectives

Overview

4.1 Watershed

4.2 Delineation and Codification

4.3 Digital Watershed Atlas of India

4.4 Manual watershed delineation

4.5 Automated or computerized watershed delineation

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to:

- Watershed Delineation and Codification
- Digital Watershed Atlas of India
- Manual watershed delineation
- Automated or computerized watershed delineation

Overview

All watershed delineation means is that you're drawing lines on a map to identify a watershed's boundaries. These are typically drawn on topographic maps using information from contour lines. Contour lines are lines of equal elevation, so any point along a given contour line is the same elevation. watershed codification is Watersheds are designated by suffixing Arabic numbers to a code of subcatchment. For example, a watershed code would be like 4G4D3, 2B2A3 etc. The watersheds codes are restricting to numerical 1 to 9 only.

4.1 Watershed

A watershed is a natural hydrologic entity governed by the terrain topography from where runoff is drained to a point. The term watershed is a general phenomenon thus its size and area depend on the scale of the base map used for delineation and codification.

Recognizing the importance of management of soil and water resources in the country following the natural system, the department of Agriculture and Cooperation developed the delineation and codification system, and “Watershed Atlas of India” has been published (Anon, 1990). The atlas has been devised based on the drainage map on 1:1 million scale following stream hierarchy where the whole country has been divided into six River Resource Region, 35 Basin, 112 Catchments, 550 Sub catchments and 3257 Watersheds. The codification has been made in a simplistic manner following alphanumeric system.

A watershed can be symbolized as 1A2B3 where “1” stands for River Resource Region, “A” designates the Basin in that river resource region, “2” indicates the Catchment within the basin, “B” indicates Sub catchment and “3” stands for the watershed number in the sequence of stream hierarchy.

4.2 Delineation and Codification

India is drained by many rivers and their tributaries where flood and droughts are frequent visitors. To combat these situations and to sustain agricultural production, the country’s land mass needs to be divided into smaller hydrological units that led to conceptualization of “Delineation and Codification of Watersheds in India”. The state of art of the delineation and codification system is that it is an open system, and the process can be extended to micro level using the drainage map of larger scale. The Watershed Atlas meets the requirement of planning at National and State level. One of the unique characteristics of the atlas is that it provides the basic framework of watersheds of the country and all the land based developmental programme of the country can be recognized with national code of watershed.

The delineation has done in seven stages starting with Water Resource Regions and their subsequent division and subdivisions into Basins, Catchments, Sub-catchments, Watershed, Sub watershed and Micrwatersheds in decreasing size of the delineated hydrologic unit.

Each of the sub-catchments is then divided into watersheds following the lower order streams, a group of tributaries or the left and right bank of higher order stream moving downstream upwards using the same base.

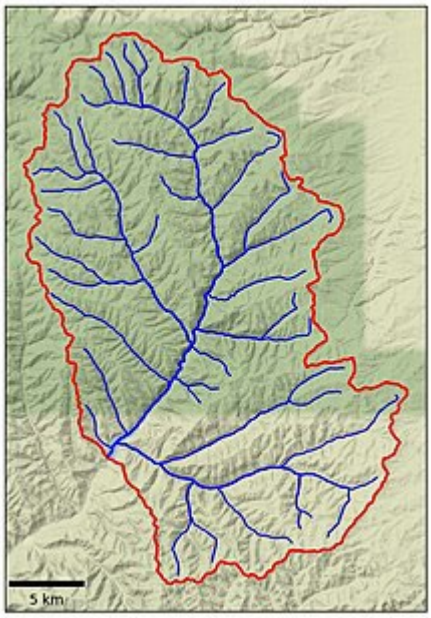


Fig. 4.1 A Model Watershed

When the left and right banks of a higher order stream are being delineated as separate watershed it would be advisable to cut across the stream at the upper end at a suitable confluence point to avoid the ambiguity in locating the ridge. The number of watersheds in a sub-catchment is restricted to 9. The size of individual subwatersheds is generally restricted to around 5,000 to 9,000 hectares, which is considered a viable working area for implemental programmes. The division of watershed is restricted to 22 alphabets (avoiding a, e, i, l and o). The code of the sub-watershed will be 2A1C1b.

1. Water Resource Regions (WPR): The six WRRs suggested by Dr. A. N. Khosla in 1949 have been adopted as such with slight modifications in their numbering which have been done in a clockwise manner, starting with Indus drainage as numbers like 1, 2, 3 etc.

- Indus drainage – 1
- Ganges drainage – 2
- Brahmaputra drainage – 3
- All drainage flowing into the Bay of Bengal except those at 2 & 3 – 4
- All drainage flowing into the Arabian Sea except Indus drainage – 5
- Western Rajasthan mostly ephemeral drainage – 6

2. Basins: Each WRR has been divided into basins which constitute individual big rivers like Krishna, Narmada, Chambal etc. or a combination of smaller ones which are contiguous to each other, such as basin between Cauvery and Krishna have been divided into lower and upper basins or left bank and right bank basins as in the case of Brahamputra. Basins are assigned letters as A, B, C……..Z.

3. Catchments: Each basin has been divided into a number of catchments, which pertain to main tributaries or a group of contiguous tributaries or individual streams. Catchments are represented by numerals suffixed to basin code as 1, 2, 3…….9.

4. Subcatchments: The catchments are further divided into a number of sub-catchments, which are mainly smaller tributaries and streams. Sub-catchments are indicated by suffixing alphabets to catchment code as A, B, C…….Z.

5. Watersheds: Each subcatchments has been divided into a number of watershed which are the smallest sized hydrologic units in the macro level category on the base of 1:1 million scale in the Watershed Atlas of India, published by SLUSI. Watersheds are designated by suffixing Arabic numbers to a code of subcatchment. For example a watershed code would be like 4G4D3, 2B2A3 etc. The watersheds codes are restricts to numerical 1 to 9 only.

6. Subwatersheds: Each watershed is further divided into sub-watersheds on 1:50000 scale (SOI topographical map) in which main tributaries and streams are taken up for delineation of sub-watersheds.

Subwatersheds designated by small English alphabets as a, b, c……z which is suffixed to watershed code. The complete code of a subwatershed appears as 2A2E8a, b, c, etc. The small alphabet ‘e’ ‘i’ ‘l’ and ‘o’ are discarded in view of the cartographic consideration and to avoid its ambiguity with the sequence of code. Hence, the total numbers of codes for subwatersheds is restricted to 22.

7. Micro Level Delineation: Planning to phase out the watershed management at catchment level and to formulate action programme needs, micro level delineation. The delineation of watershed boundary at micro level could easily be attained by superimposing the watershed boundary from Watershed Atlas on to a drainage map of 1:50000 scale.

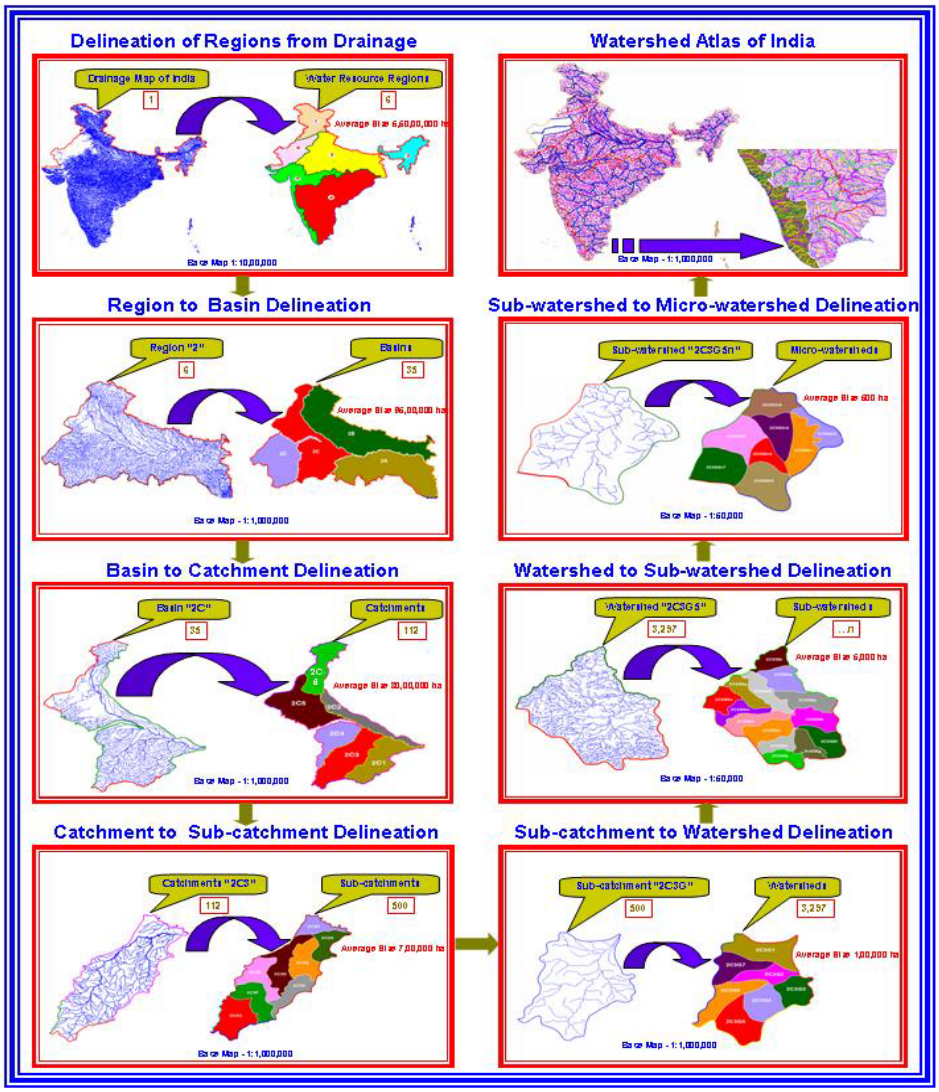
The delineation and codification would follow the similar system based on stream hierarchy and codification from downstream upward that allows to get a microwatershed of 500 to 1,500 ha size viable enough for

implementation of soil and water conservation programmes. The beauty of such delineation and codification is that it could be recognized with a national code with seven digits and symbolized as 1A2B3a1 where “a” stands for sub-watershed and “1” denotes the microwatershed. Thus 1A2B3a1 stands for a national code of a microwatershed belonging to Water Resource Region “1”, Basin “1A”, Catchment “1A2”, Subcatchment “1A2B”, Watershed “1A2B3”, Subwatershed “1A2B3a” and Microwatershed “1A2B3a1”.

Table: Average size and size ranges for each Hydrological Units

S. No.	Category of Hydrologic Units	Example of Code	Size Range (ha)	Average Size (ha)
1.	Water Resource Region	2	270,00,000-1130,00,000	5,50,00,000
2.	Basins	A	30,00,000-300,00,000	95,00,000
3.	Catchments	1	10,00,000-50,00,000	30,00,000
4.	Subcatchments	A	200,000-10,00,000	7,00,000
5.	Watersheds	2	20,000-300,000	1,00,000
6.	Subwatersheds	a	5,000-9,000	7,000
7.	Microwatersheds	2	500-1,500	1,000

Various steps involved in delineation of watersheds at micro level are illustrated and is given below: -



4.3 Digital Watershed Atlas of India

The Watershed Atlas of India published by Soil and Land Use Survey of India has now been brought under digital environment using GIS and RDBMS by the organization as step towards development of Spatial Database not only to serve the country in a sustained manner but to disseminate the information online to the users through information technology.

The digital watershed atlas is customized in such a way that it can be given to the users both River Resource Region and Basin wise or State and Catchment basis. The beauty of the atlas is that the name of the river will blink on the monitor when the cursor is placed on any part of the atlas. Subsequently, detailed description of the watershed highlighting the area and the district and State it falls and its linkages with main drainage system

will follow. The digital atlas is going to be the backbone of many future information systems and applications.

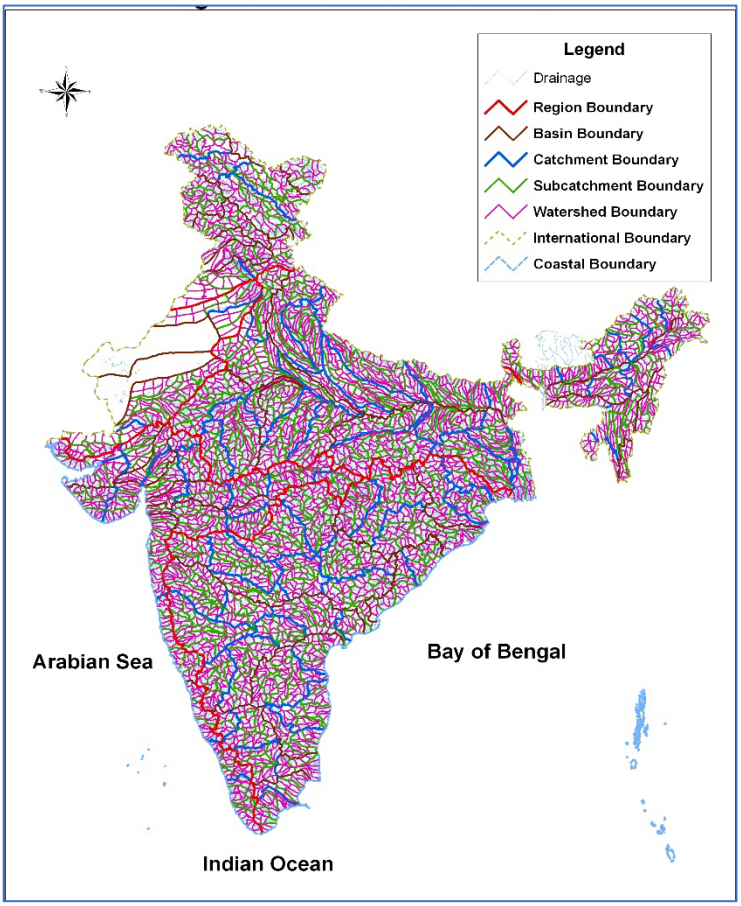


Fig. 4.2 Digital Watershed Atlas of India

4.4 Manual watershed delineation

The conventional method of finding a watershed boundary is to draw it by hand on a paper topograpic map, or on a transparent overlay. The watershed area can then be estimated using a planimeter, by overlaying graph paper and counting grid cells, or the result can be digitized for use with mapping software. The same process can be done on a computer, sketching the watershed boundary (with a mouse or stylus) over a digital copy of a topographic map. This is referred to as "heads up digitizing" or "on-screen digitizing."

Example of an idealized watershed boundary, drawn on a topographic map with elevation contours. Any precipitation that falls inside the watershed boundary will flow toward the watershed outlet at the bottom.

For "manual" watershed delination, one must know how to read and interpret a topographic map, for example to identify ridges, valleys, and the direction of steepest slope. Even in the computer era, manual watershed delineation is still a useful skill, in order to check whether watersheds generated with software are correct. Instructions for manual watershed delineation can be found in some textbooks in geography or environmental management, in government pamphlets, or in online video tutorials.

According to the US Geological Survey, there are 5 steps to manual watershed delineation:

- Find the point of interest along a stream on the map. This is the "watershed outlet" or "pour point."
- Imagine or draw surface water flow lines that point downhill perpendicular to the topographic contours (this is the steepest direction).
- Mark the location of topographical high points (peaks) around the stream.
- Mark the points along contours that divide flows towards or away from the stream (ridges).
- Connect the dots to delineate the watershed.

General Rules:

- The watershed boundary should be perpendicular to contour lines where it crosses them.
- The watershed boundary must not cross rivers or streams other than at the outlet. (In some cases, a blue line representing a man-made canal or pipeline may traverse your watershed boundary.)
- The watershed boundary should run along ridgelines and connect high points.

One disadvantage to manual watershed delineation is that it is subject to errors and the individual judgment of the analyst. The Illinois Environmental Protection Agency wrote, "bear in mind that delineating a watershed is an inexact science. Any two people, even if both are experts, will come up with slightly different boundaries.

Especially for smaller watersheds and when accurate results are important, field reconnaissance may be needed to find features that are not shown on maps. "Going out into the field allows you to identify human alterations, such as road ditches, storm sewers and culverts that could change the direction of waters flow and thus change the watershed boundaries.

4.5 Automated or computerized watershed delineation

Using computer software to delineate watersheds can be much faster than manual methods. It may also be more consistent, as it removes analyst's subjectivity. Automatic methods of watershed delineation have been in use since the 1980s and are now in widespread use in the science and engineering communities. Researchers have even used computer methods to delineate watersheds on Mars.

Automated watershed delineation methods use digital data of the earth's elevation, a Digital Elevation Model, or DEM. Typically, algorithms use the method of "steepest slope" to calculate the flow direction from a grid cell (or pixel) to one of its neighbors.

It is possible to use DEMs in different formats for watershed delineation, such as a Triangular Irregular Network (TIN), or Hexagonal tiling. However, most contemporary algorithms make use of a regular rectangular grid. In the 1980s and 1990s, digital elevation models were often obtained by scanning and digitizing the contours on paper topographic maps, which were then converted to a TIN or a gridded DEM. More recently, the DEM is obtained by aerial or satellite remote sensing, using stereophotogrammetry, lidar, or radar.

To use a rectangular grid DEM for watershed delineation, it must first be processed or "conditioned" to return realistic results. The result is sometimes referred to as a "hydro-enforced" DEM or a "HydroDEM." Most of the software packages listed below can perform these functions on a "raw" DEM, or analysts can download hydrologically conditioned DEMs such as the near-global HydroSHEDS, MERIT-Hydro, or EDNA for the continental United States. The usual steps for hydrologic conditioning of a DEM are:

- Fill sinks.
- "Burn in" the stream channels.
- Calculate flow direction.
- Calculate flow accumulation.

3D rendering of a hydrologically conditioned digital elevation model. This example from the U.S. Geological Survey shows “walling” or “fencing” using data from the Watershed Boundary Dataset (WBD) and “burning” of streams from the National Hydrography Dataset (NHDPlus).

Additionally, some methods allow for "fencing ridgelines" and burning in flow pathways through lakes. Some methods also enforce a small slope onto flat

areas so that flow will continue to move toward the outlet. The step of "burning in" stream channels involves artificially deepening the channel, by subtracting a large elevation value from pixels that represent the channel. This ensures that once flow has entered the channel, it will stay there rather than jumping out and flowing overland or into another channel. Some algorithms infer the location of channels automatically from the DEM. Better results are usually obtained by burning in mapped stream channels, or channels derived from satellite or aerial imagery.

There are several different algorithms available for calculating flow direction from a DEM. The first method, introduced by Australian geographers O'Callaghan and Mark in 1984, is referred to as D8. Water flows from a pixel to one of 8 possible directions to a neighboring cell (including diagonally), based on the direction of steepest slope. There are disadvantages to this method as water flow is limited to 8 directions, separated by 45°, which may result in unrealistic flow patterns. Also, because all of the flow is routed in one direction, the D8 method is unable to model situations where the flow diverges, such as on convex hillsides, in a river delta, or in branched or braided rivers. Alternative algorithms have been proposed and implemented to overcome this limitation, such as D. Nevertheless, the D8 algorithm remains in widespread use, and has been used to create important datasets such as HydroBasins and MERIT-Basins.

Computerized watershed delineation is not always correct. Some errors stem from incorrectly placing the watershed outlet on the digital river network, or "snapping the pour point." Another class of errors stems from inaccuracies in the digital terrain data, or where its resolution is too coarse to capture flow pathways. In general, DEMs with higher spatial resolution can more realistically describe topography of the land surface and flow direction. However, there is a tradeoff, as a finer grid with more pixels increases computing time. Nevertheless, even high-resolution data may not adequately capture flow pathways in complex environments like cities and suburbs, where flow is directed by curbs, culverts, and storm drains. Finally, some errors can result from the algorithm or the choice of parameters.

Because errors are common, some authorities insist that the results of automated delineation must be carefully checked. The US Geological Survey's standards for the US Watershed Boundary Dataset allow the use of software "to generate intermediate or “draft” boundary lines," which then must be verified by the analyst by overlaying them on a computer display over base maps (scanned topographic maps, aerial photographs) to verify their accuracy.

Let us sum up

Watershed delineation is the process of identifying the boundary of a watershed, also referred to as a catchment, drainage basin, or river basin. It is an important step in many areas of environmental science, engineering, and management, for example to study flooding, aquatic habitat, or water pollution.

The activity of watershed delineation is typically performed by geographers, scientists, and engineers. Historically, watershed delineation was done by hand on paper topographic maps, sometimes supplemented with field research. In the 1980s, automated methods were developed for watershed delineation with computers and electronic data, and these are now in widespread use.

Map of the watershed of the Lost Creek Reservoir in Morgan County, Utah, United States, showing the major streams in blue and the watershed boundary in red.

Computerized methods for watershed delineation use digital elevation models (DEMs), datasets that represent the height of the earth's land surface. Computerized watershed delineation may be done using specialized hydrologic modeling software such as WMS, Geographic Information System software like ArcGIS or QGIS, or with programming languages like Python or R.

Glossary

Catchment area: An area from which surface runoff is carried away by a single drainage system.

A river basin is the portion of land drained by a river and its tributaries.

A watershed is simply the area of land that catches rain and snow and drains or seeps into a marsh, stream, river, lake or groundwater.

Check your progress

1. What are the 5 steps to manual watershed delineation?
    - Find the point of interest along a stream on the map. This is the "watershed outlet" or "pour point."
    - Imagine or draw surface water flow lines that point downhill perpendicular to the topographic contours (this is the steepest direction).
    - Mark the location of topographical high points (peaks) around the

stream.

- Mark the points along contours that divide flows towards or away from the stream (ridges).
- Connect the dots to delineate the watershed.

2. What are the General Rules of watershed delineation?
   - The watershed boundary should be perpendicular to contour lines where it crosses them.
   - The watershed boundary must not cross rivers or streams other than at the outlet. (In some cases, a blue line representing a man-made canal or pipeline may traverse your watershed boundary.)
   - The watershed boundary should run along ridgelines and connect high points.

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.
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Block 2

Function and Structure of Watershed

Unit 5: Morphological Characteristic of Watershed

Unit 6: Climatic characteristics of Watershed

Unit 5

Morphological Characteristic of Watershed

Structure

Learning Objectives

Overview

5.1 Morphometric characteristics

5.1.1 Linear Aspect:

5.1.2 Aerial Aspect (Shape Parameters)

5.1.3 Relief Aspect

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able understand the following Morphological Characteristic of Watershed,

- Linear Aspect: one dimension
- Areal Aspect: two dimensions
- Relief Aspect: three dimensions

Overview

The physical properties of the watershed affect the characteristics of runoff and are of great interest in hydrologic analysis. The morphological characteristics such as stream order, drainage density, channel length, channel slope, watershed length and width, topography, geology and or soil characteristics, climate, vegetation and land use are all important to our understanding of the hydrology of the watershed. All of these factors are important in understanding the physical processes of the watershed and how we may understand such things as infiltration and land management.

5.1 Morphometric characteristics

Morphometric analysis is referring as the quantitative evaluation of form characteristics of the earth surface and any landform unit. It incorporates quantitative study of the various components such as, stream segments, basin length, basin parameters, basin area, altitude, slope, profiles of the land which indicates the nature of development of the basin.

In general, the watersheds are selected for the morphometric analysis in following heads:

- Linear Aspect: one dimension
- Areal Aspect: two dimensions
- Relief Aspect: three dimensions

5.1.1 Linear Aspect:

The drainage network transport water and the sediments of a basin through a single outlet, which is marked as the maximum order of the basin and conventionally the highest order stream available in the basin considered as the order of the basin. The size of rivers and basins varies greatly with the order of the basin. Ordering of streams is the first stage of basin analysis.

Stream Order (U): system of ordering streams has been followed in general because of its simplicity where the smallest, un-branched fingertip streams are designated as 1st order, the confluence of two 1st order channels give a channel segments of 2nd order, two 2nd order streams join to form a segment of 3rd order and so on. When two channel of different order join then the higher order is maintained.

Stream Number (Nu): The total number of stream segments present in each order is the stream number (Nu).

Stream Length (Lu): The total length of individual stream segments of each order is the stream length of that order.

Basin length (Lb): The basin length as the longest in the basin in which are end being the mouth.

Basin Area (A): The area of the watershed is another important parameter like the length of the stream drainage. Schumm (1956) established an interesting relation between the total watershed areas and the total stream lengths, which are supported by the contributing areas.

Basin Perimeter (P): Basin perimeter is the outer boundary of the watershed that enclosed its area. It is measured along the divides between watersheds and may be used as an indicator of watershed size and shape.

Stream Frequency (Fs): The drainage frequency means stream frequency (or channel frequency) Fs as the number of stream segments per unit area.

Length of Overland Flow (Lo): The average length of overland flow is approximately half the average distance between stream channels and is therefore approximately equals to half of reciprocal of drainage density.

Drainage Density (Dd): Drainage density is the stream length per unit area in region of watershed is another element of drainage analysis. Drainage density is a better quantitative expression to the dissection and analysis of landform.

Texture Ratio (Rt): Texture ratio is an important factor in the drainage morphometric analysis which is depending on the underlying lithology, infiltration capacity and relief aspect of the terrain. The texture ratio is expressed as the ratio between the first order streams and perimeter of the basin (Rt = Nl / P)

Drainage Texture (Dt): Drainage texture is one of the important concepts of geomorphology which means that the relative spacing of drainage lines. Drainage texture is also depending on the underlying lithology, infiltration capacity and relief aspect of the terrain. Dt is total number of stream segments of all orders per perimeter of that area. It has classified drainage texture into five different textures i.e., very coarse (<2), coarse (2 to 4), moderate (4 to 6), fine (6 to 8) and very fine (>8).

Drainage Intensity (Di): The drainage intensity, as the ratio of the stream frequency to the drainage density. Low value of drainage intensity implies that drainage density and stream frequency have little effect on the extent to which the surface has been lowered by agents of denudation. With these low values of drainage density, stream frequency and drainage intensity, surface runoff is not quickly removed from the watershed, making it highly susceptible to flooding, gully erosion and landslides.

Infiltration Number (If): The infiltration number of a watershed is defined as the product of drainage density and stream frequency and given an idea about the infiltration characteristics of the watershed. The higher the infiltration number, the lower will be the infiltration and the higher ran-off.

5.1.2 Aerial Aspect (Shape Parameters)

The areal aspect is the two-dimensional properties of a basin.

Form Factor (Ff): Form factor' may be defined as the ratio of basin area to square of the basin length. The value of form factor would always be less than 0.754 (for a perfectly circular watershed). Smaller the value of form factor, more elongated will be the watershed. The watershed with high form factors has high peak flows of shorter duration, whereas elongated watershed with low form factor ranges from 0.42 indicating them to be elongated in shape and flow for longer duration.

Elongation Ratio (Re): Elongation ratio' is defined as the ratio of diameter of a circle of the same area as the basin to the maximum basin length. Strahler states that this ratio runs between 0.6 and 1.0 over a wide variety of climatic and geologic types. The varying slopes of watershed can be classified with the help of the index of elongation ratio, i.e., circular (0.9-0.10), oval (0.8-0.9), less elongated (0.7-0.8), elongated (0.5-0.7), and more elongated (< 0.5).

Circularity Ratio (Rc): For the out-line form of watershed used a dimensionless circularity ratio as a quantitative method. Circularity ratio is defined as the ratio of watershed area to the area of a circle having the same perimeter as the watershed and it is pretentious by the lithological character of the watershed. Miller (1953) has described the basin of the circularity ratios range 0.4 to 0.5, which indicates strongly elongated and highly permeable homogenous geologic materials.

Compactness Coefficient (Cc): Compactness Coefficient is used to express the relationship of a hydrologic basin to that of a circular basin having the same area as the hydrologic basin. A circular basin is the most susceptible from a drainage point of view because it will yield shortest time of concentration before peak flow occurs in the basin (Nooka Ratnam et al. 2005).

5.1.3 Relief Aspect

Linear and areal features have been considered as the two-dimensional aspect lie on a plan. The third dimension introduces the concept of relief. By measuring the vertical fall from the head of each stream segment to the point where it joins the higher order stream and dividing the total by the number of streams of that order, it is possible to obtain the average vertical fall.

Basin Relief (H): Basin relief is the elevation difference of the highest and lowest point of the valley floor.

Relief Ratio (Rh): Relief ratio is defined as the ratio between the total relief of a basin i.e., elevation difference of lowest and highest points of a basin,

and the longest dimension of the basin parallel to the principal drainage line. The high values of Rh indicate steep slope and high relief and vice-versa. Run-off is generally faster in steeper basins, producing more peaked basin discharges and greater erosive power.

Let us sum up

morphometric parameters are discussed above, i.e., stream length ratio (Rl), bifurcation ratio (Rb), drainage density (Dd), drainage texture (Dt), stream frequency (Fs), elongation ratio (Re), circularity ratio (Rc), form factor (Rf), length of overland flow (Lg), and relief ratio (Rh).

Glossary

Basin Perimeter (P): Basin perimeter is the outer boundary of the watershed that enclosed its area. It is measured along the divides between watersheds and may be used as an indicator of watershed size and shape.

Stream Frequency (Fs): The drainage frequency means stream frequency (or channel frequency) Fs as the number of stream segments per unit area.

Length of Overland Flow (Lo): The average length of overland flow is approximately half the average distance between stream channels and is therefore approximately equals to half of reciprocal of drainage density.

Check your progress

4. What are the principal watershed characteristics?

The principal watershed characteristics are:

- Basin Area
- Basin Slope
- Basin Shape
- Basin Length

5. What are the channel characteristics of watershed?

The most common and important channel characteristics of the watersheds are:

- Channel Order
- Channel Length
- Channel Slope
- Channel Profile
- Drainage Density

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Unit 6

Climatic Characteristic of Watershed

Structure

Learning Objectives

Overview

6.1 Water Cycle

6.1.1 Evaporation

6.1.2 Condensation

6.1.3 Precipitation

6.1.4 Interception

6.1.5 Infiltration

6.1.6 Percolation

6.1.7 Transpiration

6.1.8 Runoff

6.1.9 Storage

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to be known about

- the Processes of Water Cycle and
- how water cycle impact of watershed.

Overview

Climatic Characteristics Rainfall & its movement, Intensity, Duration, Temperature, Wind Velocity, Humidity, Transpiration, and Evaporation. Climate is the major force shaping the land and controlling streamflow. The climate of an area is determined by the area's latitude, elevation, vegetation, topography and nearness to the oceans or other large water bodies.

Together these factors determine temperatures, humidity, wind, precipitation and evaporation in a watershed. Weathering factors, such as rain, snow, wind, glaciers and temperature changes, erode soil and rock formations and change the topography of the watershed. Climate also affects streamflow, which creates and changes stream channels.

6.1 Water Cycle

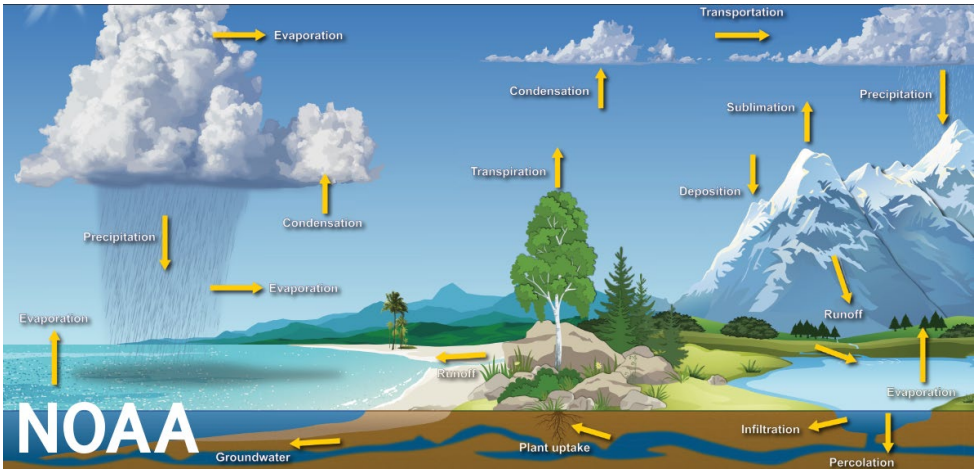


Fig. 6.1 The water cycles. (Image credit: Dennis Cain/NWS)

The water cycle shows the continuous movement of water within the Earth and atmosphere. It is a complex system that includes many different processes. Liquid water evaporates into water vapor, condenses to form clouds, and precipitates back to earth in the form of rain and snow. Water in different phases moves through the atmosphere (transportation). Liquid water flows across land (runoff), into the ground (infiltration and percolation), and through the ground (groundwater). Groundwater moves into plants (plant uptake) and evaporates from plants into the atmosphere (transpiration). Solid ice and snow can turn directly into gas (sublimation). The opposite can also take place when water vapor becomes solid (deposition).

The global water cycle can be described with nine major physical processes which form a continuum of water movement. Complex pathways include the passage of water from the gaseous envelope around the planet called the atmosphere, through the bodies of water on the surface of earth such as the oceans, glaciers and lakes, and at the same time (or more slowly) passing through the soil and rock layers underground. Later, the water is returned to the atmosphere. A fundamental characteristic of the hydrologic cycle is that it has no beginning an it has no end. It can be studied by starting at any of the following processes: evaporation, condensation, precipitation, interception, infiltration, percolation, transpiration, runoff, and storage.

6.1.1 Evaporation

Evaporation occurs when the physical state of water is changed from a liquid state to a gaseous state. A considerable amount of heat, about 600 calories of energy for each gram of water, is exchanged during the change of state. Typically, solar radiation and other factors such as air temperature, vapor pressure, wind, and atmospheric pressure affect the amount of natural evaporation that takes place in any geographic area. Evaporation can occur on raindrops, and on free water surfaces such as seas and lakes. It can even occur from water settled on vegetation, soil, rocks and snow. There is also evaporation caused by human activities. Heated buildings experience evaporation of water settled on its surfaces. Evaporated moisture is lifted into the atmosphere from the ocean, land surfaces, and waterbodies as water vapor. Some vapor always exists in the atmosphere.

6.1.2 Condensation

Condensation is the process by which water vapor changes it's physical state from a vapor, most commonly, to a liquid. Water vapor condenses onto small airborne particles to form dew, fog, or clouds. The most active particles that form clouds are sea salts, atmospheric ions caused by lightning,and combustion products containing sulfurous and nitrous acids. Condensation is brought about by cooling of the air or by increasing the amount of vapor in the air to its saturation point. When water vapor condenses back into a liquid state, the same large amount of heat (600 calories of energy per gram) that was needed to make it a vapor is released to the environment.

6.1.3 Precipitation

Precipitation is the process that occurs when all forms of water particles fall from the atmosphere and reach the ground. There are two sub- processes that cause clouds to release precipitation, the coalescence process and the ice- crystal process. As water drops reach a critical size, the drop is exposed to gravity and frictional drag. A falling drop leaves a turbulent wake behind which allows smaller drops to fall faster and to be overtaken to join and combine with the lead drop. The other sub-process that can occur is the ice-crystal formation process. It ocurrs when ice develops in cold clouds or in cloud formations high in the atmosphere where freezing temperatures occur. When nearby water droplets approach the crystals, some droplets evaporate and condense on the crystals. The crystals grow to a critical size and drop as snow or ice pellets. Sometimes, as the pellets fall through lower elevation air, they melt and change into raindrops.

Precipitated water may fall into a waterbody, or it may fall onto land. It is then dispersed several ways. The water can adhere to objects on or near the planet surface or it can be carried over and through the land into stream channels, or it may penetrate into the soil, or it may be intercepted by plants.

When rainfall is small and infrequent, a high percentage of precipitation is returned to the atmosphere by evaporation.

The portion of precipitation that appears in surface streams is called runoff. Runoff may consist of component contributions from such sources as surface runoff, subsurface runoff, or ground water runoff. Surface runoff travels over the ground surface and through surface channels to leave a catchment area called a drainage basin or watershed. The portion of the surface runoff that flows over the land surface towards the stream channels is called overland flow. The total runoff confined in the stream channels is called the streamflow.

6.1.4 Interception

Interception is the process of interrupting the movement of water in the chain of transportation events leading to streams. The interception can take place by vegetal cover or depression storage in puddles and in land formations such as rills and furrows.

When rain first begins, the water striking leaves and other organic materials spreads over the surfaces in a thin layer, or it collects at points or edges. When the maximum surface storage capability on the surface of the material is exceeded, the material stores additional water in growing drops along its edges. Eventually the weight of the drops exceeds the surface tension and water falls to the ground. Wind and the impact of raindrops can also release the water from the organic material. The water layer on organic surfaces and the drops of water along the edges are also freely exposed to evaporation.

Additionally, interception of water on the ground surface during freezing and sub- freezing conditions can be substantial. The interception of falling snow and ice on vegetation also occurs. The highest level of interception occurs when it snows on conifer forests and hardwood forests that have not yet lost their leaves.

6.1.5 Infiltration

Infiltration is the physical process involving movement of water through the boundary area where the atmosphere interfaces with the soil. The surface phenomonon is governed by soil surface conditions. Water transfer is related to the porosity of the soil and the permeability of the soil profile.

Typically, the infiltration rate depends on the puddling of the water at the soil surface by the impact of raindrops, the texture and structure of the soil, the initial soil moisture content, the decreasing water concentration as the water moves deeper into the soil filling of the pores in the soil matrices, changes in the soil composition, and to the swelling of the wetted soils that in turn close cracks in the soil. Water that is infiltrated and stored in the soil can also become the water that later is evapotranspired or becomes subsurface runoff.

6.1.6 Percolation

Percolation is the movement of water though the soil, and it's layers, by gravity and capillary forces. The prime moving force of groundwater is gravity. Water that is in the zone of aeration where air exists is called vadose water. Water that is in the zone of saturation is called groundwater. For all practical purposes, all groundwater originates as surface water. Once underground, the water is moved by gravity. The boundary that separates the vadose and the saturation zones is called the water table. Usually, the direction of water movement is changed from downward and a horizontal component to the movement is added that is based on the geologic boundary conditions.

Geologic formations in the earth's crust serve as natural subterranean reservoirs for storing water. Others can also serve as conduits for the movement of water. Essentially, all groundwater is in motion. Some of it, however, moves extremely slowly. A geologic formation which transmits water from one location to another in sufficient quantity for economic development is called an aquifer. The movement of water is possible because of the voids or pores in the geologic formations. Some formations conduct water back to the ground surface. A spring is a place where the water table reaches the ground surface. Stream channels can be in contact with an unconfined aquifer that approach the ground surface. Water may move from the ground into the stream, or visa versa, depending on the relative water level. Groundwater discharges into a stream forms the base flow of the stream during dry periods, especially during droughts. An influent stream supplies water to an aquifer while and effluent stream receives water from the aquifer.

6.1.7 Transpiration

Transpiration is the biological process that occurs mostly in the day. Water inside of plants is transferred from the plant to the atmosphere as water vapor through numerous individual leave openings. Plants transpire to move nutrients to the upper portion of the plants and to cool the leaves exposed

to the sun. Leaves undergoing rapid transiration can be significantly cooler than the surrounding air. Transpiration is greatly affected by the species of plants that are in the soil and it is strongly affected by the amount of light to which the plants are exposed. Water can be transpired freely by plants until a water deficit develops in the plant and it water-releasing cells (stomata) begin to close. Transpiration then continues at a must slower rate. Only a small portion of the water that plants absorb are retained in the plants.

Vegetation generally retards evaporation from the soil. Vegetation that is shading the soil, reduces the wind velocity. Also, releasing water vapor to the atmosphere reduces the amount of direct evaporation from the soil or from snow or ice cover. The absorption of water into plant roots, along with interception that occurs on plant surfaces offsets the general effects that vegetation has in retarding evaporation from the soil. The forest vegetation tends to have more moisture than the soil beneath the trees.

6.1.8 Runoff

Runoff is flow from a drainage basin or watershed that appears in surface streams. It generally consists of the flow that is unaffected by artificial diversions, storages or other works that society might have on or in a stream channel. The flow is made up partly of precipitation that falls directly on the stream, surface runoff that flows over the land surface and through channels, subsurface runoff that infiltrates the surface soils and moves laterally towards the stream, and groundwater runoff from deep percolation through the soil horizons. Part of the subsurface flow enters the stream quickly, while the remaining portion may take a longer period before joining the water in the stream. When each of the component flows enter the stream, they form the total runoff. The total runoff in the stream channels is called streamflow and it is generally regarded as direct runoff or base flow.

6.1.9 Storage

There are three basic locations of water storage that occur in the planetary water cycle. Water is stored in the atmosphere; water is stored on the surface of the earth, and water stored in the ground.

Water stored in the atmosphere can be moved relatively quickly from one part of the planet to another part of the planet. The type of storage that occurs on the land surface and under the ground largely depend on the geologic features related to the types of soil and the types of rocks present at the storage locations. Storage occurs as surface storage in oceans, lakes, reservoirs, and glaciers; underground storage occurs in the soil, in aquifers, and in the crevices of rock formations.

The movement of water through the eight other major physical processes of the water cycle can be erratic. On average, water the atmosphere is renewed every 16 days. Soil moisture is replaced about every year. Globally, waters in wetlands are replaced about every 5 years while the residence time of lake water is about 17 years. In areas of low development by society, groundwater renewal can exceed 1,400 years. The uneven distribution and movement of water over time, and the spatial distribution of water in both geographic and geologic areas, can cause exteme phenomena such as floods and droughts to occur.

Let us sum up

Climatology, the science of climate and its causes, becomes important in understanding regional issues in watershed science. Though sometimes used synonymously with weather, climate is actually a distinct term with important ecological ramifications. Climate refers to an aggregate of both average and extreme conditions of temperature, humidity, and precipitation (including type and amount), winds, and cloud cover, measured over an extended period. Weather refers to present-day environmental conditions; current temperatures and meteorological events make up weather, not climate. Long-term weather trends establish averages which become climatic regimes. Climate heavily influences watershed vegetation communities, streamflow magnitude and timing, water temperature, and many other key watershed characteristics.

Glossary

Evaporation occurs when the physical state of water is changed from a liquid state to a gaseous state.

Condensation is the process by which water vapor changes it's physical state from a vapor, most commonly, to a liquid.

Transpiration is the biological process that occurs mostly in the day. Water inside of plants is transferred from the plant to the atmosphere as water vapor through numerous individual leave openings.

Interception is the process of interrupting the movement of water in the chain of transportation events leading to streams.

Check your progress

1. What are the Climatic Characteristics of watershed?

Climatic Characteristics Rainfall & its movement, Intensity, Duration, Temperature, Wind Velocity, Humidity, Transpiration, and Evaporation.

2. How does process of Runoff impacts watershed?

Runoff may consist of component contributions from such sources as surface runoff, subsurface runoff, or ground water runoff. Surface runoff travels over the ground surface and through surface channels to leave a catchment area called a drainage basin or watershed. The portion of the surface runoff that flows over the land surface towards the stream channels is called overland flow. The total runoff confined in the stream channels is called the streamflow.

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Block 3

Watershed Analysis

Unit 7: Land capability classification of watershed

Unit 8: Soil resource evaluation

Unit 9: Slope analysis and vegetation cover

Unit 7

Land capability classification of watershed

Structure

Learning Objectives

Overview

7.1 Definition of Land Capability

7.2 Classification of Land Capability

7.2.1 Group 1 Lands: Generally Suitable for Cultivation

7.2.2 Group 2 Lands: Generally, Not Suitable for Cultivation

7.3 Land Capability Classification of India

7.3.1 Land Suitable for Cultivation

7.3.2 Land Not Suitable for Cultivation

7.4 Significance of Land Capability Classification in Land use Planning

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to:

- Definition of Land Capability
- Classification of Land Capability of USDA
- Land Capability Classification of India
- Significance of Land Capability Classification in Land use Planning

Overview

Land capability plays a vital role in deciding land use. In this lesson, we shall discuss land capability and its classification. The classification of the land capability is a process of assembling soils on a primary basis depending on their multiple capabilities for producing the general crops that are cultivated and the plants of the pasture without crumbling over a long

time. The classification of land capability consists of the value for assembling the soil. The process has been absorbed in several textbooks and is largely accepted by the public. The classification of the land capabilities is further subdivided into various capability subclasses and classes based on the national standard.

7.1 Definition of Land Capability

Land capability may be defined as the ability of the land surface to support natural plant growth/ wildlife habitat or artificial crop growth/ human habitat. Thus, it indicates the type of land use [viz., human habitation, agriculture, pastures, forests, wildlife habitat, etc.] that is suitable over a particular type of land. Different lands have different capabilities depending on the land characteristics like slope, soil type, soil depth and erosion conditions. If certain land characteristics are not conducive for agriculture, it is desirable to utilize or ensure the continuity of that land area for other land uses as mentioned earlier.

The goal of allocation of various land capabilities over a vast land area with varied characteristics is to achieve complete soil conservation. Complete soil conservation implies perfect soil health and zero soil erosion on a sustained basis. It also facilitates total water conservation and total vegetation conservation. Thereby it results in integrated watershed management on a long-term basis.

In the next section, we shall discuss the classification of land capability based on the land characteristics. This land capability classification should ensure appropriate land use for every land area for peaceful coexistence of different flora and fauna including human habitation and also a sustained productivity through human activities.

7.2 Classification of Land Capability

The Soil Conservation Service (SCS) of the United States Department of Agriculture (USDA) has done a pioneering work on land capability classification. According to that, the land capability is classified broadly into two groups based on the cultivability of the land. The first group consisting of all the lands which are suitable for cultivation is referred to as ‘Group 1 Lands’. The remaining group consisting of all the lands which are unsuitable for cultivation is referred to as ‘Group 2 Lands’. Each of these two groups are further classified into four classes. Thus ‘Group 1 Lands’ comprise ‘Land Classes I to IV’ which are cultivable and ‘Group 2 Lands’ comprise ‘Land Classes V to VIII’ which are non-cultivable.

The following paragraphs describe each of the two groups and eight land classes in terms of their land characteristics and land use:

7.2.1 Group 1 Lands: Generally Suitable for Cultivation

Class I Lands: These lands are nearly level with slopes generally within 1%. The soil is deep, fertile, easily workable and are not subjected to damaging overflows. There are hardly any restrictions or limitations for their use. These lands are very good lands which can be safely cultivated by using any farming method to grow any crop, even intensively also. However, proper crop rotation and green manure use should be followed to maintain soil fertility.

Class II Lands: These lands generally have gentle slope in the range of 1 to 3%. They can be easily cultivated with some conservation practices like contour farming, strip cropping, bund construction or terracing. Therefore, one or more of the following limitations exist which slightly reduce the crop choice:

1. Moderate susceptibility to erosion by wind or water;
2. Less than ideal soil depth;
3. Somewhat unfavourable soil structure and workability;
4. Slight to moderate salinity;
5. Occasionally damaging overflows;
6. Wetness existing permanently which can be corrected by drainage; and
7. Slight climatic limitations on land use and management.

Class III Lands: These lands generally have slopes in the range of 3 to 5% and therefore have severe limitations which further reduce the crop choice or require special conservation practices [like contour farming, strip cropping, cover cropping, bund construction or terracing] or both. Lands in this class have more restrictions than those in Class II Lands due to land characteristics. All the limitations of Class II Lands are applicable here also, but to a greater extent. Hay or pasture crops that completely cover the soil should be preferred. On wetlands of this Class -which usually have heavy and slowly permeable soils, a drainage system along with a suitable cropping plan to improve the soil structure is required.

Class IV Lands: These lands have good soil [i. e., having shallow soil depth and low fertility] and generally have somewhat steep slopes in the range of 5 to 8%. Therefore, they have either very severe limitations that largely

restrict the crop choice or require very careful management or both. Lands may be suitable only for two to three common crops which build and maintain soil -like the fully covering pastures, with occasional grain crops which can be grown usually once in five years. These lands may have one or more of the following permanent features:

1. Heavy susceptibility for erosion due to wind, water with severe effects of past erosion;
2. Low moisture holding capacity;
3. Frequent overflows accompanied by severe crop damage;
4. Water logging, excessive wetness and severe salinity; and
5. Moderately adverse climate.

Land Capability Sub-Classes: Lands in Classes II, III and IV are further categorised into sub-classes based on the following limitations:

1. Risk of erosion or past erosion damage is designated by the symbol ‘e’;
2. Wetness damage or overflow is designated by the symbol ‘w’;
3. Soil root zone limitations are denoted by ‘s’; and
4. Climatic limitations are designated by ‘c’.

7.2.2 Group 2 Lands: Generally, Not Suitable for Cultivation

Class V Lands: These lands generally have slopes in the range of 8 to 12%. They usually have not too little erosion hazard but have other limitations which restrict their use mainly to pastures, forests, wildlife food and cover. Controlled grazing may be permitted. Some of the examples of Class V Lands are:

1. Bottom lands subject to frequent overflows that prevent the normal production of cultivated crops;
2. Stony or rocky lands;
3. Few ponded areas where soils are suitable for grasses or trees.

Class VI Lands: The lands in this Class have shallow soils and generally have quite steep slopes ranging to 18%. They have severe limitations which restrict their use to pastures with very limited grazing, woodlands, wildlife food and cover. Some of the limitations of these lands which can’t be corrected are:

1. Severe erosion;

2. Stony texture with shallow rocks
3. Excessive wetness or overflow
4. Low moisture capacity
5. Severe climate.

Class VII Lands: The lands in this Class are generally eroded, rough, having shallow soil depth and steeper slopes ranging to 25%. The soil may be swampy or drought prone, with all the limitations of Class VI Lands even to a higher degree. If there is good rainfall, they may be used for forestry with fully green cover, gully control structures and severely restricted grazing.

Class VIII Lands: These lands are rough with probably the worst soil types and possibly the steepest slopes more than 25%. They can only be used with very sound gully control measures for forests –if conducive for tree growth, and for wildlife habitat. However, tree felling and grazing should be strictly avoided.

Certain lands in Group 2 can be made cultivable with major earthmoving or other effective and costly reclamation operations. In India, both the Class VII Lands and Class VIII Lands are combined as Class VII Lands.

7.3 Land Capability Classification of India

In India, the basic objective of the soil survey was to achieve the land capability classification. The All-India Soil and land Use Survey Organization attempted the land capability survey in 1960 which identified eight land use capability classes.

7.3.1 Land Suitable for Cultivation

Class I: Very good arable land with no specific difficulty in farming. It is nearly levelled, well drained, easily worked soil. These soils are very productive.

Class II: Good cultivation land which needs protection from erosion or floods, drainage improvement and conservation of irrigation water, intensive drainage, and protection from floods.

Class III: Moderately good cultivation land where special attention has to be paid to erosion control, conservation of irrigation water, intensive drainage, and protection from floods.

Class IV: Fairly goof land suited for occasional and limited cultivation. It needs intensive erosion control, intensive drainage and very intensive treatment to overcome the soil limitations.

7.3.2 Land Not Suitable for Cultivation

Class V: Very well suited for grazing and forestry but not for cultivation of crops. It has very good carrying capacity of livestock but needs protection from gullying.

Class VI: Well, suited for grazing and forestry but not for cultivation of crops. It has moderately carried capacity of livestock. This land has steep slopes, severe erosion, and stoniness, poor moisture retaining capacity, salinity and severe climate.

Class VII: Fairly well suited for grazing or forestry with little carrying capacity of livestock.

Class VIII: Suited only for wildlife. This land has severe climate, wet soil, stones, badlands, sandy beaches, marshes, deserts, and nearly barren lands.

7.4 Significance of Land Capability Classification in Land use Planning

The land capability classification determines the use ceiling for any piece of land and helps to define the conservation problems and possible treatments. Keeping this in mind, a piece of land can be put to the most efficient land use. For instance, all agricultural activity must be confined to classes I to IV and others, such as pasturelands, energy plantation, buildings, roads, etc., to classes V to VIII. Within this scheme, one can select the most suitable crop for a particular piece of land.

The land capability classes can change towards better classes, if the existing limitations can be permanently removed or reduced in extent by economically feasible reclamation measures, such as providing irrigation, providing proper drainage, constructing flood-control measures or stabilising, gullies. A further deterioration of existing conditions can, on the other hand, shift the capability to poorer classes.

Let us sum up

This study is appropriately based on the categorization of watersheds and the capability of the utilization of lands. From the above analysis, it can be understood that the industrial perspective is largely dependent on watershed and land use capabilities. The various objectives of watershed management have been included in this study for the readers to understand the various categorizations of watersheds and the capability of the utilization of multiple lands.

Glossary

Land capability: Land capability may be defined as the ability of the land surface to support natural plant growth/ wildlife habitat or artificial crop growth/ human habitat.

Class I Lands: These lands are nearly level with slopes generally within 1%.

Class II Lands: These lands generally have gentle slope in the range of 1 to 3%.

Class III Lands: These lands generally have slopes in the range of 3 to 5%

Check your progress

1. What are the limitations which reduce the crop choice?

    1. Moderate susceptibility to erosion by wind or water;
    2. Less than ideal soil depth;
    3. Somewhat unfavourable soil structure and workability;
    4. Slight to moderate salinity;
    5. Occasionally damaging overflows;
    6. Wetness existing permanently which can be corrected by drainage; and
    7. Slight climatic limitations on land use and management.

2. What are the severe limitations which restrict use to pasture or Grass land?

    1. Severe erosion;
    2. Stony texture with shallow rocks
    3. Excessive wetness or overflow
    4. Low moisture capacity
    5. Severe climate.

Suggested Readings

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Unit 8

Soil resource evaluation

Structure

Learning Objectives

Overview

8.1 Land Evaluation

8.2 Concept of Soil Evaluation

8.3 Objectives

8.4 Evaluation characteristics

8.5 Evaluation systems

8.6 Major systems for agricultural soil capability evaluation

8.6.1 Storie Index

8.6.2 Productivity index

8.6.3 Soil Fertility Capability Classification (FCC)

8.6.4 The FAO Framework for Land Evaluation (1976)

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to learn the following,

- Land Evaluation concept, objectives and Evaluation characteristics
- Evaluation systems and Major systems for agricultural soil capability evaluation
- Storie Index, Productivity index, Soil Fertility Capability Classification (FCC) and The FAO Framework for Land Evaluation.

Overview

As is well known, soil is a component of the natural medium that acquires its morphology and properties after a long and slow evolution after reaching

an equilibrium with environmental conditions. It is, then, a natural entity, which does not anticipate human use in its evolution. Nevertheless, ever since humans during the Neolithic shifted from hunting and gathering to farming and herding, the soil has undergone intensive exploitation. Some 10,000 years of irrational soil use by humans has transpired, with no objective beyond seeking maximum yield from every kind of soil use. As a result, the soil has reached the present day intensely degraded to the point that a great part of arable land, especially in arid and semiarid regions, is in a situation of irreversible deterioration. To stop this dramatic trend, the only solution is to institute rational soil use that is, to use each soil in a way that best suits its characteristics and to programme its management for minimal degradation. This is precisely the final aim of land evaluation.

8.1 Land Evaluation

Land evaluation is an applied classification system that assesses the capacity of the soil for its optimal use that is, to derive maximum benefits with minimum degradation. This can be defined, as “any method to explain or predict the use potential of land”.

Different types of soils present widely different properties, and therefore the response to each use differs. Land evolution is based on the idea that this response is a function of these properties, and, hence, knowing these, we can predict the behaviour of the soil under a given use. From the study of such properties, different degrees of suitability of the soil can be inferred for each end proposed. These degrees are reflected on maps of use capacity or suitability, on which the corresponding recommendations are made for the rational planning of soil use.

As land evaluation is intended to offer practical results that can be plotted on territorial maps, such endeavours cannot be limited to the analysis of the physical medium of the earth, but rather must be complemented by the corresponding socio-economic studies that enable cost-benefit analyses of the profitability of the land used. Thus, land evaluation enables predictions on the biophysical and economic behaviour of land for current and potential uses.

The above implies that land evaluation must be undertaken by an interdisciplinary team comprised of soil scientists, agronomists, ecologists, geographers, sociologists, and economists, among others. Such a multifarious team is difficult to join, given such diverse scientific training, scientific terminology and focus of each specialist. Consequently, as shown by many authors complete studies (biophysical and economic) of land evaluation have to date been scarce, the great majority being restricted to

examinations of the physical medium, thus leaving the term of land evaluation quite confused.

8.2 Concept of Soil Evaluation

Given the unquestionable importance of soil study in land evaluation, we propose the term “soil evaluation” for the assessment of soil properties as a phase prior to land evaluation. This involves understanding the soil properties in their broadest sense, including both the intrinsic ones (those of the soil itself depth, texture, etc.) as well as extrinsic ones (of the soil surface topography, climate, hydrology, vegetation, use, etc.)

At first, the terms “soil evaluation” and “land evaluation” were used interchangeably, but soon the term land evaluation became predominant. Soil evaluation fell into disuse. We propose the use of the term soil evaluation in its broadest sense, extending its meaning to all the characteristics that affect the soil, whether these be soil properties themselves or any related to the soil surface. Soil evaluation would be like what today is called land evaluation, but excluding all social, economic and political characteristics.

Soil evaluation analysed below from a political, social and economic perspective to develop the land evaluation that will serve as the basis for final land-use planning.

8.3 Objectives

In this sense, the separation of soil evaluation, as the basis for land evaluation, has the following objectives:

- To facilitate the study of soil evaluation by soil scientists, who are fully trained for soil evaluation but who find difficulty in assessing the political, social and economic aspects, such as labour costs, suitability, availability of machinery, size and situation of the parcels, costs, investment, market, infrastructure, distribution, capital, different subsidies, etc..
- To avoid the confusion that the current term “land evaluation” has generated for its use to evaluate only biophysical aspects of the soils.
- To provide documents based on biophysical data much more stable than the political, social and economic aspects, with broad variability depending on the social, political and commercial decisions.
- To enable easy adaptation of evaluations according to situational changes, given that soil-evaluation studies based on biophysical data are rather stable.

- To make soil evaluation a valid pursuit in and for itself. A land-evaluation plan without previously taking specific clients into account makes little sense. Land evaluation must be directed under the request of a client, whether individual, group (societies, cooperatives) or governmental (local or national), and must be backed by the necessary guidelines that permit the plan to be put into practice. However, this situation unfortunately is far from frequent. Thus, if we abide by this ideal, studies on land evaluation would be restricted to a limited number of occasions. However, we believe that soil-evaluation studies can be valid by themselves, without the need for prior request, as the soil evaluation can simply be undertaken at any moment as an important contribution to the knowledge of soil quality in a region, as an environmental resource, as important as knowledge of basic soil types, lithology, geology, hydrology, etc..

Land evaluation will be necessary both for the for a study of the regional planning of a territory as well as for the particular case of a change in the use of a specific parcel. Land evaluation must be capable of predicting the particular behaviour of a soil for new use.

However, to focus a land-evaluation plan, we must bear in mind that, when well formulated, the plan should provide major benefits—but, when poorly calculated, it inflicts heavy losses and, even worse, irreversible degradation. A shift in land use might could detonate “chemical time-bomb”.

8.4 Evaluation characteristics

Soil evaluation can be performed directly or indirectly. In the former case, the evaluation stems from field tests (experimental stations, random agricultural sampling in the field) or from yield production provided by individual farmers (farm records) and cooperatives, or else from agricultural statistics. These data are usually local, spotty, sometimes not reliable and are generally difficult to extrapolate. Therefore, the evaluation is normally conducted indirectly based on the soil properties, if yield of a given soil depends on its properties and its level of management. The evaluations made in this manner should be validated finally with real yield data.

In an indirect evaluation, it is evident that to define a degree of suitability, it does not suffice to choose only one property, but rather it requires a group of properties, possibly the more the better. The properties to choose from will depend on the proposed use of the soil. The values of these evaluation parameters can be derived from disparate sources (remote sensing, maps, literature and directly from the field or laboratory) and with sharply differing degrees of precision.

In addition, these evaluation characteristics can be combined in many ways in various assessment systems used in soil evaluation, so that even for the same use, the results for a given soil can differ markedly, depending on the evaluation system chosen. Therefore, we propose that it is useful to define in a general manner the properties that most influence soils as well as their degree of suitability. For the case of agricultural use, which is the most frequent, some of these, although other properties can be used, such as: genetic profile, parent material, clay mineralogy, microrelief, altitude, annual rainfall distribution, evapotranspiration, vegetation, etc.

8.5 Evaluation systems

A plethora of evaluation methods are used under diverse philosophies and techniques. Some methods value the degree of suitability of the properties, while others place more emphasis on the possible limiting factors for soil use. These latter methods are more consistent, since, as we know, the true limits on soil use are negative characteristics, regardless of the degree of suitability of the most favourable properties.

Some methods use qualitative criteria while others use quantitative. The qualitative systems are normally used in examinations aimed at a general evaluation of broad zones. The quantitative methods are used more often in detailed studies and thus need more information on the soils, both to construct the evaluation system as well as to apply it, but they are more objective, and the results are therefore more reliable. Other methods begin with qualitative data which are weighted to reach a final numerical result.

Table 8.1. Environmental indicators of the degree of suitability of soils for agricultural use.

	Very favourable	**Favourable**	**Unfavourable**	**Very unfavourable**
Intrinsic properties				
Effective depth, cm	>120	120-70	70-30	<30
Texture	balanced	moderate heavy	heavy	light
Course fragments, %	<10	10-30	30-60	>60
Structure	f, md, 3, 2	c, 1	sg, 0	ms, 0
Compact, cemen, gr, cm	Absent	md, >60	md, >20 or st, >60	st, < 30

Available water, mm	>100	100-60	60-20	<20
Internal drainage	Without hydro	Hydro > 80 cm	Hydro > 40 cm	Hydro a 0 cm
Permeability, cm/hour	>2	2-0.5	0.5-0.1	<0.1
Organic matter, %	>5	5-2	2-1	<1
CEC, cmol (+)kg-1	>40	40-20	20-10	<10
Saturation degree, %	>75	75-50	50-25	<25
pH	7.3-6.7	6.7-5.5 or 7.3-8.0	5.5-4.5 or 8.0-9.0	<4.5 or >9.0
Carbonates, %	<7	7-15	15-25	>25
Salinity, dSm-1	<2	2-6	6-12	>12
Extrinsic properties				
Slope, %	<4	4-10	10-25	>25
Surface stoniness, %	<2	2-20	2-20	>50
Surface rockiness, %	<2	2-20	2-20	>50
Flooding, months	0	<1	1-3	>3
Erosion, Tm/ha/year	<10	10-20	20-60	>60
Ploughing	no problems	limited	severe	very severe
Precipitation, mm	>1000	1000-600	600-300	<300
Frost, Tª<0º, months	<1	1-3	3-6	>6

Texture: balanced = loam, silt loam, sandy clay loam; heavy = sandy clay, clay loam, silty clay loam, silt; heavy = clay, silty clay; light = sand, loamy sand. Structure. f = fine; md = medium; c = coarse; sg = single grain; ms = massive; 3 = strong; 2 = moderate; 1 = weak; 0 = structureless. Compact = compaction, cemen = cementation, gr = degree, cm = depth at which it appears; md = moderate; st = strong. Internal drainage: hydro = hydromorphy. CEC = cation-exchange capacity. Ploughing: no problems = ploughing is possible at any time of the year; limited = not possible during wet periods, clayey soils; severe = only in dry periods, soils very

clay. Very severe = not possible due to steep slopes or high groundwater table; Precipitation = Annual precipitation.

Some systems group the classes into a series of levels of importance like order, class, subclass, type, etc., and are thus hierarchical systems. Other systems have one category, and these are frequently parametric. In these latter systems, mathematical formulae are applied so that the result is expressed in numerical terms.

Parametric System: These can be additive (Index = A + B + C + D + ...) or with a multiplicative scheme (Index = A x B x C x D x...), the latter offering better results for following the minimum law. The additive systems give an evaluation that is usually correct from the theoretic standpoint but can give evaluations that are not realistic since they do not represent the serious consequences implicit in a highly limiting soil factor.

It is generally accepted that the parametric methods are simple, objective, quantitative, reliable, easy to understand and apply, even by the non-specialist, and easy to modify and adapt to new uses. Their main disadvantage is precisely that their objectivity and precision are illusory. Their formulation is difficult and, if they are not well applied, their results can be completely erroneous. The scheme is too artificial and the relationship between the soil properties and the result of the evaluation is poorly defined. The results from this type of system, perhaps more than any other, need careful validation by values for soils under practical use.

Some systems pursue agricultural ends while others seek exclusively engineering uses (such as the support and placements for structures, roads, septic tanks, etc). Within the systems for agricultural uses, some, called land capability, evaluate the capacity of the soil for general use (crops, pasture, forestry), while others, called land suitability, evaluate the suitability of the soil for specific uses, such as a specific crop (e.g., wheat, potato) and with a particular kind of soil management. The classifications of land capability define the degrees of capacity in generally vague terms, focusing fundamentally on the limitations for the general use. Land suitability provides more practical results but needs more data both for the land as well as for concrete specifications for each type of crop. An important compilation of data on the optimal and marginal conditions of many crops is provided for the tropical and subtropical regions.

Frequently the evaluation systems consider the beneficial effects that can result from introducing certain improvements, these known as potential evaluations.

Evaluation systems use mapping methods to represent the results of the soil evaluation.

8.6 Major systems for agricultural soil capability evaluation

8.6.1 Storie Index

This represents the first parametric approach that was developed. It is an index that uses the multiplicative scheme. In addition, it uses intrinsic properties of the soils (genetic profile, parent material, profile depth, texture, drainage, nutrients, acidity an alkalinity), characteristics of the soil surface (slope and microrelief) and aspects of soil conservation (degree of erosion). The evaluation properties are grouped into four factors that are quantified in the corresponding tables. The factors are weighed a priori, the more important being related on a scale from 5 to 100 and the less important factors from 80 to 100.

With this index, general agricultural soil uses can be evaluated hence it is a soil-capability evaluation method. To formulate the index, the four factors are multiplied together, and the index is expressed as a percentage. Six classes are defined at the degree level, with decreasing values from 1 to 6. The degrees 1 to 3 are for agricultural use, degree 4 for very limited agricultural use, 5 for pasture and 6 without use. Subdegrees are established according to limiting factors: “s” for depth, “p” for permeability, “x” for texture, “t “slope, “d” for drainage and “a” for salts.

It is important to emphasize that this system does not consider climatic characteristics. Thus, the evaluation is of the soil itself, valid for comparing the soils of a certain region with the same type of climate.

8.6.2 Productivity index

The basic concept of this method is that agricultural-soil productivity, under optimal management conditions, depends on the intrinsic characteristics. This is a multiplicative parametric method to evaluate soil productivity, from a scheme like the Storie index. The concept of productivity is defined as the capacity to produce a certain quantity of harvest per hectare per year, expressed as a percentage of optimal productivity, which would provide a suitable soil in its first year of cultivation. The introduction of improvement practices leads to a potential productivity or potentiality. The quotient between the productivity and the potentiality is called the improvement coefficient.

The evaluation is made for three general types of use: agricultural crops, cultivation of shallow-rooted plants (pastures), and deep-rooted plants (fruit trees and forestation).

The determining factors of soil depth are wetness, drainage, effective depth, texture/structure, base saturation of the adsorbent complex, soluble-salt concentration, organic matter, cation-exchange capacity/nature of the clay and mineral reserves. The parameters of the soil surface (e.g., slope, erosion, flood tendency, or climate) are not considered.

8.6.3 Soil Fertility Capability Classification (FCC)

In this system, three levels or categories were established. The first, the type, was determined by the texture of the arable layer, or of the first 20 cm, if this is thinner. Its denomination and range are S, sandy (sandy and sandy loam); L, loams <35% clay (excluding sandy and sandy loam); C, clayey > 35% clay; O, organic > 30% organic matter to 50 cm or more.

The type of substrate is the second level and is used when there is a significant textural change in the first 50 cm of the soil. It is expressed with the same letters, adding “R” when a rock or a hard layer is found within this depth.

The third level is comprised of the modifiers, which are the chemical and physical parameters that negatively influence soil fertility. These are numerous and are represented by lower-case letters.

In the denomination of the soil class, the principle limitations for use are directly represented. For example, for an Orthic Solonchak, the FCC class that represents it is LCds, which signifies that it is a soil susceptible to severe erosion (L), limited drainage (C), dry soil moisture regime (d) and with salinity (s).

8.6.4 The FAO Framework for Land Evaluation (1976)

This framework is an approach, not a method. It is designed primarily to provide tools for the formulation of each concrete evaluation. The system is based on the following concepts:

- The land is qualified, not only the soil.
- Land suitability must be defined for a specific soil use (crop and management).
- Land evaluation was to consider both the physical conditions as well as economic ones.
- The concept of land evaluation is essentially economic, social and

political.

- The evaluation requires a comparison between two or more alternative kinds of use.
- The evaluation must propose a use that is sustainable.
- A multidisciplinary approach is required.

In the scheme, four categories are recognized. The highest category is the order that it reflects, in broad features, whether a soil is suitable or not for a given use. Two orders are recognized:

S = Suitable. Land in which the benefits exceed the costs and sustained use does not incapacitate the soil over a sufficiently long period of time.

N = Not suitable. Land can be classified as not suitable for a certain use for diverse reasons. The use proposed may be deemed technically impractical, as in irrigation of abrupt rocky terrain, or that it causes serious environmental degradation, as in cultivation on steep slopes. Frequently, however, the reason is economic, in that the profit expected does not justify the cost required.

The second category is the class that reflects degrees of suitability within the order. These are numbered consecutively in Arabic numerals.

For the order S, three classes are considered:

S1 = Highly suitable. Without limitations for sustained use or minor limitations that do not affect productivity nor appreciably increase costs.

S2 = Moderately suitable. Moderately serious limitations that reduce profits or involve risks of degradation in the sustained use of the soil.

S3 = Marginally suitable. The limitations for the sustained use are serious and the balance between the costs and benefits make the use only marginally justifiable. Its use is normally justified on other than economic grounds.

In the order N, three classes are also recognized:

N1 = Not currently suitable. Land with limitations that could be eliminated by technical means or investment, but that these changes are at present unfeasible.

N2 = Permanently unsuitable. Serious limitations of generally a physical nature, which are assumed to be beyond solving over the long term.

X = Land for conservation. Unsuitable for exploitation, being lands of special protection, due to their conservation, wildlife, of special scientific, ecological, or social interest (e.g., parks, reserves or recreational zones).

The limits between the orders (S and N) and between the different classes (S1, S2, S3 and N1, N2) are established by the presence of limiting factors. One limiting factor is a characteristic of the soil that hampers its use, reduces productivity, increases costs and implies degradation risk, or all of the above.

These limiting factors are used to define the third category of the system, which is the subclass. In the symbol of each subclass, the number of limitations involved should be kept to the minimum one letter, or, rarely, two. The limitations proposed include t, slope; e, erosion risk; p, depth; s, salinity; d, drainage; c, bioclimatic deficiency; r, rockiness; i, flood risk.

Finally, the fourth category is the unit that establishes the differences within the subclasses as a function of the desired use. All of the units within a subclass (S2rA, S2rM, ...) have the same degree of suitability at the subclass level (S2) and analogous characteristics of limitation at the subclass level (r). The units differ from each other in their characteristics of production or in secondary aspects of their management demands. Their examination enables a detailed interpretation at the planning level of the exploitation. The units are distinguished by upper-case letters that are placed at the end. There is no limit at all for the number of units examined within a subclass. These defined are

A, intensification in the agricultural use without need of great improvements.

M intensification in the agricultural use with need of major improvements (irrigation, etc.);

P, use for pasture for livestock;

F, forestation.

Let us sum up

The foregoing results indicate heterogeneity in the soil assessment by the different soil-evaluation systems. It would be necessary for the evaluation made for a soil and specific end (for example, agriculture) to give consistently the same capability class, regardless of the system used. Thus, it appears to be advisable for the evaluator of the soils, before developing a project of soil evaluation, to make a comparative study of three or more evaluation systems to choose the system that best matches the objectives

set, and, above all, to calibrate the system, modifying the characteristics used for the evaluation and readjusting the values.

Glossary

Land capability is the ability of land to support a given land use without causing damage.

Productivity index: The basic concept of this method is that agricultural-soil productivity, under optimal management conditions, depends on intrinsic characteristics.

Check your progress

1. What is the Parametric System of soil evaluation?

The main disadvantage is precisely that their objectivity and precision are illusory. Their formulation is difficult and, if they are not well applied, their results can be completely erroneous. The scheme is too artificial and the relationship between the soil properties and the result of the evaluation is poorly defined. The results from this type of system, perhaps more than any other, need careful validation by values for soils under practical use.

2. What are the types of Storie Index of agricultural soil capability evaluation?

The evaluation is made for three general types of use: agricultural crops, cultivation of shallow-rooted plants (pastures), and deep-rooted plants (fruit trees and forestation)

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.
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Unit 9

Slope analysis and vegetation cover

Structure

Learning Objectives

Overview

9.1 Watershed slope

9.1.1 Parent Material

9.1.2 Soil Erosion

9.2 Land Cover/ Vegetation

9.2.1 Impact of vegetation in a watershed

9.2.2 Watershed Benefits of Forest Cover

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to understand the

- Watershed slope: Impact of Parent Material and Soil Erosion
- Land Cover/ Vegetation: Impact of vegetation and Benefits of Forest Cover

Overview

The longer the slope, the steeper the grade, and the smoother the surface, the larger is the potential for erosion. Along with quantity of rainfall, slope characteristics determine the speed of flow. The faster the water flows, the greater the potential for erosion and sedimentation.

The vegetation in watersheds effects the quality of the water. Grasses, shrubs, and trees make up most of the plant cover. As they fall and decompose, they add the organic components to the soil. Trees are the protectors of the watershed. The tree litter protects the soil's surface as the roots of the tree protect the soil from erosion. Trees also provide a canopy

that keeps the water cool, so organisms don't die of thermal pollution. The canopy can also reduce the force of the rain and the velocity of wind in a watershed. Plant cover is important to a watershed to prevent the erosion of valuable soil as water rushes downstream. Plant cover also provides food and protection for many small organisms. In this unit, you will learn about the impact of slope and vegetation cover on watershed.

9.1 Watershed slope

Watershed slope reflects the rate of change of elevation with respect to distance along the principal flow path. It is usually calculated as the elevation difference between the highest and lowest elevation of the point of the watershed divided by watershed length. Watershed slope affects time of concentration, as well as time to peak.

Steepness of a watershed can produce negative effects such as **stream depletion and soil erosion to aquatic ecosystems and agricultural activities**. In this regard, assessing the effect of a watershed slope on hydrologic responses is necessary to improve watershed management including water and soil resources.

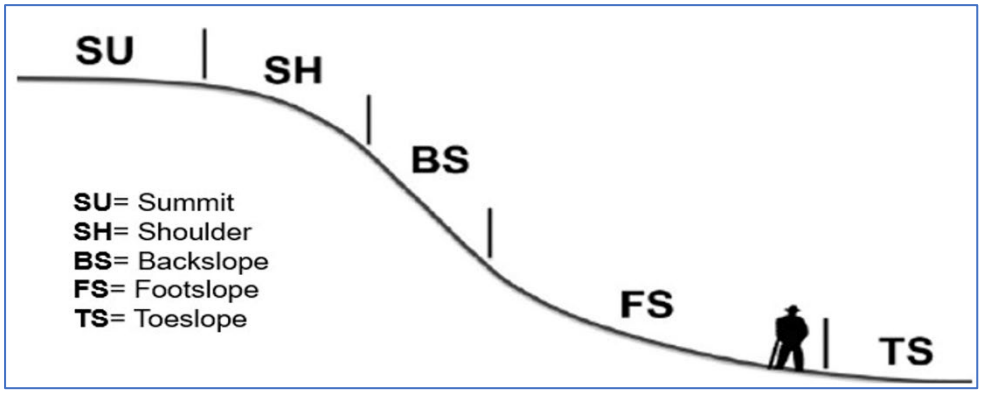


Fig. 9.1 Hillslope positions across a landscape Wysocki et al., 2000 and Schoeneberger et al., 2012

Table 9.1 Hillslope positions across a landscape

Summit	The top or highest level of an upland feature such as a hill or mountain. Usually, the summit is relatively level.
Shoulder	Forms the uppermost inclined surface at the top of a hill slope. It is the transition zone from the summit to the backslope of an upland. The surface is dominantly convex in profile and erosional in origin.

Back slope	The steepest inclined surface of the hill slope. The backslopes in profile are commonly steep and linear, or steep and convex concave slope. Backslopes are erosional forms produced mainly by mass wasting (direct gravitational action) and running water action.
Foot slope	The inner, gently inclined surface at the base of a hill slope. The surface is dominantly concave in profile. It is a transition zone between the backslope and toe slope where colluvium and alluvium accumulate.
Toe slope	The outermost, gently inclined surface at the base of a hill slope. Toe slopes are constructional surfaces forming the outermost point of a hill slope where alluvium tends to accumulate.
Stream terrace	A natural level strip of land in a stream valley, parallel to the stream channel, originally formed near the level of the stream, and representing the dissected remnants of an abandoned flood plain or stream bed.
Flood plain	The nearly level fluvial (river) plain bordering a stream and subjected to periodic inundation and sediment accumulation under flood stage conditions. It is a constructional landform built of sediment deposited during overflow and lateral migration of the stream.

9.1.1 Parent Material

When rocks are exposed to atmospheric conditions, they begin to adjust to their new environment. This adjustment (known as weathering) involves processes which cause physical disintegration and chemical decomposition of the rocks. The weathering of bedrock produces unconsolidated debris serving as the parent material for soils. The parent materials undergo continued alteration and evolve into a soil reflecting the integrated effects of climate, biotic factors (plants, animals, and microorganisms), topography, and time.

Some common parent materials of the Laramie area are alluvium, colluvium, and residuum.

Alluvium	Unconsolidated material transported and deposited by flowing water.
Colluvium	Unconsolidated material deposited on and at the base of steep slopes by direct gravitational action.
Residuum	Unconsolidated weathered mineral matter accumulating by disintegration and decomposition of bedrock in place.

The steepness of slope is determined in percent (%) calculated as [(rise/run) x 100] using an Abney level or clinometer. The Abney level or clinometer is held at eye level and parallel to the slope. The level is adjusted allowing the % slope to be read directly. The absolute percent slope is the same whether sighting up slope or down slope.

9.1.2 Soil Erosion

The amount of soil erosion is directly related to the amount of surface water runoff, which depends on the water infiltration rate and the % slope. The steeper the slope and the less rapid the water infiltration rate, the more rapid the water runoff rate for a given soil. Soils having granular structure and high porosity have slower water runoff rates than do soils with massive structure and low porosity. This occurs because more water infiltrates into granular soils with less total water being available for over the surface water runoff when compared to massive soils. Other factors including vegetative cover, the type of vegetation, and the soil moisture content influence the surface water runoff rate. The greater the extent of plant vegetation covering the soil, the less soil erosion because the plant tissue intercepts the falling rain drops and greatly dissipates the energy of the falling water mass from directly hitting the soil. If a soil is moister, it requires less rainfall to reach the point where the soil becomes saturated (all pores filled) and water runoff occurs.

Generally, as surface water runoff increases, soil erosion increases. The following table can be used to estimate the surface runoff rate using soil texture and the percentage of slope.

Table 9.2. Slopes, texture of surface horizons, and surface water runoff rates.

Slope (%)	Textural class	Water runoff rate
0-1	All textural classes	Very slow
1-2	Sands and loamy sands	Very slow
1-2	All textures except sands and loamy sands	Slow
2-6	Sands and loamy sands	Slow
2-6	All textures except sands and loamy sands	Medium
6-12	Sands and loamy sands	Medium
6-12	Sandy loams, sandy clay loams, loams, clay loams, silt loams, silty clay loams	Rapid
6-12	Silty clay, clay, sandy clay	Very rapid
12-18	Sands and loamy sands	Rapid
12-18	All textures except sands and loamy sands	Very rapid
>18	All textures	Very rapid

9.2 Land Cover/ Vegetation

The spatial distribution of vegetation directly affects the continuity of the soil erosion process, leading to the dispersion of water flow and the gradual decrease in its sediment carrying capacity. Runoff and sediment accumulation are also affected by the position of vegetation.

Plants bind the soil and help in the percolation of water in the soil and thus add in water table. The water is absorbed back by roots and maximum portion of which is lost into the atmosphere in the form of vapours by transpiration. These vapours condense and form clouds.

Depending upon the type of vegetation and its extent, this factor regulates the functioning of watersheds, for e.g., Infiltration, water retention, runoff

production, erosion, sedimentation etc. Vegetation plays vital roles in the water cycle. It intercepts rainfall, impedes overland flow, and promotes infiltration. Vegetation also uses water for growth. All these factors reduce the quantity of runoff to streams. Vegetation binds and stabilizes soil, thereby reducing the potential for erosion. Vegetation also stabilizes stream banks and provides habitat for aquatic and terrestrial fauna. Vegetation functions to slow runoff and reduce soil compaction, allowing better percolation of rainfall into soils (infiltration) and groundwater recharge, which creates better water storage for summer base flows. In addition, the patterns, sizes, and composition of the vegetation affect reduction of soil erosion. Leaves and branches intercept the falling rain and reduce the effect of raindrop splash. Vegetative litter from dead leaves and branches builds up an organic surface that provides protection of the soil layer. Root systems also help to keep soil material stable from moving down slope.

9.2.1 Impact of vegetation in a watershed

The more vegetation there is in a watershed to intercept and transpire water, the less runoff there will be. The amount and type of vegetation in a watershed influences the rates of runoff and erosion in that watershed. Plants reduce the impact of raindrops on soil and slow the speed of water flowing across the surface of the land, allowing more water to infiltrate into the soil. A forested or native grass watershed typically delivers its runoff slowly, so soil and channel erosion are usually not severe. Runoff rates and amounts are usually excessive in watersheds with too much timber clearing, row cropping, grazing, paving, urbanization and other types of development. Runoff increases in these situations because less water is allowed to infiltrate to groundwater or return to the atmosphere through transpiration. Unnaturally high runoff rates produce more frequent, larger, and faster-rising floods. Likewise, more runoffs can increase erosion of the land surface. More flooding can increase streambed and streambank erosion as channels adjust to the new channel forming flows.

Many conditions and activities in a watershed can influence a stream. Uncontrolled erosion and runoff from construction sites;

- row cropping of highly erodible land;
- construction of houses, buildings and parking lots that prevent infiltration and speed runoff of precipitation;
- removal of riparian vegetation;
- improper disposal or excessive application of industrial, agricultural or household chemicals;

- point and nonpoint sources of nutrients and runoff from strip mines are some ways receiving streams can be degraded.

These and other factors contribute to poor water quality, unstable stream channels and unhealthy biological communities. Setting up erosion control practices on construction areas and farmland, limiting the amount of impervious area associated with urban developments and maintaining or establishing areas of timber and grasses are important efforts toward reducing the negative effects on streams caused by our uses of the land.

9.2.2 Watershed Benefits of Forest Cover

Benefit	Description
Reduce stormwater runoff and flooding	Trees intercept rainfall in their canopy, reducing the amount of rain that reaches the ground. A portion of this captured rainwater evaporates from tree surfaces. Trees take up water from the soil through their roots, which increases soil water storage potential and lengthens the amount of time before rainfall becomes runoff. Trees promote infiltration by slowing down runoff and by increasing soil drainage in the root zone. The addition of organic matter (e.g. leaves) also increases storage of water in the soil, further reducing runoff. Forested land produces very little runoff, which can reduce downstream flood flows that erode stream channels, damage property and destroy habitat.
Improve regional air quality	Trees absorb pollutants such as nitrogen dioxide, carbon monoxide, ozone, and particulate matter from the atmosphere. Trees reduce air temperature, which reduces formation of pollutants that are temperature dependent, such as ozone Trees indirectly improve air quality by cooling the air, storing carbon, and reducing energy use, which reduces power plant emissions

Reduce stream channel erosion	Trees growing along a stream bank prevent erosion by stabilizing the soil with root systems and the addition of organic matter
Improve soil and water quality	Trees prevent erosion of sediment by stabilizing the soil, and by substantially dispersing raindrop energy Trees take up stormwater pollutants such as nitrogen from soil and groundwater Forested areas can filter sediment and associated pollutants from runoff Certain tree species break down pollutants commonly found in urban soils, groundwater, and runoff, such as metals, pesticides and solvents
Provide habitat for terrestrial and aquatic wildlife	Forests (and even single trees) provide habitat for wildlife in the form of food supply, interior breeding areas, and migratory corridors Streamside forests provide habitat in the form of leaf litter and large woody debris, for fish and other aquatic species Forest litter such as branches, leaves, fruits, and flowers, form the basis of the food web for stream organisms
Reduce summer air and water temperatures	Riparian forests shade the stream and regulate summer air and water temperatures, which is critical for many aquatic species Trees and forests shade impervious surfaces, reducing temperature of stormwater runoff, which can minimize the thermal shocks normally transmitted to receiving waters during storms.

The important link between forests and watershed health has been documented by several researchers. Booth (2000) found that at least 65% watershed forest cover is needed for the presence of a healthy aquatic insect community in a Puget Sound, Washington study. Other researchers have evaluated the relative impact of streamside forest cover on various indicators of stream health. In a Montgomery County, Maryland study,

Goetz and others (2003) found that for streams to have a health rating of Excellent, at least 65% of the length of the stream network in the watershed must be forested (within 100 feet of the stream). At least 45% streamside forest cover was required for streams to have a health rating of Good (Goetz and others, 2003). Other researchers such as Kathy Wolf from the University of Washington are documenting the human dimension of urban forestry with compelling research on the psychological and sociological benefits of trees.

Let us sum up

Watershed slope reflects the rate of change of elevation with respect to distance along the principal flow path. It is usually calculated as the elevation difference between the highest and lowest elevation of the point of the watershed divided by watershed length.

Vegetation cover plays a very important role on protecting the soil surface from raindrop splashing, increasing soil organic matter, soil aggregate stability, water holding capacity, hydraulic conductivity, retarding and reducing surface water runoff, etc.

Glossary

Back slope: The steepest inclined surface of the hill slope. The backslopes in profile are commonly steep and linear, or steep and convex concave slope. Backslopes are erosional forms produced mainly by mass wasting (direct gravitational action) and running water action.

Foot slope: The inner, gently inclined surface at the base of a hill slope. The surface is dominantly concave in profile. It is a transition zone between the backslope and toe slope where colluvium and alluvium accumulate.

Toe slope: The outermost, gently inclined surface at the base of a hill slope. Toe slopes are constructional surfaces forming the outermost point of a hill slope where alluvium tends to accumulate.

Check your progress

1. What is the relationship between slope and runoff in a watershed?

   The amount of soil erosion is directly related to the amount of surface water runoff, which depends on the water infiltration rate and the % slope. The steeper the slope and the less rapid the water infiltration rate, the more rapid the water runoff rate for a given soil.

2. What are the factors affecting vegetation cover?

   The factors affecting natural vegetation are the slope of the land,

thickness of the soil, moisture, and temperature. Of these, temperature and moisture are the most important factors. Temperature is determined by the sunlight and moisture is determined by rainfall.

Suggested Readings

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2. Murty, J.V.S., 1994. Watershed Management in India, Wiley Eastern Ltd, New Delhi.
3. Rajesh Rajora, 2002. Integrated Watershed Management, R. Rawat Publications, New Delhi.

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1. https://www.fao.org/3/t0165e/t0165e00.htm#cont
2. https://cwp.org/urban-watershed-forestry/#
3. https://www.epa.gov/watershedacademy/online-training-watershed-management

Block 4

Integrated Watershed Management

Unit 10: Objectives of IWM

Unit 11: Land and Water conservation practices

Unit 12: Importance of land use planning in watershed development

Unit 13: Water harvesting and afforestation, population and livestock development.

Unit 10

Objectives of Integrated Watershed Management (IWM)

Structure

Learning Objectives

Overview

10.1 Watershed Development Programmes- Paradigm Shift

10.2 WDC-PMKSY 2.0

10.3 Objectives of Watershed Development Projects

10.4 Expected End Results

10.4.1 Economic Front

10.4.2 Ecological Perspective

10.4.3 Equity Aspect

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10.5 Other Natural Resources Development Programmes

10.6 Convergence with Horticulture in WDC-PMKSY 2.0

10.6.1 Horticulture crops cultivated in watershed project areas

10.7 Spatial Technology Inputs in Watershed Programmes

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After learning this unit, you will be able know the following,

- Watershed Development Programmes- Paradigm Shift
- WDC-PMKSY 2.0, Objectives of Watershed Development Projects
- Other Natural Resources Development Programmes
- Convergence with Horticulture in WDC-PMKSY 2.0

- Spatial Technology Inputs in Watershed Programmes

Overview

Land is a critically important national resource. Its efficient management is vital for economic growth and development of rural areas. The per capita availability of land and its yield is very low in India.

This is one of the main factors behind low productivity and high cost of Indian agriculture. Ministry of Rural Development has hence taken up various initiatives to make the optimum utilization of land and improve its quality. Concerted efforts are being made through Area Development Programmes of Department of Land Resources to regenerate and rejuvenate wastelands and degraded lands.

The Department of Land Resources of MoRD is implementing three long-term Watershed Development Programmes i.e., Integrated Wasteland Development Programme (IWDP), Drought Prone Areas Programme (DPAP) and Desert Development Programme (DDP) on watershed basis.

Each project under the programme is a micro level effort to achieve this objective through treating the underproductive and hitherto unproductive land and taking up allied activities for the benefit of landless. The programmes adopt a common strategy of multi-resource management involving all stakeholders within the watershed, who together as a group, co-operatively identify the resource issues and concerns of the watershed as well as develop and implement a watershed plan with solutions that are environmentally, socially and economically sustainable. Watershed development programme is considered and adopted as an effective tool to address problems of rain-fed areas in the country.

The 11th Plan has stressed upon developing concerted action plans for rainfed areas in close consultation with the State Governments. Accordingly, the Common Guidelines for Watershed Development, 2008 have been issued and made effective from 1.4.2008. The three watershed programmes of the Department of Land Resources namely DPAP, DDP and IWDP have been consolidated as a comprehensive programme named ‘Integrated Watershed Management Programme (WMP). In 2015, the IWMP along with On-Farm Water Management (OFWM) scheme and Accelerated Irrigation Benefit Programme (AIBP) was subsumed into Pradhan Mantri Krishi Sinchayee Yojana (PMKSY).

10.1 Watershed Development Programmes- Paradigm Shift

The rainfed regions of the country need an accelerated pace of growth, comprehensive time-tested interventions, benefitting people and sustaining ecology. This paradigm goes well with the recommendation of the Inter-Ministerial Committee on Doubling Farmers Income (DFI) that invites attention to the specific needs of three principal stake holders:

a. the consumers need nutrition security

b. the farmers need income security and

c. the production environment needs ecological security.

The past experiences of watershed approach for development of land prove that watershed is an appropriate platform for converging financial resources from various programs to supplement the project funds. This will accelerate the rate of development of land within a short period of about three to five years.

However, the implementation experiences also necessitate redesigning the treatment approach, to transit from current over-emphasis on engineering-centric soil & water conservation at the cost of agronomic and other biological treatments. Secondly, it needs to recognize the importance of creating farm and non-farm- based livelihood options and enrol every member in the project area as a stake holder. Thirdly, peoples` responsibility for resolving demands side management, as they learn to own the Project in the interest of long-term sustainability.

10.2 New Generation Watershed Development Projects - WDC-PMKSY2.0

The new watershed programme (WDC-PMKSY 2.0) is envisaged to affect the following shifts:

- A clear transition from the current predominant practice of mechanical/engineering treatments to more agriculture engineering measures. This implies overwhelming focus on trees, cropping systems, soil moisture conservation & management and soil organic matter.
- Emphasis on realizing effective use of rainwater by relying more on water productivity. This involves integrated measures to enhance water percolation for storage of rainfall in the soil profile for longer periods; and supplementing of moisture deficits in crops with water that is harvested by constructing small water harvesting structures like field bunds, trench cum bunds, contour trenches, continuous contour trenches, farm ponds,

diversion weirs, embankments, percolation tanks, check dams etc.

- Diligent planning for crop systems diversification for risk management; enhancing productivity by adopting water use efficient crops (like nutria-cereals, pulses, and oil seeds, besides dryland horticulture); and opting for crop alignment as a principle. An important initiative would include aligning crop growth phase with water availability, to provide protective irrigation at critical stages of crop growth.
- Clear risk management plans for adaptation and mitigation of adverse impacts of climate variability and change; diversification of the watershed economy by adopting integrated farming systems – agronomic and horticulture crops, livestock, agro-forestry, fishery, poultry etc.; enlarging livelihood portfolios; building climate change projections into water harvesting designs; precision based use of water for managing drought spells; and timely agro-met farm advisories to deal with climate induced uncertainties. Access to technology 5 suite that forecasts events like pest & disease, price & demand etc. would further help in risk negotiation.
- Economically vibrant institutions, like Farmers Producers Organization (FPO), to promote agri-business services and impart efficiency to transactions at both input and output management stages. Such organizations are to be people – owned, people – managed and people-centric approach.
- Setting up and nurturing of community groups that will take interventions beyond mere creation of assets and promote responsible ownership and management. Formation of User Groups based on common identities and interests around natural resources and work consciously towards integrating the principle of sharing.
- Focus on decentralization, flexibility, community empowerment and greater role for village-level institutions in the planning process with a view to accommodating the local social and traditional strengths.
- Rejuvenation of springs by taking up appropriate watershed activities in the watershed development projects.

10.3 Objectives of Watershed Development Projects

Objectives of watershed development projects are to improve productive potential of rainfed / degraded land through integrated watershed management; to strengthen community based local institutions for promotion of livelihoods & watershed sustainability, and to improve the

efficiency of watershed projects through cross learning and incentive mechanism.

At macro - level, the vision of WDC-PMKSY2.0 projects is to accelerate the economic growth rate of agriculture in the less endowed rainfed areas of the country. Moreover, this should be achieved by adopting harmony with ecological principles of development for ensuring sustained transformation of economy and ecology. The guiding principles shall be a better Economy, Ecology and Equity in the rainfed regions of the country.

At watershed level, the development plan shall be guided by the need to achieve higher incomes for farmers, expanded livelihood options for landless, equity in distribution of benefits, community ownership and management, and ecologically sustainable action plan.

Enhancing economic growth of village community dependent on watershed by:

a. Securing production and farmers' income against climate variability and its risks of drought spells through diversification of crop systems & animal husbandry, and varied livelihood portfolios; efficient water harvesting and retentions of rainwater in soil profile; and entitling all project members to ground and surface water resources for life saving irrigation on equitable basis.

b. Improving intensity and productivity of various crops, livestock, fisheries and biomass production systems through optimal, integrated, sustainable and efficient use of natural resources in project areas.

c. Recognising the stake of non-land holding project members and promoting alternate livelihood opportunities. d) Building an ecosystem of enterprises for facilitating efficient scales of operations, access to credit, and market linkages; knowledge sharing; and resource convergence led by vibrant member managed farmers' institutions.

Ecological restoration and sustainable management of natural resources across the project area through:

a) Sustained community action in management of natural resources/assets such as groundwater, soil, community resources, etc. by way of building community organisations like User Groups (UG) and transferring maintenance responsibilities to them. Further, supporting UGs with regulatory norms, that are institutionalized through the Gram Panchayats.

b) Promotion of simple, easy to use and affordable technologies and

practices, that builds upon local knowledge and available materials.

Improving the economic and social conditions of the resource poor, asset-less, differently abled and women in particular through:

a) Shared and equitable access to the land, water and biomass resources developed.

b) Greater access to income generating opportunities.

c) Facilitating co-option of members of these categories in various community institutions i.e. FPO, User Groups etc.

10.4 Expected End Results

Each watershed development project shall deliver following results by the end of the project period:

10.4.1 Economic Front

a. Increase in the total agricultural production of the project area substantively, because of higher productivity, enhanced cropping intensity and scientific use of land, water and other natural resources.

b. This would result in increased livelihood opportunities for landless and marginal farmers. These positive changes in production and job portfolio would reflect higher average annual incomes for all families in project areas.

c. Farmers and other non–farm producers (practitioners of livestock, fishery, cottage industries etc.) using project resources would effectively be organized into common interest groups, to facilitate sustainable management of resources, that would enable them to manage the resources sustainably.

d. There would be healthy FPOs with an adequate number of shareholders, a good capital base and professional management. The FPO would promote well established subsidiary units like demand base services, Custom Hiring Centre (CHC)etc, which would serve the local community. It would succeed in integrating local agricultural produce into alternate market channels.

e. There would be active Self-Help Groups (SHGs) playing an important role in micro-credit management.

f. Secondary agriculture, promotinguse of by-products, value addition (e.g., husk, straw, stalk etc.) and supplementary/complementary activities like beekeeping, mushroom etc. would be in place, which will create additional jobs and incomes for stakeholders.

10.4.2 Ecological Perspective

d. Ridge to valley approach in project area would result in soil and water conservation. This in turn would enhance soil health by increasing Soil Organic Carbon (SOC).

e. There would be an increase in green cover on account of rich green mulch (crops and trees), and higher moisture in the soil. The forests on the upper reaches would show improved density, canopy cover and pastures capable of supporting the local livestock.

f. Water-budget based crop-planning would be done every season, and all the farmers would adhere to their agreed plans. Crop alignment and diversification in favour of less water consuming crops, in tune with the available water would be the norm.

g. There would be satisfactory systems of monitoring the health of natural resources such as soil, ground water, biomass cover, and their rejuvenation. The status of natural resources would be audited at intervals, and the Gram Panchayats (GPs) and UGs would be enforcing the norms relating to use regulation and sharing of usufructs rights.

h. All members of GPs and UGs would be trained to maintain and monitor all the natural resources and assets created during the project implementation period.

10.4.3 Equity Aspect

a. The landless, the poor and the women would find a place in watershed units like watershed committee and would have stake in management of the project activities. They would be active members of FPO, SHG, Village Level Institutions (VLIs) and various UGs, and would have access to usufruct rights.

b. The investments made in various project works and activities would consider the poorer sections, the members of SC/ST community and the women.

c. The norms for regulating the benefit access to community assets would be fairly ensured to address the concerns of marginal farmers and the landless people.

d. The landless and asset-less poor would benefit from activities that promote alternate livelihood options.

10.4.4 Success Criteria

Notwithstanding inter-project variations in terms of natural resources, human and livelihood status, it is necessary to outline some common, and measurable success criteria for performance appraisal and impact assessment. Given the high regional diversity in rainfed areas, defining generic indicators across the program can also be counterproductive. It is, therefore, imperative that the decentralized participatory watershed development project plans must define the local success criteria, that are measurable and are in consonance with the end results.

At the macro-level, the framework of success criteria relating to the new generation watershed projects may be described by the following:

i. Increase in average productivity of crops, including horticulture plantations, livestock and other agricultural enterprises.

j. A rise in cumulative output of all agricultural produce from the project area.

k. Diversified production system, that helps to minimize risks and uncertainties at both production and marketing stages.

l. Increase in the average income of the farmers.

m. Popularity of sustained production technologies & farm management practices among the farmers.

10.5 Other Natural Resources Development Programmes

Various available schemes along with eligible activities and resources may be taken into consideration. Some of the important schemes that can be together with Watershed development Programmes.

a) Pradhan Mantri Krishi Sinchayee Yojana (PMKSY): District Irrigation Plans (DIPs) prepared under PMKSY provide a master plan for water sector development in the district. PWDPs prepared as per these Guidelines need to be harmonised with the DIPs. The convergence to be promoted here is incorporating ‘Per Drop More Crop’ component of PMKSY, with a view to enhance water use efficiency of the water sources created in the project area. This is done by integrating micro-irrigation systems to the water bodies and also promoting low water duty crops and varieties.

b) Mahatma Gandhi National Rural Employment Guarantee Scheme

(MGNREGS): Mechanisms of integration of watershed development projects with MGNREGS activities have been developed in various States. MGNREGS that promotes labour intensive works, from a variety of activities is highly suited to watershed development. There is scope to take up water harvesting structures, land development, soil & water conservation treatments and plantation, all of which are required in a watershed project. The convergence between PWDP and MGNREGS can happen to mutual advantage. All activities that are proposed to be taken up through MGNREGS may be indicated, and approved by GP.

c) National Food Security Mission (NFSM): This program has, of late, brought sharper focus on pulses through NFSM (Pulses), oilseeds and nutri-cereals. These crops are important for diversifying rainfed crop systems, besides promoting resilience to vagaries of weather. Strategic convergence between watershed projects and NFSM will help in promoting these crops in the project area. Improved status of soil, water, quality inputs and good agricultural practices can contribute substantially increase in yield of these crops grown in rainfed areas.

d) Integrated Farming System (IFS): Rainfed agriculture can generate higher income and better resilience when integrated with trees (amenable to agroforestry), horticulture, livestock, fisheries etc. The IFS approach in a project area can benefit from several ongoing government schemes. These encompass Mission Integrated Development of Horticulture (MIDH), National Mission for Sustainable Agriculture (NMSA), National Mission on Edible Oils – Oil Palm (NMEO-OP), Sub Mission on Agro-forestry (SMAF), National Bamboo Mission (NBM), National Afforestation Programme (NAP), Compensatory Afforestation Fund Management and Planning Authority (CAMPA), Green India Mission (GIM), Rashtriya Krishi Vikas Yojana (RKVY), National Rural Livelihood Mission (NRLM), National Livestock Mission, National Gokul Mission and schemes related to pisciculture, apiculture, sericulture etc.

10.6 Convergence with Horticulture in WDC-PMKSY 2.0

Horticulture crops are integral part of rainfed agriculture system which include wide range of water use efficient crops of fruits, vegetables, ornamental, medicinal and aromatic, tuber, spices and plantation crops, etc. growing in diversified agro climatic situations. Convergence with horticulture crops in new generation watershed development projects (WDC-PMKSY 2.0) will be a thrust area because the prime objective is doubling farmers income in the rural areas. Some of the advantages of horticultural crops

include enrichment of diets, premium on economic returns and more suitable for small & marginal farmers. The horticulture has been further widened with the introduction of mushroom, bamboo, beekeeping and other related activities.

Horticulture may be treated as an integral component of watershed development along with other components such as agro-forestry, sericulture, pisciculture and livestock. Though the main objective of watershed development is resource conservation, resource generation and resource utilization, the livelihood improvement through enhanced farm income is vital for sustainability of any watershed project. In this regard, horticultural component in watershed development plays a pivotal role in achieving the overall objective of the watershed project.

10.6.1 Some important horticulture crops cultivated in watershed project areas:

a) Fruits: Mango, Jack fruit, tamarind, sapota, cashew, Jamun, Guava, Ber, Anona, Dragon fruit, Dates (Kujoor), Pomegranate, cashew, orange, lemon Amla, Carmbola, Phalsa, Bael, Kokum, Fig, papaya, Roseappleetc,.

b) Vegetables: Tomato, Brinjal, Chillies, Drumstick (Moringa), Curryleaf and Cucurbits, Cow pea,

c) Flower crops: Roses, Marigold, Jasmine, Crossandra, Barlaria, Champaka, Chrysanthimum, Nerium.

d) Medicinal and aromatic Plants: Amla, Aloevera, Ashwagandha, Bael, Neem, Annota, Bursera, Lemon grass, Citronella, Khus grass, Palma rosa, Henna, Tylophora, Periwinkle, Sappan wood, Karonda, Terminaliaarjuna.

Depending on local suitability other horticulture crops may also be taken up based on their appropriateness and farmers interest.

Sustainable Techniques for enhancement of crop yields in watershed project areas. The techniques used for receiving sustainable yield under watershed areas consists of growing of drought tolerant varieties, in-situ grafting, root stock selection, planting techniques, application of organic matter, mulching and other soil and water conservation practices.

Border planting, alternate row irrigation for closely spaced vegetables, micro irrigation techniques, nursery production, kitchen garden / nutritional and medicinal garden, value addition and processing, vermi compost & compost making by using horticulture waste etc. may also play important role.

10.7 Spatial Technology Inputs in Watershed Programmes

Technology enables, inter-alia, to strengthen program management and coordination, undertake activity-based project planning, formulate action plans, streamline sanctions and release of funds, create useful data bases, assess actual impacts of projects, make effective prioritizations, prepare scientific DPRs, document best practices and case studies, as also facilitate free and seamless flow of information and data.

The new vision of watershed programs is best achieved by integrating new technologies with indigenous technologies, besides harnessing the knowledge &experiences of local communities. Availability of digital maps at Bhuvan platforms, open-source tools integrating mobile apps, Geographic Information System & Remote Sensing (GIS&RS) and web platforms enable smooth integration of field data with maps/spatial data.

The National Rainfed Area Authority (NRAA) as a Knowledge Partner of Department of Land Resources (DoLR) would facilitate the States/UTs with available new technologies, innovations and tools for participatory planning, implementation, and assessment.

There are new technological advances and tools available in climate monitoring and weather-based advisories developed by various agencies like India Meteorological Department (IMD), Indian Space Research Organization (ISRO), Network for Information on Climate (Ex) Change (NICE) platform, Climate Information Centres and others including, various Ministries / Departments. NRAA would facilitate to avail these technologies in the project planning and implementation.

GIS&RS technologies may be used for scientific planning and monitoring performance of projects. This will require the National Level Nodal Agency (NLNA) and State Level Nodal Agency (SLNA) to establish core GIS facilities by using satellite imagery data received from National Remote Sensing Centre (NRSC) and ISRO.

All the GIS layers on various themes may be overlaid on a geo-referenced base layer up to the level of village boundaries at the first instance and field agencies, like Project Implementing Agencies (PIAs) may be given controlled access to operate for project planning. Application software for web-enabled integrated project development, spatial & non-spatial data standards and meta-data can also be put in place. This knowledge base would make it possible to define project boundaries with a unique-identification (unique-id) for each project. It would also be possible to map

treatment areas with reference to their respective administrative units like villages, blocks and districts.

Both GIS and RS technologies can be generously used in assessing the actual impact of various interventions in a given area and period. These include periodic changes in geo-hydrology, regeneration of biomass, soil cover etc. Further, the proposed interventions at specific locations can be geo-tagged prior to their implementation.

To promote data and technology in area development approach like a watershed, the importance of a national portal is well understood. The NLNA may strengthen existing Management Information System (MIS) and develop data base related to watershed projects. Relevant and desired data updated from the field levels can be hosted by implementing and supervising agencies [Watershed Committee (WC), PIA, Watershed Cell cum Data Centre (WCDC) etc.] from across the country which will ultimately be collected and reflected at national portal.

Let us sum up

The objectives of Watershed Development Programmes are:

- Developing wastelands or degraded lands, drought-prone and desert areas on watershed basis, keeping in view the capability of land, site conditions and local needs.
- Promoting the overall economic development and improving the socioeconomic condition of the resource poor and disadvantaged sections inhabiting the programme areas.
- Mitigating the adverse effects of extreme climatic conditions such as drought and desertification on crops, human and livestock population for their overall improvement.
- Restoring ecological balance by harnessing, conserving and developing natural resources i.e. land, water, vegetative cover.
- Sustained community action for the operation and maintenance of assets created and further development of the potential of the natural resources in the watershed.
- Simple, easy and affordable technological solutions and institutional arrangements that make use of, and build upon, local technical knowledge and available materials.
- Employment generation, poverty alleviation, community empowerment and development of human and other economic resources of the village.

- To minimize the adverse effects of drought on the production of crops, livestock and productivity of land, water and human resources for drought proofing of the affected areas.
- To promote the overall economic development and improve the socioeconomic condition of the resource poor and disadvantaged sections inhabiting the programme areas.
- To take up development works by watershed approach for land development, water resource development and afforestation or pasture development.

Glossary

WDC-PMKSY: Prime Minister Krishi Sinchayee Yojna (Watershed Development Component

NLNA: National Level Nodal Agency

SLNA: State Level Nodal Agency

NRSC: National Remote Sensing Centre

Check your progress

1. What are the three principal stake holders of Doubling Farmers Income (DFI) ?
    - the consumers need nutrition security.
    - the farmers need income security and
    - the production environment needs ecological security.
2. Name some other few Natural Resources Development programmes in India?
    - Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)
    - Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS)
    - National Food Security Mission (NFSM)
    - Integrated Farming System (IFS)

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.
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Unit 11

Soil and Water conservation practices

Structure

Learning Objectives

Overview

11.1 Agronomic measures

11.2 Land configuration techniques

11.3 Mechanical measures

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to:

- Agronomic measures
- Land configuration techniques
- Mechanical measures

Overview

Watershed management is a term used to describe the process of implementing land use practices and water management practices to protect and improve the quality of the water and other natural resources within a watershed by managing the use of those land and water resources in a comprehensive manner. Soil conservation practices are tools the farmer can use to prevent soil degradation and build organic matter. These practices include crop rotation, reduced tillage, mulching, cover cropping and cross-slope farming. farmers to increase soil organic matter content, soil structure and rooting depth. Herewith let us see various Soil and Water conservation practices as fallowed in watershed programmes in India.

11.1 Agronomic measures

Agronomic measures are applicable in the landscape of ≤2% slope. Agronomic measures reduce the impact of raindrops through the covering of soil surface and increasing infiltration rate and water absorption capacity of the soil which results in reduced runoff and soil loss through erosion. These measures are cheaper, sustainable, and may be more effective than structural measures. Important agronomic measures are described below.

11.1.1 Contour farming

Contour farming is one of the most used agronomic measures for soil and water conservation in hilly agro-ecosystems and sloppy lands. All the agricultural operations, viz. plowing, sowing, inter-culture, etc., are practiced along the contour line. The ridges and furrows formed across the slope build a continual series of small barriers to the flowing water which reduces the velocity of runoff and thus reduces soil erosion and nutrient loss. It conserves soil moisture in low rainfall areas due to increased infiltration rate and time of concentration, while in high rainfall areas, it reduces soil loss. In both situations, it reduces soil erosion, conserves soil fertility and moisture, and thus improves overall crop productivity. However, the effectiveness of this practice depends upon rainfall intensity, soil type, and topography of a particular locality.

11.1.2 Choice of crops

The selection of the right crop is crucial for soil and water conservation. The crop should be selected according to the intensity and critical period of rainfall, market demand, climate, and resources of the farmer. The crop with good biomass, canopy cover, and extensive root system protects the soil from the erosive impact of rainfall and creates an obstruction to runoff, and thereby reduces soil and nutrient loss. Row or tall-growing crops such as sorghum, maize, pearl millet, etc. are erosion permitting crops which expose the soil and induce the erosion process. Whereas close growing or erosion resisting crops with dense canopy cover and vigorous root system viz. cowpea, green gram, black gram, groundnut, etc. are the most suitable crops for reducing soil erosion. To increase the crop canopy density, the seed rate should always be on the higher side.

11.1.3 Crop rotation

Crop rotation is the practice of growing different types of crops in succession on the same field to get maximum profit from the least investment without impairing the soil fertility. Monocropping results in exhaustion of soil nutrients and deplete soil fertility. The inclusion of legume crops in crop

rotation reduces soil erosion, restores soil fertility, and conserves soil and water. Further, the incorporation of crop residue improves organic matter content, soil health, and reduces water pollution. A suitable rotation with high canopy cover crops helps in sustaining soil fertility; suppresses weed growth, decreases pests and disease infestation, increases input use efficiency, and system productivity while reducing soil erosion.

11.1.4 Cover crops

The close-growing crops having high canopy density are grown for protection of soil against erosion, known as cover crops. Legume crops have good biomass to protect soil than the row crops. The effectiveness of cover crops depends on crop geometry and development of canopy for interception of raindrops which helps in reducing the exposure of soil surface for erosion. It has been reported that legumes provide better cover and better protection to land against runoff and soil loss as compared to cultivated fallow and sorghum. The most effective cover crops are cowpea, green gram, black gram, groundnut, etc.

Advantages

- Protection of soil from the erosive impact of raindrops, runoff, and wind.
- Act as an obstacle in water flow, reduce flow velocity, and thereby reduce runoff and soil loss.
- Increase soil organic matter by residue incorporation and deep root system.
- Improve nutrients availability to the component crop and succeeding crops through biological nitrogen fixation.
- Improve water quality and water holding capacity of the soil.
- Improve soil properties, suppress weed growth, and increase crop productivity.

11.1.5 Intercropping

Cultivation of two or more crops simultaneously in the same field with definite or alternate row pattern is known as intercropping. It may be classified as row, strip, and relay intercropping as per the crops, soil type, topography, and climatic conditions. Intercropping involves both time-based and spatial dimensions. Erosion permitting and resisting crops should be intercropped with each other. The crops should have different rooting

patterns. Intercropping provides better coverage on the soil surface, reduces the direct impact of raindrops, and protects soil from erosion.

Advantages

- High total biomass production.
- Efficient utilization of soil and water resources.
- Reduction of marketing risks due to the production of a variety of products at different periods.
- Drought conditions can be mitigated through intercropping.
- Reduce the weed population and epidemic attack of insect pests or diseases.
- It improves soil fertility.

11.1.6 Strip cropping

Growing alternate strips of erosion permitting and erosion resistant crops with a deep root system and high canopy density in the same field is known as strip cropping. This practice reduces the runoff velocity and checks erosion processes and nutrients loss from the field. The erosion resisting crops protects soil from the beating action of raindrops, reduces runoff velocity, and thereby increased time of concentration which results in a higher volume of soil moisture and increased crop production. Strip cropping is practiced for controlling the run-off and erosion and thereby maintaining soil fertility.

Types of strips cropping

1. Contour strip cropping: The growing of alternate strips of erosion permitting and erosion resisting crops across the slopes on the contour is known as contour strip cropping. It reduces the direct beating action of raindrops on the soil surface, length of the slope, runoff flow and increases rainwater absorption into the soil profile.
2. Field strip cropping: In this practice the field crops are grown in parallel strips across fairly uniform slopes, but not on exact contours. It is useful on regular slopes and with soils of high infiltration rates, where contour strip cropping may not be practical.
3. Wind strip cropping: It consists of the planting of tall-growing row crops (such as maize, pearl millet, and sorghum) and close or short growing crops in alternately arranged straight and long, but relatively narrow, parallel strips laid out right across the direction of the prevailing wind, regardless of the contour.

4. Permanent or temporary buffer strip cropping: It is the growing of permanent strips of grasses or legume or a mixture of grass and legume in highly eroded areas or in areas that do not fit into regular rotation, i.e., steep or highly eroded, slopes in fields under contour strip cropping. These strips are not practiced in normal strip cropping and generally planted on a permanent or temporary basis.

11.1.7 Mulching

Mulch is any organic or non-organic material that is used to cover the soil surface to protect the soil from being eroded away, reduce evaporation, increase infiltration, regulate soil temperature, improve soil structure, and thereby conserve soil moisture. Mulching prevents the formation of hard crust after each rain. The use of blade harrows between rows or inter-culture operations creates “dust mulch” on the soil surface by breaking the continuity of capillary tubes of soil moisture and reduces evaporation losses. Mulching also reduces weed infestation along with the benefits of moisture conservation and soil fertility improvement. Hence, it can be used in high rainfall regions for decreasing soil and water loss, and in low rainfall regions for soil moisture conservation. Organic mulches improve organic matter and consecutively improve the water holding capacity, macro and micro fauna biodiversity, their activity, and fertility of the soil.

Inorganic mulches have a longer life span than organic mulches and can reduce soil erosion, water evaporation losses, suppress weeds but cannot improve soil health. This practice is costly and labor intensive therefore, suitable for cash crops such as fruits and vegetables. Polyethylene mulch is commonly used for the conservation of soil and water resources to increase crop productivity [.

11.1.8 Conservation tillage

In this practice at least 30% of soil surface should remain covered with crop residue before and after planting the next crop to reduce soil erosion and runoff, as well as other benefits such as C sequestration. This term includes reduced tillage, minimum tillage, no-till, direct drill, mulch tillage, stubble-mulch farming, trash farming, strip tillage, etc. The concept of conservation tillage is widely accepted in large scale mechanized crop production systems to reduce the erosive impact of raindrops and to conserve the soil moisture with the maintenance of soil organic carbon. Conservation tillage improves the infiltration rate and reduces runoff and evaporation losses. It also improves soil health, organic matter, soil structure, productivity, soil fertility, and nutrient cycling and reduces soil compaction.

11.1.9 Organic farming

Organic farming is an agricultural production system that devoid the use of synthetic fertilizers or pesticides and includes organic sources for plant nutrient supply viz. FYM, compost, vermicompost, green manure, residue mulching, crop rotation, etc. to maintain a healthy and diverse ecosystem for improving soil properties and ensuring a sustained crop production. It is an environmentally friendly agricultural crop production system.

The maintenance of high organic matter content and continuous soil surface cover with cover crops, green manure, and residue mulch reduces the soil erosion in organic farming. It leads to the addition of a large quantity of organic manures which enhances water infiltration through improved bio-physico-chemical properties of soil, and eventually reduces soil erodibility. Organic materials improve soil structure through the development of soil binding agents (e.g., polysaccharides) and stabilizing and strengthening aggregates which reduce the disintegration of soil particles and thus reduce soil erosion. Soil erosion rates from soils under organic farming can be 30–140% lower than those from conventional farming.

11.2 Land configuration techniques

Adoption of appropriate land configuration and planting techniques according to crops, cropping systems, soil type, topography, rainfall, etc. help in better crop establishment, intercultural operations, reduce runoff, soil and nutrient loss, conserve water, efficient utilization of resources and result in higher productivity and profitability. Ridge and furrow raised bed and furrow, broad bed and furrow, and ridging the land between the rows are important land configuration techniques.

11.2.1 Ridge and furrow system:

Raising rainy season crops on ridges and rabi season crops in furrows reduces the soil crusting and ensures good crop stand over sowing on flat beds. Moreover, inter-row rainwater can be drained out properly during the monsoon period and collected in farm ponds, for life-saving irrigations and profile recharging for the establishment of rabi crops. It leads to an increased moisture content in soil profile which reduces moisture stress on plants during the drought period. This method is most suitable for wide-spaced crops viz. cotton, maize, vegetables, etc.

11.2.2 Broad bed and furrow system:

This system has been developed by the ICRISAT in India. It is primarily advocated for high rainfall areas (>750 mm) having black cotton soils

(Vertisols). Beds of 90–120 cm width are formed, separated by sunken furrows of about 50–60 cm wide and 15 cm depth. The preferred slope along the furrow is between 0.4 and 0.8% on Vertisols. Two to four rows of the crop can be grown on the bed, and the width and crop geometry can be adjusted to suit the cultivation and planting equipment.

Advantages

- Increase in-situ soil moisture conservation
- Safely dispose of excess runoff without causing erosion
- Improved soil aeration for plant growth and development
- Easier for weeding and mechanical harvesting
- It can accommodate a wide range of crop geometry.

11.2.3 Agroforestry measures

Agroforestry is a sustainable land management system which includes the cultivation of trees or shrubs with agricultural crops and livestock production simultaneously on the same piece of land. It is an emerging technology for effective soil and water conservation and comprises a wide range of practices for controlling soil erosion, developing sustainable agricultural production systems, mitigating environmental pollution, and increasing farm economy. The leaf litter addition acts as a protective layer against soil erosion improves soil health and moisture retention capacity of the soil and increases crop productivity. It has been reported that different agroforestry practices can reduce up to 10% of soil erosion. Agroforestry not only controls soil erosion but also produces tree-based several marketable products.

Types of agroforestry systems

Agri-Silviculture: It is the growing of agricultural crops as a primary component with the secondary component of multipurpose trees (MPTs) on the same managed land unit. The tree species bind soil particles in the root zone and increase water infiltration and reduce runoff.

Agri-Horticulture: Growing of agricultural crops and fruit trees on the same managed land unit is known as agri-horticulture. Fruit tree species like lemon (Citrus limon), mango (Mangifera indica), ber (Ziziphus mauritiana), and aonla (Phyllanthus emblica) can be successfully planted in agricultural fields and on degraded and low fertile lands with some restoration measures.

Alley Cropping: Growing of agricultural crops in the alley formed between the hedge rows of leguminous nitrogen-fixing tree species. This system is one of the effective measures for soil and water conservation in hilly areas.

Silvi-pasture System: Raising grasses or livestock with MPTs on the same managed land unit is known as silvi-pasture system. This system has the potential to reclaim eroded and degraded lands. Mechanical measures combined with grass species cultivation are more effective for controlling soil erosion processes. The grass species such as Cenchrus ciliaris (buffel grass), Cenchrus setigerus (birdwood grass), Dichanthium annulatum (marvel grass), Panicum antidotale (blue panicgrass), Panicum maximum (Guinea grass), Brachiaria mutica (para grass) and Pennisetum purpureum (elephant grass) are important in ravine restoration.

11.3 Mechanical measures

Mechanical measures or engineering structures are designed to modify the land slope, to convey runoff water safely to the waterways, to reduce sedimentation and runoff velocity, and to improve water quality. These measures are either used alone or integrated with biological measures to improve the performance and sustainability of the control measures. In highly eroded and sloppy landscape biological measures should be supplemented by mechanical structures. A number of permanent and temporary mechanical measures are available such as terraces, contour bunding, check dams, gabions, diversion drains, geo-textiles, etc.. The mechanical measures are preferred based on the severity of erosion, soil type, topography, and climate.

11.3.1 Bunding

1. Contour bunding: Contour bunding is used to conserve soil moisture and reduce erosion in the areas having 2–6% slope and mean annual precipitation of <600 mm with permeable soils. The vertical interval between two bunds is known as the spacing of bunds. The spacing of bund is dependent on the erosive velocity of runoff, length of the slope, slope steepness, rainfall intensity, type of crops, and conservation practices.
2. ii. Graded bunding: Graded bunds are made to draining out of excess runoff water safely in areas having 6–10% land slope and receiving rainfall of >750 mm with the soils having infiltration rate < 8 mm/h.
3. iii. Peripheral bunds: Peripheral bunds are constructed around the gully head to check the entry of runoff into the gully. It protects the gully head from being eroded through erosion processes. It creates a favorable

condition for the execution of vegetative measures on gully heads, slopes, and beds.

11.3.2 Contour trenching

Trenches are constructed at the contour line to reduce the runoff velocity for soil moisture conservation in the areas having <30% slope. Bunds are formed on the downstream side of trenches for the conservation of rainwater. Trenches are of two types:

Continuous contour trenches: Continuous contour trenches are constructed based on the size of the field in the low rainfall areas with the 10–20 cm trench length and 20–25 cm equalizer width without any discontinuity in trench length (10–20 m).

Staggered contour trenches (STCs) Generally, these trenches are constructed in alternate rows directly beneath one another in a staggered manner in the high rainfall areas, where the risk of overflow is prominent. SCTs are 2–3 m long with 3–5 m spacing between the rows. Planting of tree species is done based on the land slope. It is highly effective in forestalling extension of gully head, soil loss, and arrest the overflow.

11.3.3 Terracing

Terraces are earthen embankments built across the dominant slope partitioning the field in uniform and parallel segments. Generally, these structures are combined with channels to convey runoff into the main outlet at reduced velocities. It reduces the degree and length of slope and thus reduces runoff velocity, soil erosion and improves water infiltration. It is recommended for lands having a slope of up to 33% but can be adopted for lands having up to 50–60% slope, based on socio-economic conditions of a particular region. Where plenty of good-quality stones are available, stone bench terracing is recommended. Sometimes, semi-circular type terraces are built at the downstream side of the plants, known as half-moon terraces. Based on the slope of benches, the bench terraces are classified into the following categories:

Bench terraces sloping outward: These types of terraces are used in low rainfall areas having permeable soils. A shoulder bund is provided for stability of the edge of the terrace and thus has more time for rainwater soaking into the soil.

Bench terraces sloping inward (hill-type terraces): These types of bench terraces are suitable for heavy rainfall areas where a higher portion of rainfall is to be drained as runoff. For this, a suitable drain should be

provided at the inward end of each terrace to drain the runoff. These are also known as hill-type terraces.

Bench terraces with level top: These types of terraces are suitable for uniformly distributed medium rainfall areas having deep and highly permeable soils. These are also known as irrigated bench terraces because of their use in irrigated areas.

11.3.4 Contour wattling

Wattling is a technique of dividing the length of the slope into shorter sections and in these sections, the wattles are constructed at a vertical interval of 5–7 m up to 33% slope and 3 m up to 66% slope. It is not effective on slopes steeper than 66% and on very loose or powdery rocks.

11.3.5 Crib structures

Crib structures are used to stabilize the steep slopes of >40% by constructing log wood structures filled with stone/brushwood. Eucalyptus poles with 2–3 m length and 8–12 cm diameter can be used for the construction of crib structures. These poles are joined together with the help of 20–25 cm long nails. The height of the structure is kept 1.5–2 m above the ground depending upon the land slope.

11.3.6 Geo-textiles

Geo-textiles are made up of natural fibers of jute or coir, which are used for stabilization of degraded slopes in mine spoil and landslides areas along roadsides. It facilitates the initial establishment of vegetation on highly degraded sloping lands by holding the vegetation in place and conserving moisture. The open mesh size of geo-textiles varies from 3 to 25 mm. The biodegradability of geo-textiles was reported for 2–3 years. It can absorb 12–25% water under 65 and 95% humidity, respectively and when fully soaked in water it can absorb 40% moisture.

11.3.7 Loose boulder/stone/masonry check dams

Check dams are effective for preventing runoff rate and severe erosion in steep and broad gullies, and most suitable for high elevation areas of the catchment. These structures are cheap, have a long life, and fewer maintenance requirements. The depth of gully bed is kept about o.3 m and flat stones of 20–30 cm size are used for the construction of dams. A spillway is provided in the middle of the dam to allow the safe discharge of runoff water. Similarly, gabion check dams are also used for drainage line treatment in sharp slanted gullied areas to check sedimentation, erosion, and to conserve soil moisture.

11.3.8 Brushwood check dams

Branches of tree and shrub species are staked in two rows parallel to each other filled with brushwood and laid across the gully or way of the flow. These are usually built to regulate the overflow in small and medium gullies which are supplemented with vegetative barriers for long term effectiveness. There is enough soil volume to establish the vegetation. The tree species are planted in 0.3 m × 0.2 m trenches across the way of gullies. It reduces the runoff velocity, soil loss, and improves soil moisture which helps in the successful establishment of vegetative barriers.

11.3.9 Diversion drains

The channels are constructed to protect the downstream area and for safe draining and diverting of runoff water. It is applicable in high rainfall areas to control runoff losses during the initial stage. The gradient of diversion drain should preferably be kept within 0.5%. Generally, a narrow and deep drain does not get silted up as rapidly as a broad and shallow drain of the same cross-sectional area. Soil dug from the drain should be dumped on the lower side of the drain. Outlet ends should be opened at natural drainage lines.

11.3.10 Conservation bench terrace

In the conservation bench terrace (CBT) system, the land is divided into 2:1 ratio along the slope in which the upper 2/3 area (Donor area) contributes runoff to the lower 1/3 runoff collecting area (recipient area). The donor area is left in its natural slope condition. It is also known as the zingg terrace and developed by Zingg and Hauser in 1959. The runoff contributing area is used for cultivation of kharif while the lower 1/3 area with conserved soil moisture is used to cultivate rabi crops. This mechanical measure can be successfully applied in a semi-arid climate on mild sloppy lands (2–5%) for erosion control, water conservation, and improvement of crop productivity. This system can be used in silty loam to silty clay loam soils. CBT system resulted in the reduction of runoff from 36.3 to 7.4% and soil loss from 10.1 to 1.19 Mg ha−1 as compared to the conventional system of sloping border. An average reduction of 78.9 and 88.0% in runoff and soil loss, respectively reported in the CBT system over the conventional system.

Let us sum up

In brief various control measures are:

- Vegetative measures (Agronomical measures): Strip cropping,Pasture cropping, Grass land farming and Wood lands

- Engineering measures: Contour bunding, Terracing, Construction of earthern embankment, Construction of check dams, Construction of farm ponds, Construction of diversion, Gully controlling structure and Rock dam
- Establishment of permanent grass and vegetation and Providing vegetative and stone barriers
- Construction of silt tanks dentension

Glossary

Mulch is any organic or non-organic material that is used to cover the soil surface to protect the soil from being eroded away, reduce evaporation, increase infiltration, regulate soil temperature, improve soil structure, and thereby conserve soil moisture.

Terraces are earthen embankments built across the dominant slope partitioning the field in uniform and parallel segments.

Geo-textiles are made up of natural fibers of jute or coir, which are used for stabilization of degraded slopes in mine spoil and landslides areas along roadsides.

Check your progress

1. Define: Alley Cropping

Growing of agricultural crops in the alley formed between the hedge rows of leguminous nitrogen-fixing tree species. This system is one of the effective measures for soil and water conservation in hilly areas.

2. What are the advantages of intercropping practice?
    - High total biomass production and Efficient utilization of soil and water resources.
    - Reduction of marketing risks due to the production of a variety of products at different periods.
    - Drought conditions can be mitigated through intercropping.
    - Reduce the weed population and epidemic attack of insect pests or diseases.

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.
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Unit 12

Watershed Based Land Use Planning

Structure

Learning Objectives

Overview

12.1 Impact on Watershed due to Land Use

12.2 Planning the Land Use

12.2.1 Goals of Land Use Planning

12.2.2 Trade-offs among Conflicting Goals of Land Use Planning

12.2.3 The Focus of Land Use Planning

12.2.4 Land Use Planning at Different Levels

12.2.5 Land Use in Relation to Sectoral and Development Planning

12.2.6 People Involved in Land Use Planning

12.3 Applications of RS and GIS in Watershed Planning

13.3.1 Management Planning

13.3.2 Watershed Restoration

13.3.3 Watershed Policy Analysis and Decision Support

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to:

- Impact on Watershed due to Land Use
- Planning the Land Use
- Applications of RS and GIS in Watershed Planning

Overview

Land Use Planning and Watershed Management is responsible for directing, coordinating, monitoring and monitor processes using protection and conservation of watersheds and water systems, land management, geographic information systems, biological corridors and environmental education. In this Unit, we shall discuss the impact on watershed due to land use and spatio-temporal changes in it. Then we shall move on to the planning of land use to ensure a sustainable watershed.

12.1 Impact on Watershed due to Land Use

Watershed and land use are quite inter-dependent. Watersheds with a healthy aquatic system -in the form of adequate streams & wetlands, and an equally healthy biotic system -in the form of adequate flora and fauna, are generally sustainable systems. Once they are subjected to large scale human interventions and/ or natural calamities, the land use gets altered significantly. This in turn causes major impact in the watershed in terms of its hydrology, flora and fauna.

In many parts of the world, extensive areas of native forests and grasslands have been converted into croplands or urban areas or road/ railway systems/ networks. This has resulted in the alteration of riparian corridors, drainage of wetlands and modification of natural river systems. These changes in land use have resulted in hydrologic changes in the watersheds, their stream systems and surface water-groundwater linkages. Changes in water quantity and quality can affect people and ecosystems in both upstream and downstream areas of watersheds.

If these changes occurring in watersheds are not managed properly, they may become unsustainable in the long run. Therefore, to avoid any undesirable consequences, increased attention is being paid to maintaining or restoring natural stream channel systems, riparian communities, wetland ecosystems and floodplains which can restore the good hydrologic conditions of watersheds.

If watersheds are not sustainably managed, they may show cumulative watershed effects i.e., combined environmental effects of activities in a watershed that can adversely impact beneficial uses of lands. Individually these environmental effects may not appear to be relevant. But collectively, they may become significant over time and space.

For example, the conversion of forest to crop lands in one part of a watershed can cause an increase in the water and sediment flow. Likewise, road construction and drainage can also have effects in a watershed like

drainage of a wetland at some other location. Similarly, removal of dense shrubs to increase forage production may also increase water yield in some cases, benefit certain wildlife species and reduce fire hazards. However, the same shrub removal may be detrimental to other types of wildlife. Changes in vegetation composition/ density/ age structure/ continuity across the landscape can affect evapo-transpiration losses and thereby influence antecedent soil moisture conditions, water yields & their timings, stream flow volumes & their peaks, at different parts of watersheds. Overgrazing -which results in excess trampling in a watershed and excessive soil compaction, reduces infiltration capacity and increases surface runoff. Roads and trails possibly increase soil erosion due to the exposure to erodible soil and subsoil during their construction. This reduces infiltration and concentrates overland flow from precipitation excess which erodes the increased gradients in the side slopes of cuts and fills.

The increase in flooding due to the creation of finished impervious surfaces as well as due to the filling up of water bodies, especially in the urbanized areas leading to a drastic reduction in infiltration or surface storage is very well known. On the other hand, forest and wild land watersheds are frequently affected by wildfires. This results in increased soil erosion due to the loss in the vegetation cover and an increased surface runoff due to the formation of water repellent layers in the soil.

These are some of the examples wherein a change in land use has impacted the watersheds and made them ecologically unsustainable. There are many other examples of land use changes which also disturb the watersheds in terms of water quality, geomorphic and hydrologic effects. To overcome these undesirable effects, an interdisciplinary approach involving hydrology, geomorphology and ecology into watershed management and land use planning is needed to understand and appreciate the impacts of cumulative watershed effects on water yield, other stream flow characteristics and water quality. The next section will deal with the planning the land use to ensure sustainability in watershed management.

12.2 Planning the Land Use

There are conflicts over land use, many a times. The demands for arable land, grazing, forestry, wildlife, tourism, and urban development are greater than the land resources available. In the developing countries, these demands become more acute every year. The population dependent on the land for food, fuel and employment is expected to double within the next 25 to 50 years. Even where land is still available in plenty, many people may have inadequate access to land or to the benefits from its use. In the face

of scarcity, the degradation of farmland, forest or water resources are visible for all to see but individual land users lack the incentive or resources to stop it.

Land-use planning is the systematic assessment of land and water potential, alternatives for land use and economic and social conditions to select and adopt the best land-use options. Its purpose is to select and put into practice those land uses that will best meet the needs of the people while safeguarding the resources for the future. The driving forces in land use planning are the needs for change, improved management or quite different patterns of land use dictated by the changing circumstances.

All kinds of rural land use like agriculture, pastoral lands, forestry, wildlife conservation and tourism are involved in land use planning. It also provides guidance in cases of conflict between rural land use and urban or industrial expansion, by indicating which areas of land are most valuable under rural use.

The following two conditions must be met if the land use planning is to be useful:

- The need for changes in land use or the action to prevent some unwanted changes must be accepted by the people involved.
- There must be the political will and ability to put the plan into effect.

Wherever these two conditions are not met and the problems are pressing, it may be appropriate to mount an awareness campaign or set up demonstration areas with the aim of creating the conditions necessary for effective planning. Our basic needs of air, water, food, clothing, shelter and fuel must be met by the land which is in limited supply. As population and aspirations increase, the land becomes an increasingly scarce resource.

Land must change to meet new demands which may bring new conflicts among the competing uses of the land and among the interests of individual land users and the common good. Land taken for towns and industry is no longer available for farming. Likewise, the development of new farmland may compete with forestry, water supplies and wildlife.

Planning to make the best use of land is an established idea. Over the years, farmers have made plans season after season, deciding what to grow and where to grow it. Their decisions have been made according to their own needs, knowledge of the land & the technology, labour and capital available. As the size of the area, the number of people involved and the complexity of the problems increase, the need for information and rigorous methods of analysis and planning also increase.

However, land-use planning is not just farm planning on a different scale. It has a further dimension, namely the interest of the whole community. Planning involves anticipation of the need for change as well as reactions to it. Its objectives are set by social or political requirements which consider the existing situation. In many places, the existing situation cannot continue because the land itself is being degraded. Examples of unwise land use include the following:

- The clearance of forest on steep lands or on poor soils for which sustainable systems of farming have not been developed so far
- Overgrazing of pastures
- Industrial, agricultural and urban activities that produce pollution.

Degradation of land resources may be attributed to human greed, ignorance, uncertainty, or lack of an alternative but essentially, it is a consequence of using land today without investing in tomorrow. Land-use planning aims to make the best use of limited resources by the following actions:

- Assess the present and future needs and systematically evaluating the land's ability to supply them.
- Identify and resolve the conflicts among competing uses, the needs of individuals and those of the community, and among the needs of the present generation and those of future generations.
- Seek sustainable options and choose those which fully meet identified needs.
- Plan to bring about desired changes; and,
- Learn from experience.

There can’t be a blueprint for change. The whole process of planning is iterative and continuous. At every stage, as better information is obtained, a plan may have to be changed to take account of it.

12.2.1 Goals of Land Use Planning

Goals of land use planning define what is meant by the "best" use of the land. They should be specified at the outset of a particular land use planning project. Goals may be grouped under the following three headings of efficiency, equity & acceptability, and sustainability.

Efficiency: Land use planned must be economically viable. Therefore, one goal of development planning is to make efficient and productive use of the land. For any land use, certain areas are better suited than others. Efficiency

is achieved by matching different land uses with the areas that will yield the greatest benefits at the least cost, i.e., maximum benefit cost ratio.

Efficiency might mean different things to different people. To the individual land user, it means the greatest return on capital and labour invested or the greatest benefit from the land area available. Government objectives are more complex: they may include improving the foreign exchange situation by producing for export or for import substitution.

Equity & Acceptability: Land use must be socially acceptable. It should ensure food security, employment, and income security in rural areas. Land improvements and redistribution of land may be undertaken to reduce inequality or to attack absolute poverty. One way of doing this is to set a threshold standard of living to which the target groups should be raised. Living standards may include levels of income, nutrition, food security and housing. Planning to achieve these standards involves the allocation of land for specific uses as well as the allocation of financial and other resources.

An example of acceptability is given here. Following the drought of 1973-74 and the subsequent famine, the Government of Ethiopia became more aware of the serious degradation of soil in the highlands.

An ambitious soil conservation programme which concentrated on protecting steep slopes by bunding and afforestation was launched. This had made a substantial impact on soil erosion but has not contributed much to increasing agricultural production. Large-scale afforestation was also unpopular with local people because it reduced the area available for livestock grazing while forest protection implied denying access to the public for fuel wood collection. A balance between the competing requirements of conservation and production was clearly needed if popular support for soil conservation work was to continue without inducements such as the Food-for-Work Programme.

A land-use plan to conserve steeper slopes by restoring good vegetative cover through closure, followed by controlled grazing, was found to be more acceptable to the local people than large-scale afforestation applied in isolation.

Sustainability: Sustainable land use is that which meets the needs of the present while simultaneously conserving resources for future generations. This requires a combination of production and conservation. The production of the goods required by the people now need to be combined with the conservation of natural resources on which that production depends so as to ensure a continued production in the future.

A community that destroys its land will forfeit its future. Land use has to be planned for the community as a whole because the conservation of soil, water and other land resources is often beyond the means of individual land users.

12.2.2 Trade-offs among Conflicting Goals of Land Use Planning

Clearly, there are conflicts between these various goals of land use planning. More equity may lead to less efficiency. In the short term, it may not be possible to meet the needs of the present without consuming resources such as burning oil or clearing areas of natural forest. Decision-makers need to consider the trade-offs between different goals. But if the system is to survive, the use of natural assets must be compensated for by the development of human or physical assets of equal or greater worth.

Good information such as information about the needs of the people, about land resources and about the economic, social, and environmental consequences of alternative decisions is always essential. The job of the land use planner is to ensure that decisions are made based on consensus or acceptable degree of disagreement.

In many cases, planning processes like introducing appropriate new technology can reduce the costs in trade-offs. It can also help in resolving the conflict by involving the community in the planning process and by revealing the rationale and information on which decisions are based.

12.2.3 The Focus of Land Use Planning

The following points constitute the focus of land use planning.

Land Use Planning is for the People: People's needs are the driving forces in the land use planning process. Local farmers, other land users and the wider community who depend on land must accept the need for a change in land use, as they will have to live with its results.

Land use planning must be positive and needs to be for the people’s betterment. The planning team must find out about people's needs and about the local knowledge, skills, labour and capital that they can contribute. It must study the problems of existing land use practices and seek alternatives while drawing the public attention to the hazards or inconveniences of continuing with the present practices and to the opportunities for change.

Regulations to prevent people doing what they now do for pressing reasons are most likely to fail. Local acceptability is readily achieved by local participation in land use planning. The support of local leaders is essential.

At the same time, the participation of agencies that have the resources to implement the plan is also important.

Land is not the Same Everywhere: Land is the other focus of land-use planning. Capital, labour, management skills and technology can be moved to where they are needed. On the other hand, land cannot be moved, and different areas present different opportunities and different management problems. The land resources are generally changing as is obvious in the case of climate and vegetation. But examples such as the depletion of water resources or the loss of soil by erosion or salinity remind us that resources can be degraded, many a times irreversibly. Good information about land resources is thus essential to land use planning.

Technology: A third element in land use planning is the knowledge of technologies like agronomy, silviculture, livestock husbandry and other means by which land is used. The technologies recommended must be appropriate technologies for which the users have the capital, skills, and other necessary resources. New technologies may have social and environmental implications that should be addressed by the land use planner(s).

Integration: A mistake in early attempts at land use planning was to focus too narrowly on land resources without enough thought given to their possible use. Good agricultural land is usually also suitable for other competing uses. Land use decisions are not made only based on land suitability but also according to the demand for products and the extent to which the use of a particular area is critical for a particular purpose. Planning has to integrate information about the suitability of the land, the demands for alternative products or uses and also the opportunities for satisfying those demands on the available land, now as well as in the future.

Hence, land use planning is not sectoral. Even where a particular plan is focused on one sector, e.g., small holder tea development or irrigation, an integrated approach must be carried down starting from the strategic planning at the national level to the details of the individual projects and programs at district and local levels.

12.2.4 Land Use Planning at Different Levels

Land use planning can be applied at three broad levels: national, district and local. These are not necessarily in that order. They correspond to the levels of government at which decisions about land use are taken.

Different kinds of decisions are taken at each level, where the planning methods and plan types also differ. However, at each level there is a need for a land use strategy, policies that indicate planning priorities, projects that tackle these priorities and operational planning to get the work done smoothly, swiftly and cost-effectively.

The greater the interaction between the three levels of planning, the better for all. The flow of information should be in both directions. At each successive level of planning, the degree of details needed as well as the direct participation of the local people increase.

National Level Land Use Planning:

At the national level, land use planning is concerned with the national goals and the allocation of resources. In many cases, national land use planning may not involve the actual allocation of land for different uses. In place of them, it may establish the priorities for district level projects. A national land use plan may cover:

- Land-Use Policy related to balancing the competing demands for land among different sectors of the economy such as food production, export crops, tourism, wildlife conservation, housing & public amenities, roads, industry.
- National Development Plans and Budget consisting of project identification and the allocation of resources for development.
- Coordination of sectoral agencies involved in land use.
- Legislation on such subjects as land tenure, forest clearance and water rights.

National goals are complex while policy decisions, legislation and fiscal measures affect a large population and wide areas. Decision makers can't possibly be specialists in all facets of land use. So, the planners' responsibility is to present the relevant information so that the decision makers can both understand and act on it.

District Level Land Use Planning:

District level refers not necessarily to administrative districts but also to land areas that fall between national and local levels. Development projects are generally at this level, where planning first comes to the grips with the diversity of the land and its suitability to meet the project goals. When planning is initiated nationally, national priorities need to be translated into local plans. Conflicts between national and local interests should be resolved. The kinds of issues tackled at this stage include:

- The site of developments such as new settlements, forest plantations, irrigation schemes, etc.
- The need for improved infrastructure such as water supply, roads, marketing facilities, etc.
- The development of management guidelines for improved types of land use on each type of land.

Local Level Land Use Planning:

The local planning unit may be the village, a group of villages or a small watershed or a catchment. At this level, it is very easy to fit the plan to the people, making use of local people's knowledge and contributions. Wherever the planning is initiated at the district level, the programme of work to implement changes in land use or management has to be carried out locally. Alternatively, this may be the first level of planning, with its priorities drawn up by the local people. Local level planning is about getting things done on areas of land including what shall be done where and when, and who will be responsible.

Some of the examples of local level land use planning are:

- Layout of drainage, irrigation and soil conservation works.
- Design of infrastructure - road alignment and the siting of crop marketing, fertilizer distribution, milk collection or veterinary facilities.
- Sitting of specific crops on suitable land.

Requests at the local level, e.g., for suitable areas to introduce tobacco or coffee, must be met with firm recommendations. Planning at these different levels needs information at different scales and levels of generalization. Much of this information may be available in maps. The most suitable map scale for national level land use planning is one by which the whole country fits on to one map sheet, which may call for a scale ranging from 1:5 million to 1:1 million or larger. District level land use planning requires details to be mapped at about 1:50000, although some information may be summarized at smaller scales ranging to 1:250000.

For local level land use planning, maps in the scales ranging between 1:20000 and 1:5000 are found to be the best. Reproductions of air photographs can be used as base maps at the local level land use planning, since field workers and experience can show that local people can recognize where they are in the photos.

12.2.5 Land Use in Relation to Sectoral and Development Planning

Land use planning is non-sectoral by definition but, unless a special planning authority is set up, a land use plan must be implemented by sectoral agencies - in agriculture, forestry, irrigation, etc. Implementation will call for help from the different extension services.

There will be no clear boundary between land use planning and other aspects of rural development. For example, a desirable change in land use may be the introduction of a cash crop. Successful management may require the use of fertilizers. This cannot be done unless there are local centres for fertilizer distribution, effective advice on its use and a system of credit for its purchase.

Local services will be of no use without an adequate national distribution system and the sufficient manufacture or allocation of foreign currency for imports. Building a fertilizer factory and organizing national distribution are not part of land use planning but they may be essential for the success of planned land use. On the other hand, the siting of local distribution centres in relation to population and suitable land could well be part of the work of a land use planner.

Hence, there is a spectrum of activities ranging from focus on the interpretation of the physical qualities of the land for which the land use planner will be largely responsible to activities that need a combined input with other technical specialists. Furthermore, where matters of national policy such as adequate prices for crops are prerequisites for successful land use, the job of the planner(s) is to mention it clearly.

12.2.6 People Involved in Land Use Planning

Land use planning involves getting many different people to work together towards common goals. The following three groups of people are directly involved:

Land Users: These are the people living in the planning area whose livelihood depends wholly or partly on the land. They include not only the farmers, herders, foresters and others who use the land directly but also those who depend on these people's products such as operators in crop or meat processing, sawmills and furniture factories. The involvement of all land users in planning is very essential. Ultimately, they have to put the plan into practice and must therefore believe in its potential benefits as well as in the fairness of the planning process.

The experience and determination of local people in dealing with their environment are generally the most neglected in spite of being the most important resource. People will grasp development opportunities that they themselves have helped to plan more readily than any other schemes that are imposed on them. Without the support of local leaders, a plan is not likely to succeed.

Achieving effective public participation in planning is a real challenge. Planners have to invest the time and resources needed to secure participation through local discussions, by broadcasting and newspaper articles, through technical workshops and extension services. Imagination, a sincere interest in people and the land as well as a willingness to experiment mark the more successful efforts by the land users.

Decision Makers: Decision makers are those responsible for putting plans into effect. At the national and district levels, they will usually be government ministers. At the local level, they will be members of the local self-government or other authorities.

Generally, the planning team provides information and expert advice. The decision makers guide the planning team on key issues and goals while also deciding whether to implement plans and if so, which of the options presented needs to be chosen. Although the leader of the planning team oversees the day-to-day planning activities, the decision maker(s) should be involved at regular intervals. Decision makers also have a key role in encouraging public participation through their willingness to expose their decisions and the way they are reached to public scrutiny.

Land Use Planning Team: An essential feature of land use planning is the treatment of land and land use. This involves crossing boundaries between disciplines like natural resources, engineering, agriculture, and social sciences. Therefore, teamwork is essential. Ideally a team needs a wide range of special expertise such as a soil surveyor, a land evaluation specialist, an agronomist, a forester, a range and livestock specialist, an engineer, an economist, and a sociologist.

Such a range may be available only at the national level. At the local level, a more typical planning team may consist of a land use planner and one or two assistants. Each member must tackle a wide range of jobs and will subsequently need specialist advice. Government agency staff and universities may be useful sources of such advice or assistance.

12.3 Applications of Remote Sensing and Geographical Information System (GIS) in Watershed Planning

Remote sensing and GIS are two of the important modern tools which have many applications in watershed planning. In this section, the remote sensing applications in watershed planning are discussed followed by the GIS applications.

RADAR (i.e., Radio Amplification Detection and Ranging) is used in the enhanced meteorological collection of data such as wind speed and direction within weather systems. By measuring the bulges of water caused by gravity, features on the seafloor to a resolution of about a mile are mapped. By measuring the height and wavelength of ocean waves, the altimeters measure wind speeds and direction and surface ocean currents and directions. Light detection and ranging (LIDAR) are used to detect and measure the concentration of various chemicals in the atmosphere, while airborne Heights of objects and features on the ground can be measured more accurately by LIDAR than radar technology.

Remote sensing of vegetation cover is a principal application of LIDAR. Simultaneous multispectral platforms such as the images from the Landsat remote sensing satellite have been in use since the 1970s. Maps of land cover and land use from thematic mapping can be used to find minerals, detect or monitor land usage and deforestation and examine the health of indigenous plants and crops, including entire farming regions or forests.

Within the scope of the combat against desertification, remote sensing allows to follow up and monitor risk areas in the long term, to determine desertification factors, to support decision-makers in defining the relevant measures of environmental management and to assess their impact on watershed planning. After the successful launching of India's remote sensing satellites viz., Bhaskara 1 and Bhaskara 2 in 1979 and 1981, respectively, India began developing an indigenous Indian Remote Sensing (IRS) satellite program to support the national economy in the areas of agriculture, water resources, forestry and ecology, geology, watersheds, marine fisheries and coastal management.

The Indian Remote Sensing satellites are the mainstay of National Natural Resources Management System (NNRMS) for which Government of India's (GoI) Department of Space (DOS) is the nodal agency, providing operational remote sensing data services. Data from the IRS satellites are received and disseminated. With the advent of high-resolution satellites, new applications in the areas of urban sprawl, infrastructure planning and

other large-scale applications for mapping have been initiated. Remote sensing applications in the country, under the umbrella of NNRMS, now cover diverse fields within the domain of watershed planning and management such as pre-harvest crop area and production estimation of major crops, drought monitoring and assessment based on vegetation condition, flood risk zone mapping etc.

GIS has been widely used in characterization and assessment studies which require a watershed-based approach. Basic physical characteristics of a watershed such as the drainage network and flow paths can be derived from readily available Digital Elevation Models (DEMs) and data such as the United States Geological Survey's (USGS) National Hydrography Dataset (NHD) Program. This, in conjunction with precipitation and other water quality monitoring data from sources such as the Environmental Protection Agency's (EPA) BASINS (i.e., Better Assessment Science Integrating Point & Non-point Sources) database and USGS, enhances development of a watershed action plan and identification of existing and potential pollution problems in the watershed.

Data gathered from Global Positioning System (GPS) surveys and from environmental remote sensing systems can be fused within a GIS for a successful characterization and assessment of watershed functions and conditions.

13.3.1 Management Planning

When faced with challenges involving water quality and quantity due to natural as well as human-induced hazards (e.g., droughts, hazardous material spills, floods, and urbanization), planning becomes extremely important so as to mitigate their impacts and ensure optimal utilization of the available resources. Information obtained from characterization and assessment studies, primarily in the form of charts and maps, can be combined with other datasets to improve understanding of the complex relationships between natural and human systems as they relate to land and resource use within watersheds. GIS provides a common framework [i.e., spatial location] for watershed management data obtained from a variety of sources. Because watershed data and watershed biophysical processes have spatial dimensions, GIS can be a powerful tool for understanding these processes and for managing potential impacts of human activities.

The modeling and visualization capabilities of modern GIS, coupled with the explosive growth of the Internet and the World Wide Web, offer fundamentally new tools to understand the processes and dynamics that shape the physical, biological and chemical environment of watersheds.

The linkage between GIS, the Internet, and environmental databases is especially helpful in planning studies where information exchange and feedback on a timely basis is very crucial and more so when there are several different agencies and stakeholders involved.

13.3.2 Watershed Restoration (Analysis of Alternative Management Strategies)

Watershed restoration studies generally involve evaluation of various alternatives and GIS provides the perfect environment to accomplish that efficiently and accurately. GIS has been used for restoration studies ranging from relatively small rural watersheds to heavily urbanized landscapes. Coupled with hydrodynamic and spatially explicit hydrologic/water quality modeling, GIS can assist in unified source water assessment programs including the total maximum daily load (TMDL) program. As an example, alternatives for restoring a waterbody or a watershed can be studied by creating digital maps that show existing conditions and comparing them to maps that represent the alternative scenarios. GIS can also provide a platform for collaboration among researchers, watershed stakeholders, and policy makers, significantly improving consensus building and offering the opportunity for collaborative work on interdisciplinary environmental policy questions. The integrating capabilities of a GIS provide an interface to translate and emulate the complexities of a real-world system within the confines of a digital world accurately and efficiently.

13.3.3 Watershed Policy Analysis and Decision Support

The field of watershed science, particularly watershed planning, is experiencing fundamental changes that are having profound impact on the use of computer-based simulation models in resource planning and management. On one hand, the dramatically increased availability of powerful, low-cost, and easy-to-use GIS software, and more extensive spatially referenced data, are making GIS an essential tool for watershed planning and management tasks. However, with this increased use has come an increased realization that GIS alone cannot serve all the needs of planning and managing watersheds. This realization has renewed resource planners' interest in development of decision support systems that combine GIS, spatial and non-spatial data, computer-based biophysical models, knowledge-based (i.e., expert) systems, and advanced visualization techniques into integrated systems to support planning and policy analysis functions. As a component of a spatial decision support system, GIS provides very powerful visualization facilities for display and manipulation,

giving immediate intuitive evaluation capabilities to which, a wide range of non-technical users and decision makers can relate to.

GIS can assist the decision maker in dealing with complex management and planning problems within a watershed, providing geo-processing functions and flexible problem- solving environments to support the decision research process.

Let us sum up

Land-use planning is the systematic assessment of land and water potential, alternatives for land use and economic and social conditions in order to select and adopt the best land-use options. Its purpose is to select and put into practice those land uses that will best meet the needs of the people while safeguarding the resources for the future. The driving forces in land use planning are the needs for change, improved management or quite different patterns of land use dictated by the changing circumstances. Goals of land use planning define what is meant by the "best" use of the land. They should be specified at the outset of a particular land use planning project. Goals may be grouped under the following three headings of efficiency, equity & acceptability and sustainability. Land use planning can be applied at three broad levels: national, district and local. These are not necessarily in that order. They correspond to the levels of government at which decisions about land use are taken.

Glossary

Land Users: These are the people living in the planning area whose livelihood depends wholly or partly on the land.

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RADAR: Radio Amplification Detection and Ranging

Check your progress

1. What is a Watershed?

The body of water (e.g., rivers, lakes, ponds, streams, and estuaries) has a watershed. The watershed is the area of land that drains or sheds water into a specific receiving waterbody, such as a lake or a river. As rainwater or melted snow runs downhill in the watershed, it collects and transports

sediment and other materials and deposits them into the receiving waterbody.

2. What is Watershed Management?

Watershed management is a term used to describe the process of implementing land use practices and water management practices to protect and improve the quality of the water and other natural resources within a watershed by managing the use of those land and water resources in a comprehensive manner.

3. What is Watershed Management Planning?

Watershed management planning is a process that results in a plan or a blueprint of how to best protect and improve the water quality and other natural resources in a watershed. Very often, watershed boundaries extend over political boundaries into adjacent municipalities and/or states. That is why a comprehensive planning process that involves all affected municipalities located in the watershed is essential to successful watershed management.

Suggested Readings

1. Elango, L., Jayakumar, R., 2001. Modeling in Hydrology, UNESCO, and New Delhi.
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Unit 13

Water harvesting, Afforestation, Population and Livestock Development

Structure

Learning Objectives

Overview

13.1 Need for Water Harvesting Structures

13.2 Design of Tanks and Ponds for Water Harvesting

13.1.1 Classification of Tanks

13.2 Afforestation

13.2.1 Conservation and Development Strategies

13.2.2 Significances

13.2.2 National Afforestation and Eco-Development Board (NAEB)

13.3 Population and livestock development

13.3.1 Management of common property resources (CPR)

13.3.2 Reduced migration

13.3.3 Women empowerment

13.3.4 Impact on landless community and livelihood improvement

13.3.5 Employment Generation

13.3.6 Poverty Alleviation

13.3.7 Improvement in Standard of living

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to understand the

- Need for Water Harvesting Structures
- Afforestation: Conservation and Significances
- Population and livestock development

Overview

As the world faces an increasingly critical need to address climate change, the impact that water conservation has on a sustainable environment is undeniable. Groundwater is the primary source of freshwater that caters to the demand of ever-growing domestic, agrarian and industrial sectors of the country. In this context, adopting rainwater harvesting and recharging groundwater is one of the simplest and best measures in conserving water globally. This practice can efficiently be implemented in lieu of traditional water supplies that are currently on the verge of tapping out.

Forests and trees play crucial roles in hydrological processes in watersheds. Forested mountains and upland watersheds supply an estimated 70 percent of the world's accessible freshwater resources for domestic, agricultural, industrial and ecological needs. In this unit you will learn about water harvesting, afforestation, population and livestock development in watershed development.

13.1 Need for Water Harvesting Structures

Although we generally get a lot of rain in India, we do not have adequate water at all the times and at all the places. The reason for this is the lack of the realization of the value of each rain drop. The average annual rainfall in India is 1,170 mm, which is higher than all the continents except South America as well as the global annual average of around 850 mm. However, the rainfall in India occurs during short spells of high intensity. Because of such high intensities and short durations of heavy rain, most of the rainfall in India tends to flow away rapidly leaving very little amount for groundwater recharge. This results in most parts of India experiencing water scarcity even for domestic uses.

Ironically, Cherrapunji in Meghalaya –historically regarded as the wettest place on earth with a normal annual rainfall of over 11,000 mm, also suffers from acute drinking water shortage. This is because rainwater is not conserved adequately and is allowed to drain away. Thus it does not matter how much rain we get, if we don't harvest or capture it.

This highlights the need to implement water harvesting through appropriate structures, to ensure that the rain falling over a region is conserved as fully as possible. This can be done by either storing it for direct use on/below/above ground or recharging it into the groundwater.

13.2 Design of Tanks and Ponds for Water Harvesting

Percolation Tank: A percolation tank is generally constructed in low level wasteland or a small drain. It has well defined catchment and the water spilling over is diverted to a nearby natural drain. It consists of earthen embankment and an overflow type masonry waste weir. Permeable formation in the reservoir bed is an essential requirement of percolation tank. The tank acts as storage of intercepted runoff, which percolates down to phreatic aquifer creating a recharge mound. The percolation time depends upon the permeability of bed formation. Normally it is expected that between two consecutive rain spells, most of the storage percolates down. Thus, during 2 to 3 rainfall cycles, the actual recharge gain is two to three times the storage capacity of the tank. The shape and size of the recharge mound depends on the nature of phreatic aquifer underlying the surface. This type of recharge structure is useful in area having sandstone and limestone formation underlying.

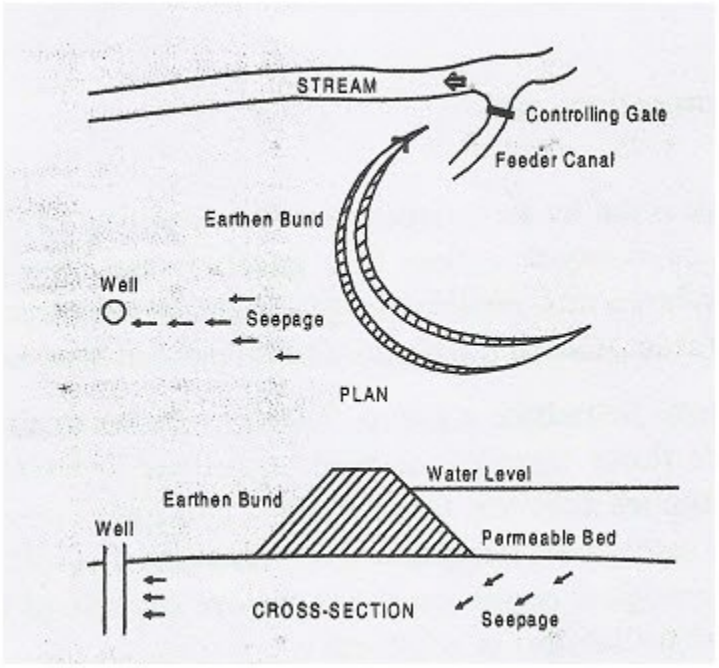


Fig. 13.1. Schematic Diagram of a Percolation Tank. (Source: Patel and Shah, 2008)

Tank is a general term used for surface water storage of moderate size. The storage may have come into being due to interception of rainwater in a natural depression or a manmade excavation. Such water bodies are popularly called ponds. Alternatively, water storage may be done by closing the openings of a natural saucer shaped landform by constructing bund sized embankments. The storage so created is called a tank. The tank bunds are mostly constructed with earth to keep cost of construction low and are commensurate with the benefits envisaged.

Water storages of large size are not called tanks, but they are referred to as reservoirs. Such reservoirs are formed in the river valleys by constructing a barrier or a dam using masonry, concrete or earth depending upon site conditions. Technically, bund is a miniature form of a dam.

Due to simplicity in construction, it was a very popular mode of conserving rainwater. In South India where rivers are monsoon fed, tanks assume special importance. In the plains of Uttar Pradesh, West Bengal, and Orissa as also in the plateaus of Madhya Pradesh, Chhattisgarh tanks have been extensively practiced.

13.1.1 Classification of Tanks

Bunds which are generally less than 12 m high, results in tanks. From the consideration of height of bunds, up to 4.5 m high bunds generate a small tank. Medium sized tanks are formed by bunds up to 9 m height. It may however be noted that, this classification is very approximate because the shape and size of the tank is not dependent on the height of the bund alone. It is equally influenced by the topographical features of the region.

Network of Tanks

The tank system may exist with each tank as a separate entity or in the form of a group of tanks in a series or tanks with inter-connection. In a tank system, following types of networks exist:

(i) Isolated tanks;

(ii) Tanks with inter-connections; and

(iii) Tanks in series.

Isolated Tanks:

When a tank is fed by an independent free draining catchment and when the surplus flows do not form network inflow into another tank, the tank system is called an isolated tank system. Mostly large and medium sized

tanks are constructed as isolated tanks with independent catchment area. Also, in the plains and on plateau lands, tanks exist in isolation.

Tanks with Inter-Connections:

Sometimes a group of tanks may be so situated that they could be inter-connected to receive flows through, as well as deliver flows to other tanks in the group severally. It thereby implies that the tanks have a combined catchment. Any surplus water received by a tank from the catchment lying above it is transferred to other tanks. Depending upon the prevailing hydro-meteorological conditions, the tanks are capable of feeding either each other or one another. Thus, optimum water utilization and storage is achieved.

Tank in Series:

Such tanks are located alongside the river drainage channels. They are fed by inflow drains and serviced by escape or outflow drains. The tanks in upper reaches get their supplies from the catchment through inflowing drains. It then lets its surplus flow down through an escape or outflow drain, which contributes to the inflow of the tank lower down in the series. Thus, while the uppermost tanks have substantially free catchment, the tanks lower down have limited free catchment falling between two tanks. The tanks lower down in the series get inflows immediately after rainfall from their free draining catchments. But supplies from already intercepted catchments are received only after the upper tanks get filled. The advantage of this system is that surplus water from the upper tank is picked up by the lower tanks rather than allowing any wastage.

However, there is a safety related disadvantage. In case of breach in the upper tank, lower tanks also become prone to severe flooding endangering the safety of the tank bunds. To avoid this, breaching sections are provided at appropriate locations in each tank.

Water Harvesting by Ponds

The traditional practice of collection rainwater in village ponds suffers from following limitations:

- Large open surface is subjected to high evaporation losses.
- Large bed area is subjected to high seepage losses.

In Spite of these limitations in the village ponds, the bulk water requirement for about half the year is met from them. Small groundwater recharge mound formed under the pond bed used to partially supply through wells dug in the pond bed. This traditionally practiced rainwater harvesting

structures have served water supply requirements since ages. During the scarcity years, some deepening of ponds is made through de-silting. But if it is done in a haphazard manner, it does not help much and rather it may lead to other problems like water logging, mosquito breeding, etc.

Pond Lining

The limitations of traditional ponds, especially the quality deterioration and seepage losses could be completely stopped by using plastic lining. This simple technique has proved to be effective. The evaporation loss could be checked by use of chemical retardants and adopting system of multiple ponds system (i.e., compartments). Limitation of the depth factor must be accepted and required storage could be built by increasing the length and breadth of the pond. Of course, in places where salty water is available is at greater depths, the pond storage can be increased even with less open surface and thereby reducing the evaporation losses. Such favorable locations may not be provided with plastic lining, provided the rate of salt contamination from the sediments is within the limits. Plastic lining is a rather costly proposition; however, it envisages better use of local water resources and regular maintenance is almost negligible.

This technology involves lining of the walls and floor of the pond, tank reservoir with tough, wide-width low-density polyethylene (LDPE) film. These LDPE films are available in widths of 4 to 12 m and thickness of 100 to 250 microns. These films meet specifications as per the Bureau of Indian Standards Code IS: 2508 - 1984. This film has excellent water barrier properties, very good blend of physical properties such as tensile impact strength coupled with good weathering capability and chemical resistance properties. These films also prevent the inherent salinity of the soils and saline groundwater from seeping into the pond or tank and start contamination.

The construction of plastic lined pond includes excavation work, screening work for removing big boulders and sharp-edged gravels which could damage the plastic film, dressing of sides and beds so that lining is not punctured, laying plastic film on beds and sides, brick lining on the sidewalls, soil filling on the floor, inlet system and distribution system. By 2008, 19 villages of Bhal area of Gujarat, India had 20 such plastic lined ponds.

13.2 Afforestation

Afforestation is the preferred measure for policymakers to solve a range of ecological problems, as it is generally believed that forests help to improve

soil health and quality, reverse land degradation, halt land desertification, and provide clean water through an intensified water cycle.

13.2.1 Conservation and Development Strategies

Forest conservation and development are primarily accomplished through three strategies: afforestation via natural/artificial regeneration, protection, and management. The Ministry of Environment is implementing three major schemes for forest development: the National Afforestation Programme (NAP), the National Mission for a Green India (GIM), and the Forest Fire Prevention and Management Scheme (FFPM). While NAP focuses on reforestation of degraded forest lands, GIM focuses on improving forest quality and increasing forest cover, as well as cross-sectoral activities on a landscape scale.

The FFPM is in charge of forest fire prevention and management. For scientific forest management, states prepare a management plan known as a Working Plan, which highlights various activities to be undertaken in a forest division for effective forest management. To achieve the targeted objectives of national afforestation programmes, the Ministry has approved the merger of NAP into Green India Mission (GIM), and accordingly financial allocation for both schemes is provided under one budgetary head, thereby augmenting overall greening efforts. Funds collected as compensatory levies from states under the Compensatory Afforestation Fund Management and Planning Authority (CAMPA) are used in plantation activity, including compensatory afforestation by states/UTs.

13.2.2 Significances

NAP and GIM, along with other afforestation schemes, have helped to restore degraded forest areas throughout the country. This has contributed to the stabilisation and expansion of forest cover, as evidenced by the India State of Forest Report (ISFR) published by the Forest Survey of India (FSI).

According to the most recent India State of Forest Report (ISFR 2021), the country's total forest and tree cover is 8,09,537 square kilometres (24.62 percent of the country's geographical area), up from 7,94,245 square kilometres (24.16 percent) in ISFR 2015. This represents an increase of 15,292 square kilometres in the country's forest and tree cover. The ISFRs also stated that the positive change could be attributed to conservation measures or management interventions such as afforestation activities, participation of local people for better protection in both plantation and traditional forest areas, expansion of trees outside of forests, tree plantation drives, and so on.

Through the School Nursery Yojana and the Nagar Van Yojana, the Ministry also promotes tree planting as a people's movement. Tree plantation/afforestation is a cross-sectoral effort undertaken by various departments of the Centre and State Governments, Non-Governmental Organizations, civil society, corporate bodies, and others to achieve the targeted objectives of national afforestation programmes.

13.2.2 National Afforestation and Eco-Development Board (NAEB)

In August 1992, the National Afforestation and Eco-development Board (NAEB) was established by the Ministry of Environment and Forests. Through a participatory planning process called Joint Forest Management and microplanning, the National Afforestation and Ecodevelopment Board has developed specific schemes for promoting afforestation and management strategies. That assists the states in creating tailored afforestation and management strategies and eco-development packages for increasing biomass production.

Promoting afforestation, tree planting, ecological restoration, and eco-development initiatives across the nation is the responsibility of the National Afforestation and Eco-Development Board (NAEB). It lays emphasis on environmentally vulnerable regions including the Western Himalayas, Aravallis, and Western Ghats, as well as degraded forest areas, lands next to degraded forest areas, national parks, sanctuaries, and other protected places.

13.3 Population and livestock development

13.3.1 Management of common property resources (CPR)

Common Property Resources refer to the resources in which all the villagers have equal rights. The villagers maintain, protect and enjoy the usufructs with equal rights and responsibility. They, however, do not have any legal right over the resources. Several common property resources are developed under watershed development programme such as pastures, development or renovation of water bodies, plantations in common land. While these resources are being developed these provide employment to the folks and once, they are developed it contributes directly towards livestock management and non-farm activities besides agricultural production. The Common Property Resources are analyzed and presented here to understand the position regarding the way the community land is used for fodder, fuel wood, etc., their maintenance and the role played by the community with regard to CPR.

13.3.2 Reduced migration

The latent aim of this project was also to reduce migration and generate sufficient employment opportunity. Migration had completely stopped during the project implementation stage. Since employment opportunities in form of labor were available. But after implementation, though productivity and income has increased to some extent, it was unable to reduce or stop migration altogether. Promotion of non-farm sector activities like dairy, poultry, goatry was found to be negligible, which could have provided some employment opportunities. Tree based farming or agroforestry and horticulture can also form a major source to provide employment opportunity and reduce migration. But the projects attempted it on a meager scale. Attempt to mitigate migration is required on a major scale.

13.3.3 Women empowerment

Empowerment is a multi-faceted, multi-dimensional and multi-layered concept. Women empowerment is a process in which women gain greater share of control over resources such as material resource, human and intellectual resource, information, and financial resources amongst others. According to the Country Report of Government of India, "Empowerment means moving from a position of enforced powerlessness to one of power". Since empowerment is a latent phenomenon and cannot be measured directly so aspects like participation, mobility, voice in decision making in home, community, society were taken into consideration.

13.3.4 Impact on landless community and livelihood improvement

Landless community should not be ignored in the developmental process. These marginal families can be part of indirect benefits and can be included in the users groups, SHGs and other institutions. Most of them are part of SHGs but not included the users groups in the project area. It seems that watershed activities have not improved conditions of landless community significantly. Apart from some minor labour work, there has not been much to improve their livelihood.

13.3.5 Employment Generation

According to the watershed guidelines, livelihood promotion is a very important outcome of the project. Under this study, additional employment as labor days in considered as a parameter for livelihood promotion.

13.3.6 Poverty Alleviation

The data with respect to poverty alleviation as directly observed from the field was only made available in the state of Assam. According to which there was reduction of poverty level from 45 percent to 36 percent in the project area. The income generating activities carried out under the project helped in providing some regular income. Farrington et al. (1999) also observed that the impact of these projects on poverty alleviation and the long-term sustainability was however less clear. Even though the results indicate that successful projects have in fact reduced rainwater runoff and recharged ground and surface water aquifers, improved drinking water supply, increased the irrigated area, changed cropping patterns, crop intensity and agricultural productivity, increased availability of fuel and fodder, improved soil fertility and changed the composition of livestock.

13.3.7 Improvement in Standard of living

Successful implementation of the watershed programme is realized in the fact that it brings more lands under cultivation, improves the quality of the land thereby the productivity. All the positive impacts of WSD are expected to culminate in an improved standard of living at the household level. People are 43 able to get some regular income, perhaps some additional income which leads to additional expenditure. Raised income enables a better life in terms of better food, clothes, education, health, more spending at the time of festivals and marriages, physical assets and amenities acquired. Findings of the study revealed that most of the households acquired.

Let us sum up

Harvesting and collection of rainwater is an adequate strategy that can be used to address the problem of water crisis globally. The use of a rainwater harvesting system provides excellent merits for every community. This simple water conservation method can be a boost to an incredible solution in areas where there is enough rainfall but not enough supply of groundwater. It provides the most sustainable and efficient means of water management. Watersheds should be understood as dynamic systems characterized by diverse interactions and spatial relations between humans and the environment that manifest as mosaics of different land-use systems. Successful implementation of the watershed programme is realized in the fact that it brings more lands under cultivation, improves the quality of the land thereby the productivity.

Glossary

Water harvesting is the activity of direct collection of rainwater. The rainwater collected can be stored for direct use or can be recharged into the groundwater. Rain is the first form of water that we know in the hydrological cycle, hence is a primary source of water for us.

Afforestation is the planting of saplings and trees to increase forest cover. This is a measure taken to protect and conserve our environment from the harmful effects of pollution and human development.

Check your progress

1. What are the Advantages of implementing rainwater harvesting?
    - Reduced Water Bills
    - Ecological benefit
    - Reduces erosion and flooding around buildings.
    - An adequate means for Irrigation purpose
    - Reduces demand on Ground Water
2. What are the techniques of rainwater harvesting?

There are two major techniques of rainwater harvesting.

- Surface runoff harvesting
- Groundwater recharge

Suggested Readings

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Block 5:

Development and Management of Watershed

Unit 14:	Participatory Rural Appraisal in Watershed Programme
Unit 15:	Empowerment of Women and Gender issues
Unit 16:	Problems and prosperity of watershed management.

Unit 14

Participatory Rural Appraisal (PRA) in Watershed Programme

Structure

Learning Objectives

Overview

14.1 Participatory Rural Appraisal and Peoples Participation

14.2 Basic Principles and Fundamentals of PRA

14.2.1 Basic Principles

14.2.2 Fundamentals of Participatory Rural Appraisal

14.2.3 Assumptions and Basics of Participatory Rural Appraisal

14.2.4 The PRA need to include the following basic approaches:

14.3 Linkage between People and Policy Makers

14.3.1 Tips for PRA Practitioners

14.3.2 Myths of PRA Techniques

14.4 Different Tools Employed in PRA

14.4.1 Social and Resource Mapping

14.4.2 Essential Data Collection in a Tabular Form

14.4.3 Ranking Matrix

14.4.4 Historical Time Line

14.4.4 Transect or Group Walk

14.4.5 Seasonal Analysis

14.4.6 Venn Diagram

14.4.7 Indigenous Technical Knowledge

14.4.8 Women's Participation

14.4.9 Participation of NGO

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to understand the following,

- Participatory Rural Appraisal and Peoples Participation
- Basic Principles and Fundamentals of PRA
- Linkage between People and Policy Makers

Overview

A close relationship exists between resources such as land, water, forest and minerals and the community, particularly living in the rural areas of the watershed. Therefore, participation and awareness of the community about the development and management program of the watershed are very important. Two participatory and learning action techniques are adopted for community. These are rapid rural appraisal (RRA) and participatory rural appraisal (PRA). These techniques have enabled local people to express, share, emphasize and examine their knowledge. RRA had been in practice till the late 1970s and 1980s. Some limitations and flaws have been observed in participation of village community in RRA. This is due to some outsiders who used to enter the village area to obtain data from the village people and thus they finally become the central executing members. In the late 1980s, PRA technique evolved. In PRA, investigators or members are all villagers. They control the whole project. They are learners, catalysts, and facilitators. They do all the work of mapping, diagramming, viewing, and analyzing. They identify the priorities and give shape to their information, knowledge, attitudes, and aptitudes. Thus, their efforts become a creative approach to information sharing and a challenge to prevailing preconceptions about the rural people. Mukherjee (1993, 1995, and 1997) presented ideas on participatory rural appraisal methodology, PRA analysis through questionnaire survey and PRA on national resources.

14.1 Participatory Rural Appraisal and Peoples Participation

Participatory rural appraisal is already defined in the Introduction Section earlier. The people who participate in watershed management are villagers, farmers, and common people. They are the participants, beneficiaries and promoters of any development works in the watershed. Their full cooperation and participation are at the root of the success of any project. They may participate in different modes. According to Pretty (1988), these participation modes are as follows:

- Passive Participation: It is the indirect participation of people in the event

which is going to happen or has already happened.

- Participation to Supply Information: This is the people's participation to supply information by answering questions through questionnaire, survey(s) or other methods.
- Participation through Consultation: People participate through consultation and the agencies who hear the people's views may modify the program as per the views of the people.
- Participation for the Material Incentive: This is the participation of people by providing resources such as labour in return for food, money and other material benefits.
- Functional Participation: People participate by forming groups to meet the pre-determined objectives related to a project.
- Participation through Interaction: People participate with the implementing agency through interaction.
- Participation by Self-Mobilization: People participate in self-initiated mobilizations and corrective actions.

The project implementing agency should keep in mind the community's participation for the successful completion of the watershed management project(s).

14.2 Basic Principles and Fundamentals of PRA

14.2.1 Basic Principles

PRA is a reversal of learning. It is an informal way of learning from the local, physical, technical, social and psychological knowledge of the people. PRA is a way to understand and analyze the peoples' living conditions, to share the outcomes and to plan for their activities.

PRA is conducted to establish rapport with the people. It also aims to identify and define their problems for prioritization in the village itself. PRA is the technique of immediate analysis and survey of village resources, based on principle of listening and progressive learning. Thus, the main principle is to gather information about the villagers, their willingness to participate and resources of the area through patient listening and interaction.

14.2.2 Fundamentals of Participatory Rural Appraisal

- Sharing: It is the sharing of information, ideas, knowledge, and experience between facilitators (i.e., policy makers) and villagers (i.e., stake-holder population).

- Villagers as Performers: The facilitators should initiate a process so that villagers can work as performers, taking up the task of facilitating investigation, analysis, presentation, and learning.
- Self-Critical Awareness: Facilitators examine critically and continuously their own behavior.
- Personal Responsibility: Personal responsibility should be taken up by the facilitators for what is done rather than relying on the authority or authorities for the rigid set of rules.
- Maximizing Diversity: By ensuring maximum diversity, information is enriched. It is essential to notice and investigate the differences, contradictions, and anomalies. The objective should be to seek variability rather than objectives.
- Triangulation: It is the process of cross-checking and progressive approximation of truth. Here, investigators assess from findings from different methods, places, times and disciplines.

14.2.3 Assumptions and Basics of Participatory Rural Appraisal

PRA is based on the following assumptions.

- It is assumed that it is quite possible and desirable to involve the local community in the development projects of the watershed.
- It is also assumed that active participation of the local people can be increased with time in the ongoing works.
- It is assumed that learning from the local people is possible.
- It is assumed that informal approaches and discussions with local people are more effective as the projects progress.
- In the execution of the project, multidisciplinary teams are more effective in completing the work in time smoothly.
- The issues that may be involved in the developmental works should be investigated from different perspectives with the help of different approaches.
- The circumstances and systems can be explored instead of adhering to statistical findings.

14.2.4 The PRA need to include the following basic approaches:

- Due respect for behavior, attitudes, aptitude and knowledge of the village people should be given.
- Facilitators should have full confidence on the ability of the community

to do things.

- There exists a lot of scope for learning from the community.
- Facilitation to the community should be recommended to encourage them to do all the investigations, planning and analysis.
- The community should be empowered to own the outcome as an incentive.
- Information and field experience are to be collected and shared by both the facilitators and the community.

Thus, in the basic approaches of PRA, more emphasis is given to the establishment of a cordial relationship between the community and the facilitators. It is also attempted to let the community feel more empowered during the whole process. Local people should be involved as active agents.

14.3 Linkage between People and Policy Makers in Watershed Management

To ensure an effective linkage between the people and policy makers in watershed management, the tips listed below should be followed by the PRA practitioners. At the same time, the myths of PRA listed below need to be properly understood.

14.3.1 Tips for PRA Practitioners

- For the successful completion of the watershed projects, project implementation agency (PIA) and the community should act as a unified team to handle all areas.
- A checklist should be made every day before going to the field. This may help in encompassing techniques and progress of the work.
- The facilitators should keep time for the use of PRA processes and techniques in the field.
- Participatory rural appraisal techniques should be applied to different parts of the watershed. This may facilitate cross-checking and triangulation. This will further help in rapport building among different sections of society.
- It is better to explain the objective(s) and methodology in detail to the group before starting the PRA techniques.
- Participatory rural appraisal is a continuous process. Hence, its techniques should be in the first 4 years of the watershed development.

- There are various participatory rural appraisal techniques and tools. A judicious selection of the technique and tool for a particular project helps to produce better results of the work.
- PIA members and community should be accommodative and innovative in handling the available, suitable and adoptable techniques and tools.
- The team building culture in the community is to be promoted. Identification of self-help groups (SHGs) and user groups (UGs) should be an automatic outcome.
- Approach to a project should be flexible to suit the needs and demands of the community.
- The community should control the techniques and tools so that they can modify, rectify, evolve, and include relevant aspects of their own.
- The community members may be allowed to start the work in the morning hours at their convenience.
- Community should not feel left out or ignored. Regular consultation and facilitation should be made with them.
- It is advisable to select a permanent and spacious place in the village for discussion / meeting between villagers and PIA members.

14.3.2 Myths of PRA Techniques

Participatory rural appraisal is a simple process, yet the facilitators should be aware of the myths of the following PRA techniques.

- “It is quick” means all stakeholders need to devote time to each technique.
- It is easy when skills of communication, facilitation, conflict, and negotiation are imparted.
- “Anyone can do it” when one gives insight into various organizational management methods.
- It is a fancy if one is aware of the complicated and unnecessary innovations and makes procedure, process, and outcomes very simple.
- It has no theoretical basis. Participatory rural appraisal is based on action and research approach.
- It is just an old wine in a new bottle. PRA techniques are flexible and hence their innovations and modifications may be promoted.

- Training is necessary. But complex training to the members of PIA and the community should be avoided.
- People involved are neutral. People involved should be free from the influences of political and social biases.
- It is useful only for the need assessment. PRA is used for all-round need assessment of the watershed.
- It is universal. To respect heterogeneity, it is preferable to use and apply PRA techniques in different groups.

14.3.3 Benefits of Participatory Rural Appraisal

PRA offers lots of benefits to the community. The poor and weaker section of the community is empowered through PRA techniques. This section of the people can take actions on the need-based proposals. PRA plays an important role in improving the outlook of the community when they are allowed to involve in diverse fields of watershed management. PRA helps both the community and the PIA members in appraisal, identification, planning, implementation, monitoring and evaluation of all types of management works. Research priorities and initiation of participatory research are identified through PRA. PRA helps in indicating changes and modifications in the organization, and thus it becomes closer to community's aspirations. Policy reviews in the watershed programmes and management are possible in PRA.

14.4 Different Tools Employed in PRA

There are many tools and techniques employed under PRA. A few of the important tools are discussed here.

14.4.1 Social and Resource Mapping

In this tool, local people are involved in preparing the map of the village areas to show the village resources and different parts of the villages such as their living areas, wastelands, agricultural lands, grazing fields, forests, ponds, wells, fisheries, rivulets, streams or rivers, flood- and erosion-prone areas, schools, village libraries, development clubs, other public institutions, village roads, side drains, railway lines if any near the village and irrigation canal system in the agricultural fields. Village people draw the map to solicit their knowledge and participation. This is done by the villagers, and thus they gain confidence to participate in other development activities. Their map helps to explore and analyze the spatial information, particularly to study the problems and opportunities. The map can give a quick impression about the existing village setup.

Soil and Hydrology Mapping

This map is also prepared by the villagers in addition to the social and resource mapping. In this map, detailed data on soil and hydrology of the village such as types of soil in different areas of the map and hydrological information showing stream, rivulets and drains to carry the runoff produced by rainfall are shown.

14.4.2 Essential Data Collection in a Tabular Form

The village people collect all the essential data and present them in different tabular forms. These data are the most important tools to plan the watershed development project in the community as they give vivid pictures of the status of development. The data collected and presented by the villagers in different tabular forms include the population of the village, distribution of families on caste/class basis, family distribution according to land holdings, total employed persons in the village and categories of employment such as public/private etc., the number of qualified persons, total population of villagers and the number of cattle, goats, sheeps, etc. A table on the information about the miscellaneous items in the village such as its connectivity with pitched road, electrical lines, banks, telephone lines, mobile telephone towers, water supply lines, cable TV connections, and cooperative milk collection centres is also presented.

14.4.3 Ranking Matrix

Ranking matrix of preference is performed by villagers to know about their attitude, valuation, etc. to a particular item of daily livelihood. For example, rabi crop wheat is placed in the first rank in terms of its use as food, taste, market value, etc., and so wheat occupies the matrix ranking of 1. Similarly, if mustard is placed in the second highest position, it will be shown in the matrix as ranking of 2. This ranking matrix analysis is normally presented in a tabular form.

14.4.4 Historical Time Line

This is another tool to collect the information on community or village about their stages and trends of development with a historical time line. In other words, it is the information about the sequence of changes in the village with respect to social, economical, educational, agricultural and other aspects of living standards. It reveals the trends of periodical development. The older people are witness of this development by virtue of their age. These sections of elderly people may be contacted to know about the past history of the village, indicating systematically the period when those changes took place. The time line for agriculture has indicated occurrence of floods,

droughts, adaptation of new crop varieties, use of fertilizers and years of major crop production and crop failure. Similarly, timeline will indicate the year of irrigation facilities, years of major water harvesting tanks and ponds, years of major soil erosion in the rivers and hills, land degradation by landslides, water logging, etc.

14.4.4 Transect or Group Walk

The group walk is a process of participatory rural appraisal that involves travelling across the village from one corner to other along with villagers for verification of items supplied for the social and resource mapping. This travel or walk has a lot of purposes such as to have a clear concept on the farm practices, tree plantation, forest cover, wasteland, water table, ponds, water holding capacity of the soil, slope of land, common lands and land use, grassland, grazing fields, recreational grounds, available water resources and other resources. Thus, full picture of the village with existing facilities could be gathered and accordingly this tool paves the way for future planning.

14.4.5 Seasonal Analysis

Seasonal analysis is a tool to provide insight into rainfall pattern, average rainfall, extent of cultivated kharif and rabi crops, income of farmers in each season, seasonal functions in the village, seed sowing and harvesting periods, seasonal human and animal disease, etc. All these items and the time required to complete are indicated by the villagers in a simple table.

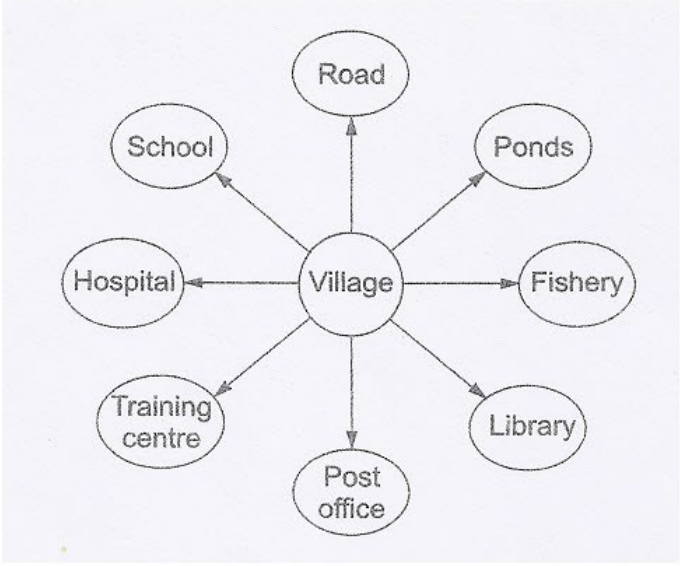


Fig. 14.1. A typical Venn diagram used in participatory rural appraisal (PRA). (Source: Das and Saikia, 2013)

14.4.6 Venn Diagram

A typical Venn diagram is shown in Fig. 25.1. It gives an approximate idea about some existing infrastructure of the village and their relevance to the village community. The size and distance of the circle from the centre indicates the importance of facilities to the village. The item in the biggest circle indicates the highest need. Thus, the Venn diagram is another tool to learn the existing infrastructure and their need for the community.

14.4.7 Indigenous Technical Knowledge

Indigenous technical knowledge (ITK) available in rural areas is one of the main tools in PRA exercise. Villagers are the storehouse of ITK. These people are rich sources of traditional wisdom/practices/skills, beliefs and local resources. The villagers can express their proverbs, idioms, drama, dance, local history, etc. through traditional wisdom. The ITK tool helps them to express their views. ITK can also be gathered for sharing the views with the community.

14.4.8 Women's Participation

Women's participation in the development of watershed management is considered as one of the major tools. Women in the villages take part in various activities such as agricultural production in the field, collection of fuel wood, fetching drinking water, preservation of seeds, milk processing, sharing the knowledge on traditional foods/medicines, besides their exclusive role in housekeeping. Women's committees may be formed in the village to contribute in a systematic way to the PRA activities.

14.4.9 Participation of NGO

Non-governmental organizations (NGOs) can be actively involved in the developmental works of watershed. NGOs can create awareness of watershed development education in the rural masses. They can also arrange field training for the villagers. They can evaluate and monitor the project activities. NGQs can act as facilitators towards watershed management efforts.

Let us sum up

A close relationship exists between resources and the community living in the rural areas of the watershed. Therefore, participation of the community in the development and management of the watershed is very important. The two techniques viz., rapid rural appraisal (RRA) and participatory rural appraisal (PRA), help the local people to express, share, emphasize and examine their knowledge. The RRA has become almost obsolete owing to

some of its major drawbacks. In late 1980s, participatory rural appraisal technique was evolved. In the PRA, villagers monitor the whole project. The participants in PRA are villagers, farmers and common people. They are the beneficiaries and promoters of any developmental works in the watershed. The principle of PRA is based on listening and progressive learning. So it is the technique of immediate analysis and survey of village resources. For the successful completion of the watershed project, a project implementation agency (PIA) is formed. This PIA and the community should act as a unified team to handle all areas. Both PRA and PIA can offer lots of benefits to different sections of the community. Various tools and techniques have been brought under PRA. Some of the important tools are social and resource mapping, soil and hydrology mapping, essential data collection in the tabular form, ranking matrix, historical timeline, seasonal analysis, transect or group walk, Venn diagram to show an approximate idea about the existing infrastructures, indigenous technical knowledge (ITK), women's participation in watershed project and participation of NGOs.

Glossary

Sharing: It is the sharing of information, ideas, knowledge, and experience between facilitators (i.e., policy makers) and villagers (i.e., stake-holder population).

Villagers as Performers: The facilitators should initiate a process so that villagers can work as performers, taking up the task of facilitating investigation, analysis, presentation and learning.

Self-Critical Awareness: Facilitators examine critically and continuously their own behavior.

Check your progress

1. What is the PRA technique in watershed management?

In the late 1980s, participatory rural appraisal techniques evolved. In the PRA, villagers monitor the whole project. The participants in PRA are villagers, farmers, and common people. They are the beneficiaries and promoters of any developmental works in the watershed.

2. What are the benefits of participatory appraisal?

Participatory Appraisal is a broad empowerment approach that seeks to build community knowledge and encourages grassroots action. It uses a lot of visual methods, making it especially useful for participants who find other methods of participation intimidating or complicated.

3. What is the concept of PRA techniques?

PRA is an assessment and learning process that empowers practitioners / people to create the information base they need for participatory planning and action. Outsiders contribute facilitation skills and external information and opinions. Many different tools have been developed for use in PRA.

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Unit 15

Empowerment of Women and Gender issues

Structure

Learning Objectives

Overview

15.1 Women Empowerment

15.2 Significant Participation

15.3 Gender Issues

15.4 Women-friendly Technologies

15.5 The Road ahead

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to learn the following,

- Women Empowerment and Significant Participation
- Gender Issues
- Women-friendly Technologies and The Road ahead

Overview

Agriculture is an engine of growth and poverty reduction worldwide and women are an integral part of it. In India, women's involvement in agriculture is complex and diverse. They participate in wide range of activities in farming. There is a considerable variation in women's participation across regions from north to south, from one socio-cultural group to another and even two production systems. As per census data, share of total women workers in agriculture was 71.8% in 2001 and has come down to about 65% in 2011. However, the multifarious nature of their involvement in farm and households makes it very difficult to assess the exact contribution both in

terms of magnitude and nature. At present reliable and relevant gender disaggregated data is absent.

15.1 Women Empowerment

Empowerment is a multi-faceted, multi-dimensional and multi-layered concept. Women empowerment is a process in which women gain greater share of control over resources such as material resource, human and intellectual resource, information, and financial resources amongst others. According to the Country Report of Government of India, "Empowerment means moving from a position of enforced powerlessness to one of power". Since empowerment is a latent phenomenon and cannot be measured directly so aspects like participation, mobility, voice in decision making in home, community, society were taken into consideration.

The analysis of this aspect reveals that women participation was not adequate. They were part of Self-Help Groups (SHGs), UGs, WCs, WAs also but it was nothing more than mere presence. Mere presence of women members on the watershed committee had no real impact as they were not effective in decision-making process in the committee. Women in SHG did not feel confident to interact with people, officers, panchayats, bank, etc. Only in case of Nagaland women seemed confident enough. A hearting change resulted from the awareness created is that the community has started paying attention to girl's education.

The World Bank defines empowerment as "the process of increasing the capacity of individuals or groups to make choices and to transform those choices into desired actions and outcomes. Central to this process is actions which both build individual and collective assets and improves the efficiency and fairness of the organizational and institutional context which govern the use of these assets." To enlist active participation of women and vulnerable groups targeted activities benefiting these groups economically are suggested.

Harnessing gender power by balancing activities for men and women, farmers and landless people was found effective to enhance the impact of community watershed programs. Thus, as the World Bank (2001) report confirms, societies that discriminate based on gender pay the cost of greater poverty, slower economic growth, weaker governance, and a lower living standard of their people.

15.2 Significant Participation

In the agricultural sector, women participate in several agro-production systems that govern the nature and extent of their involvement. Most

significant agricultural activities undertaken by women include farming, post-harvest management, horticultural crop production, livestock management, fisheries and homestead resources. In paddy, women are mainly involved in transplanting, weeding, harvesting, drying harvest, winnowing and seed storage. As AICRP Report, total workload is concerned, women spend 40.2 percent of their time per season, performing transplanting (39.1 hours), harvesting (29.8 hours) and weeding (19.0 hours) as the major activities.

In sugarcane based cropping system, women participate in activities like manure and fertilizer application at first step, preparation of sugarcane sets for sowing, placing these sets into the ridges, irrigation, weeding, harvesting, tying the bundles, carrying sugarcane bundles and loading it into the vehicle. Again, these are not women dominant or exclusive activities and are performed jointly with males. The data on role profile indicates that joint participation of rural women with men was higher than independent participation of women in all activity areas.

Gender analysis in various sub activities in livestock management reveals that women play significant role in milk production, processing, and marketing of milk products, acting as care providers, feed gatherers, and birth attendants. Women are typically responsible for milking, processing, and selling milk products, providing fodder and water, caring for newborn calves and sick animals. The data reflects that independent participation of women is highest in homestead garden related activities followed by livestock management. Similar is the case for responsibility profile as activities in these two areas are confined to household vicinity.

A critical analysis of participation of women in different cropping systems has shown that sorghum based cropping system is the only one where participation of women as equal partners is relatively higher. Though the largest participation of women in percentage terms is observed in paddy and wheat-based systems, they play a subordinate role in both these. Participation of women was the minimum in cotton based cropping system as 58.6% of the activities were exclusive to men.

The rural Indian women and the forests are inter-related and inter-woven. They are the major collectors, processors, and users of Non-timber Forest Products (NTFP) than men. Forest is vital for them in providing NTFPs as the main source of livelihood. There are 66.5 million tribals in India and with few exceptions; most of them are forest dwellers. The women collect fodder, fuel, food, building materials, household materials, Sal and Tendu leaves and farm implements. The tribal women obtain maximum earnings from fuel

wood and salia leaves. Women also collect medicinal plants as well as possess the traditional knowledge on fodder collection of properties and potential uses of these products. Hence, they are also carriers of indigenous and traditional knowledge. Women are involved in fish production, drying/curing, marketing, shrimp processing and net mending in the coastal areas including Maharashtra, Tamil Nadu, Andhra Pradesh and Odisha. In addition, in Andhra Pradesh, women are engaged in mollusk and shell collection on seasonal basis. Among the mangroves of Bhitarkanika on the Odisha coast, both women and men fish in the freshwater.

Gender disaggregated data on various aspects of women in agriculture show that the independent role of women in all the activities has decreased over the years. Apart from the homestead activities that are traditionally considered to be women's domain, their independent role is the highest in livestock production management, followed by post-harvest management. The joint participation of men and women is the maximum in fisheries, followed by horticulture and post-harvest management. Analysis of the sub activities in this area reflects that women are involved in home-based post-harvest operations whereas more men were involved not only in field-based activities but also activities outside home such as processing, sale, etc. The livestock management performed by women were more laborious and time consuming than the activities performed by men.

Women are involved in fish production, drying/curing, marketing, shrimp processing and netminding in the coastal areas including Maharashtra, Tamil Nadu, Andhra Pradesh, and Odisha. In addition, in Andhra Pradesh, women are engaged in mollusk and shell collection on seasonal basis. Among the mangroves of Baitarani on the Odisha coast, both women and men fish in the freshwater.

Entrepreneurship is the process of starting a business by acquiring financial, physical, technical, and human resources in relevant areas and developing the required management skills to run the business successfully. Broadly speaking, entrepreneurship involves two things; an enterprise and capacity building to establish and run the enterprise. There is a big gap to bridge between a Women SHG and Group Entrepreneurship.

15.3 Gender Issues

1. Variation among Households on Private Resources

Land owning status, house structure and household assets decide to great extent the attitude and motivation of the people to participate in the watershed. The more is the diversity the greater will be the difference in

motivation. In rural areas, the type of farm decides the sway of a household in programme planning and implementation. The success of the watershed depends to a great extent the attitude and motivation of one and all.

2. Cultural Compatibility

Each society has contained well agreed upon norms to celebrate / observe / perform some rituals / activities. The interventions / technologies are judged by the people basing on how is it compatible / desirable? The values and cultural patterns are employed to decide this issue. Generally, programmes / innovation antagonizing the societal norms are seldom accepted and sustained. Perceptions of the people are an important component of culture that may oppose or favor the project.

3. Conflict of Demands and Interests

The rural communities have many social groups created on common interest who often do not accept proposals that go against their interest. Contradiction within a system may leads to conflicts among farmers. Without identifying links between various groups, joint determination of the use of and access to, resources is not sustainable.

4. Co-operation of People

There is a general complain in the watersheds that people do not understand the importants and do not co-operate with the programme. Wherever a resource (land / water / forest) to be managed has some of the characteristics of a public good, co-operation of the people is a vital and necessary component. Some kind of adjustment and accommodation in social decision-making system are required to accommodate conflicting demands.

5. Technological Appropriability

It refers to how for agricultural technology, be it seed, fertilizer or improved machinery suits the social and infrastructural situation of the end users. It should also ensure efficiency over the current practice being followed by the farmers.

6. Traditional / Social norms in Management of Common Property Resources

The number of resources varies across states, but agriculturally less progressive state is found having more resources (water, land, forest) which provides income in terms of (irrigation, fodder, fuel and grass). Social norms in managing resources and distribution of benefits go a long way for the

success and absence of this would hamper the implementation in developing a conducive social environment.

7. Recognition to Women Folk

Women in rural areas perform triple role i.e., production, reproductive and community welfare but remain invisible in the family, community, and society. They work hard in farm sectors namely crop production, vegetable farming, livestock, pisciculture, post-harvest and value addition and market. They also perform household chores and render services for welfare of community. The cultural sanction given to women is deplorable for which they remain stagnant and de-motivated. With due recognition to their contribution, they will rise and play a vital role in the watershed programmes.

8. Participation

The issue of participation especially the gender participation is crucial for the success of the watershed. Participation of households belonging to the lower strata remains in question. Similarly, women do not get opportunity to plan, implement, monitor, and evaluate the program, even though they have knowledge, experience, and capacity to perform. At household level, they do not enjoy freedom to decide and their access to and control over productive resources is very limited. They fail to show their talents because of their subordinate position in the society. Gender equality in opportunity is right way to bring women to main stay of development. Even women's mobility especially journey to distance places to participate in trainings, undertake jobs etc is limited.

9. Sharing of Benefits

Every one of the social systems should get benefit from the watershed otherwise it would not be sustainable. Owning the programme, a positive behavior emanates from benefit sharing. Landless farmers and poorest of the poor should not be left out of the programme and must get benefit in accordance with their contribution. Usually, women support different productive activities in the home and community, but they are paid less than men. Gender equity is an important issue to be looked upon while implementing the program. It is not only the monetary benefit but other social benefits like nutrition, employment, fuel, fodder, and leadership should be kept in view while compensating their contribution.

10. Differences between Common Property Resources (CPRs) and Private Property Resources (PPRs)

Participation as a process can be promoted if there is increasing adherence to a set of rules. It is furthered if there is equal access to CPRs and a high

level of development at the village level. Linking the CPRs and PPRs is very much wanted to define how different social homogenous group would equal access to CPRs. Grouping of households, considering the links between CPRs and PPRs and levels of benefits derived is complex problems both statistically and interpretationally.

15.4 Women-friendly Technologies

Poor nutrition is one of the damaging outcomes of gender inequality. Women and children are worst affected due to nutritional insecurity. As per 'The State of Food Insecurity in the World 2012', India remains home to the largest number of undernourished people in the world: 217 million (17.5% of its population) as of 2012. However, the status of hunger and malnutrition in India varies according to different sources/ estimates and goes up to 67% to 77%. Nevertheless, 75% Indians suffer from hunger to varying degrees, nearly 50% of them acutely.

Hunger and malnutrition are, to a large extent, two sides of the same coin. The recent Global Survey Report (2012), released by Save the Children, says that 42% children in India are underweight and 58% are stunted by two years of age. Moreover, under nutrition among women increases reproductive and maternal health risks and lowers productivity. This situation contributes to women's diminished ability to gain access to other assets later in life and undermines attempts to eliminate gender inequalities. Though the government has tried to address the issue through nutritional supplements and some other indirect measures, sustainable solutions to the problem still elude us. Improving the performance of agricultural sector through gender focused agricultural innovations at micro level would hold the key to overcoming the problem of household food and nutritional insecurity.

15.5 The Road ahead

Efforts for gender mainstreaming are required to bring social, cultural and attitudinal changes which not only strive for ending the invisibility of women's contribution to agriculture, but of eliminating the drudgery that blights the lives of millions of working women in India. It is important to recognize that women's empowerment through technologies can raise their status only through meaningful stimulation. There is therefore, needed to have the participation of women at every level in decision making, program formulation and implementation.

There is a need to tailor technologies to meet the needs of women agricultural workers- and to make them cheap enough for women to access.

It should be ensured that technologies are accessible, income generating, poverty reducing, women friendly and drudgery eliminating, both for pre- and post- harvest operations. The technologies that are reducing drudgery of women must be publicized through the media and as an extension activity. These need to introduce them into the capacity building programs for reducing drudgery and increasing output of women workers.

Agricultural development programs are usually planned by men and aimed at men. Our extension system is male dominated and the extension service given to women is very limited. Expanding the sphere of women extension workers is required who can better identify the women's needs and constraints, priorities, and opportunities to meet their requirements. Adequate funds need to be provided to women extension functionaries for field activities and its recognition. For effective extension service, a greater number of change agents, training programs, infrastructural facilities, allocation of gender budgets, sensitization of extension functionaries, etc. should be made and need-based extension models may be developed.

Let us sum up

Accumulating evidence show that empowering women is not only important, but also often highly conducive to improving agricultural productivity, food security, and nutrition. Research and development of agricultural technologies and interventions should begin with an understanding of how men's and women's interests as producers and consumers work to address the needs of both as equal partners. Only gender mainstreaming in agriculture can lead to productivity gains and improve developmental outcomes for the next generation. Let us all make a concerted effort to improve the might of women in agriculture today for more productive and sustainable agriculture.

Glossary

Self-Help Groups (SHGs) are informal associations of people who choose to come together to find ways to improve their living conditions.

Social integrity – SHGs encourages collective efforts for combating practices like dowry, alcoholism etc.

Gender Equity – SHGs empowers women and inculcates leadership skill among them. Empowered women participate more actively in gram sabha and elections.

Check your progress

1. What is gender mainstreaming in water sector?

Gender mainstreaming in development projects refers to the incorporation of a perspective that promotes gender equality and women's empowerment in all stages of a project (i.e., planning, implementation, and monitoring and evaluation).

2. What are the problems associated with watershed management?

The final effects of watershed degradation - soil erosion, landslides, heavy sedimentation, water pollution, floods and droughts, etc. - must be identified as quickly as possible.

3. What are the objectives of gender equality?

Gender equality is a goal and a prerequisite for sustainable and peaceful development. Gender equality is achieved when women, men, girls and boys have equal rights, conditions and opportunities, and the power to shape their own lives and contribute to the development of society.

Suggested Readings

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Unit 16

Problems and Prospects in Watershed Management

Structure

Learning Objectives

Overview

16.1 Problems and Constraints in Watershed Management

16.2 New Prospects and Opportunities Associated with Watershed Management

16.3 Key Challenges for Watershed Management

16.3.1 Impacts of management interventions are difficult to assess:

16.3.2 Areas of decision making are not identical with watersheds:

16.3.3 Watershed management must face competition and even conflict:

16.3.4 Global Climate Change

Let us sum up

Glossary

Check your progress

Suggested Readings

Web Sources

Learning Objectives

After studying this unit, you should be able to learn the following,

- Problems and Constraints in Watershed Management
- New Prospects and Opportunities Associated with Watershed Management
- Key Challenges for Watershed Management

Overview

India occupies approximately 2.4% of the total geographical area of the world, while it supports over 15% of the world population. Unprecedented population pressure and demand of society on scarce land, water and biological resources and the increasing degradation of these resources is

affecting the stability and resilience of our ecosystems and the environment. Therefore, the productive agriculture lands in the country are in constant process of varying degrees of degradation and are fast turning into wastelands. It is precise to restore this ecological imbalance by developing the degraded non-forest wastelands. To harness the full potential of the available land resources and prevent its further degradation, wasteland development is of great significance. The problem of degraded land, water and its management are complex and multi-dimensional, and its development aims to develop human resource in watershed development and management and generate awareness about the importance of sustainable development and maintenance of existing work force working in the watershed development and develop skill in the rural youth to work in the watershed development based on watershed management approach and developing natural resources on sustainable basis.

16.1 Problems and Constraints in Watershed Management

Watershed management encounters many problems and constraints. Some of the major problems and constraints in watershed management are listed below.

a. Land degradation in rain fed areas due to soil erosion from runoff is one of the major problems. In India it was estimated that the soil erosion in the 1990s was almost double that of soil erosion in the 1980s. Rainfall uncertainty and poor economic conditions act as a major constraint and thus prevent the farmers in rainfed areas from making investments. This leads to improper watershed management.

b. Equitable benefit sharing of watershed management within the farming communities as well as within the different locations of watershed is a huge problem. Generally, women, marginal farmers and landless laborers gain very little or nothing at all from the watershed management activities. Several case studies in water scarce states of Gujarat and Madhya Pradesh in India have showed that overdevelopment of water harvesting structures in the upstream portion of watersheds had significantly reduced the inflows into the downstream reservoirs. On the other hand, it is also noticed that building of large reservoirs resulted in the submergence and hardship in the upstream parts and benefits for people in the downstream parts of the same watershed or a neighboring watershed generally having an urban or an industrial area.

c. Acute shortage of water in general and drinking water especially in summer has been observed in many watersheds with inadequate

watershed management which may result in severe/ recurrent droughts. It may often result in limited and temporary food productivity gains.

d. Many a times, common lands do not get treated adequately and re-vegetation does not take place as expected in spite of the watershed management programs. As a result of this, domestic/ ecosystem water needs and livestock water/ fodder needs are either inadequately addressed or are made to suffer due to increased water withdrawals by other uses or due to overgrazing.

e. Problems exist or new problems crop up due to improper understanding of the interaction between biophysical and socio-economic processes in watershed management.

f. Conflict among various government ministries such as those related to agriculture [with emphasis on food production], rural development [with emphasis on employment generation & poverty alleviation], forests [with emphasis on maintaining biodiversity & wildlife], as well as conflict between government bureaucracy and elected representatives in their zeal to control funds, is a major problem in watershed management programs -which requires to be resolved on a priority basis.

g. It is hard to conduct meaningful impact assessment studies on watershed management programs for lack of baseline data for monitoring and comparison of the current conditions. The whole exercise of watershed management is undertaken without properly estimating the water supply scenarios under drought/ normal/ surplus years as well as without proper demand management especially during drought years.

h. Large areas inhabited with tribal population lack facilities to harvest water and to stabilize their food/ crop/ fodder production due to reduced forest yields, deterioration in land quality, lack of tribal agriculture policy and population pressure. This leads to sustained misery, socio-political unrest, and insurgency among the tribal population.

16.2 New Prospects and Opportunities Associated with Watershed Management

In spite of the above-mentioned problems and constraints as well as some other problems and constraints, watershed management is associated with new prospects and opportunities. Some of them are listed below:

a. There is a need to produce more and better food without further

undermining the environment/ ecology, especially the land, water, forests, wildlife and atmosphere. This may include adoption of best management practices (BMPs) such as organic farming, de-silting for reservoir capacity restoration as well as for crop productivity increase, sprinkler and/ or drip irrigation to avoid excess use of water, no tree felling policy, afforestation and arboriculture through high oxygen yielding & other medicinal plants etc.

b. There is a need to ensure that gains due to groundwater recharge are not dissipated by excess groundwater extraction. To achieve this, groundwater over-extraction should be avoided through public awareness and also through regulation.

c. There is a need to consider the downstream impacts of intensive upstream water conservation. For this, watershed associations with representations from all the stakeholders in the watershed should be made operational. These associations can take decisions in the best interest of all the people concerned.

d. Decreasing the costs at which the gains are achieved and thereby increasing the modest benefit-cost ratio should offer new prospects and opportunity in watershed management. To realize this, low-cost technologies which may involve local materials, labour at practically no cost, technologies which are traditional, and time tested should be employed to generate more benefits spread over the entire watershed among all the stakeholders.

e. Increasing all sections of people's participation beyond the project implementation stage to ensure sustainable watershed management should be a top priority. Only this can ensure progress on a sustained basis overcoming the hydro-geological, socio-political, and other uncertainties.

f. Many successful watershed management programs -especially in India, have been implemented on a small scale in a few villages by collaborated efforts among the government departments, non-governmental organizations (NGOs) and research organizations. They represent sporadic BMPs. Hence there is a need to scale up the watershed management activities over large areas which could include remote and/or difficult terrains, so that many problems affecting our agricultural, rural and forest sectors can be effectively addressed.

g. Since there have been no or very few institutions built for research & development on collective management of watersheds, there is a need

to build centers of advanced learning employing the modern tools of remote sensing, geographic information systems, decision support systems, computer-based planning tools, poverty & socio-economic analysis etc.

h. There is a need to preserve and improve common pool resources (CPRs) of land, water, fodder, forest, fisheries, wildlife and agriculture which significantly contribute towards people's livelihood especially in the rural areas.

i. There is a need to minimize migration to urban areas by creating opportunities in agriculture, natural disasters like floods/ droughts, forest/ mountain economies and by arresting fall in agricultural prices, gap in urban/ rural wages, gaps in urban/ rural employment opportunities.

16.3 Key Challenges for Watershed Management

From an organisational or a managerial point of view, there are some key challenges for watershed management:

16.3.1 Impacts of management interventions are difficult to assess:

Watershed management addresses issues related to very complex ecological processes. These depend on many physical parameters that vary in space and time, such as rainfall intensity, soil retention capacity and vegetation types. The complex interactions between upstream causes and downstream impacts inevitably mean that the consequences (impacts) of human activities and management interventions are difficult to monitor, evaluate and quantify. This in turn means that it is often difficult to identify "best" watershed management practices.

16.3.2 Areas of decision making are not identical with watersheds:

Watersheds are drainage units, delineated by topographic features of the landscape. On the other hand, watershed management is directed at stakeholders and land use systems that are a part of administrative units, which do not usually coincide with watersheds. Decisions made by stakeholders are usually made with respect to administrative units (provinces, districts, communes), or ecological zones, or land tenure units. Because these locally relevant spatial units often cut across watersheds, there can be conflicting goals and priorities. Watershed management must accommodate this fact.

16.3.3 Watershed management must face competition and even conflict:

Often, there is disagreement between upstream and downstream stakeholders and administrative units at various levels. Often downstream stakeholders set goals for management intervention measures which must be implemented upstream, but which are potentially incompatible with the needs and priorities of the upstream water users. This often expresses itself as economic, political, or cultural disputes. Such upstream downstream conflicts are less obvious in small watersheds, where the same community shares both upstream and downstream resources. In larger watersheds or river basins, economic and political power is almost always concentrated in the lowlands and downstream regions. This can lead to a unilateral downstream dominated approach, with low levels of consultation with upstream and upland stakeholders.

16.3.4 Global Climate Change

Global climate change refers to the gradual warming of the earth caused by the so-called greenhouse effect. This effect is believed to be the result of man-made emissions of carbon dioxide, chlorofluorocarbons (CFC's) and methane. The anticipated consequences of global climate change are a rise in mean sea level, changes in weather patterns and in global atmospheric circulation. The assessment of the potential impacts upon hydrology and water resources has received increased attention as part of the wider examination of the consequences of climate-induced changes in water availability. The question is how the range of possible changes in temperature, precipitation and runoff is likely to affect water use and what the socio-economic implications might be.

It is difficult to predict or quantitatively assess the impacts of climate change on water management. Despite the uncertainties, however, the impact of climate change on water availability is likely to create serious social problems in most vulnerable regions of the world. There is a growing body of evidence that points towards an increased incidence of climatic extremes such as cyclones and intense weather systems which lead to floods. In addition, the increased frequency of anomalies in global circulation is seen as the major cause of long-term drought conditions in East Africa and Central Asia. Although the world is anticipated to become "moister", some regions will experience a decrease in precipitation, accompanied by increased evapo-transpiration due to higher temperatures. Surface and groundwater supplies would be reduced, demand would increase and lead to increased competition for a decreasing resource as well as the

development of conflict potential. The most serious problems are likely to arise in the agricultural sector, globally the largest consumer of water.

As human populations in developing countries increase during the next century, the area of irrigated croplands will have to increase to guarantee food security. Some recent studies have indicated that for each 1 °C increase in long term average temperature, a 12 to 25 % increase in irrigation demand is likely.

Let us sum up

A balance between economic and environmental objectives and consideration of all interactions of the watershed system are important criteria in watershed management. This balance is necessary for countries at various stages of development. Conflicts are increasing over shared water resources between agriculture, industry, and urban domestic use as well as between State governments. Sustainable water management is thus crucial for economic development and the livelihood of the people. In a country like India, where a lot of running water goes waste, it becomes very important to apply the technology of watershed management to solve its annual problems of droughts and floods.

Glossary

Climate change refers to long-term shifts in temperatures and weather patterns.

Watershed management is a term used to describe the process of implementing land use practices and water management practices to protect and improve the quality of the water and other natural resources within a watershed by managing the use of those land and water resources in a comprehensive manner.

Ecology is the study of organisms and how they interact with the environment around them.

An ecologist studies the relationship between living things and their habitats.

Check your progress

1. What are the Socio-economic Problems of watershed Management?
    - Rural poverty in the uplands, causing migration to crowded urban centres Food insecurity.
    - Degradation of land through improper land use and unsustainable farming reducing productivity and income

- Rapid population growth (in the LMB estimated to be 2 - 2.5 % per year during the next 20 - 30 years), which increases pressure on natural resources and land scarcity.
- Increasing conflicts over land, forest and water resources
- Increased competition for resources and resource allocation difficulties
- Lack of access to knowledge and decision-making powers for women
- Poor infrastructure and therefore limited access to markets, health care and education
- Expanding irrigation leading to increased water demands

2. What are the Ecological Problems of watershed Management?
   - Decreasing amount and quality of water resources
   - Damages caused by natural disasters (heavy storms, landslides, wildfire, etc.)
   - Extensive and rapid deforestation and forest degradation through legal and illegal
   - exploitation, which can reduce the replenishment of groundwater and increase the amount of sediment in surface water
   - Clearing and conversion of forests mainly into agriculture land uses
   - Increased run-off and erosion
   - Siltation and sediment discharge declining on the mainstream, through deposits of sediment in reservoirs and irrigation systems on the tributaries.
   - Reduced biodiversity with known and unforeseeable impacts on ecological integrity and food production

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Document Information

Analyzed document	MGEOSE-31 Watershed Management_Book.pdf (D166677684)
Submitted	2023-05-12 07:22:00
Submitted by	
Submitter email	kumarnano@gmail.com
Similarity	10%
Analysis address	kumarnano.tnou@analysis.urkund.com

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About Tamil Nadu Open University



Tamil Nadu Open University (TNOU), with its Headquarters at Chennai was established in 2000 by an Act of Tamil Nadu Legislature at the State level for the introduction and promotion of Open University and Distance Education in the educational and for the co-ordination and determination of standards in such system. The salient features of TNOU are , relaxed entry rules, maintenance of standards, individualized study, flexible in term of place, duration of the study, use of latest information and communication technology, well-knit student support services network, cost effective programmes, collaboration and resource sharing with other Universities.

School of Sciences

School of Sciences, established in 2004, has been offering the B.Sc. and M.Sc. programmes in Mathematics since 2005 and B.Sc., Mathematics with Computer Application since 2007. In 2017, B.Sc. programmes in Physics, Chemistry, Botany, and Zoology were introduced, while M.Sc. programmes in Physics, Chemistry, Botany, and Zoology were launched in 2018. As per the academic restructured, the Department of Geography and Apparel & Fashion Design were merged in the School of Science in 2020 and these departments are offering B.Sc., and M.Sc., Programmes.

The main objective is to excite the brains and hearts of rural students through constant inquiry and active participation in Science. The School of study has blazed a trail of information transmission and generation, graduating over 25000 Science students across the Nation. It has built a niche for itself in the core areas of teaching, research, consultation, administration, and community services over the last 17 years.

The School of study consists of the Departments of Physics, Chemistry, Mathematics, Botany, Zoology, Geography and Apparel & Fashion Design. Moreover, all the above said. Departments offer various academic Programmes from Certificate to Research degree level (M.Phil., & Ph.D.) in their concerned disciplines.

The Department of Geography offers the following Programmes

- **B.Sc., Geography (Semester - Both Tamil & English Medium)**
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- **M.Phil., Geography (Full Time & Part-Time)**
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